

Econ 220 The Economics of Human Flourishing

Numbered Transcript

ECON 220 • Dr. Marian L. Tupy & Dr. Gale L. Pooley

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Lecture 1: Ingenious Economics

At a glance — Tupy introduces 'ingenious economics' and human flourishing.

- 1.1** Hello and welcome to our course. My name is Marian Tupy. I'm the founder and editor of HumanProgress.org and a senior fellow at the Cato Institute in Washington, D.C. The course introduces the new economics of knowledge, learning, and time. In this course, you will learn about the ideas, principles, and analytical framework in Superabundance, including the relationship between population growth and resource abundance, how to use time prices to measure resource abundance, the importance of innovation, as well as how and why innovation happens, the reasons for rational optimism about the future, and present threats to human flourishing. So, thank you once again for taking this course and allowing Gale and me to share our new approach to understanding how people create value for one another and lift the world out of poverty.
- 1.2** Let me begin with a quotation. The quotation comes from an immensely popular 2018 movie, Avengers: Infinity War. You will recall that Thanos is an intergalactic warlord searching for superpowers that will allow him to destroy half the life in the universe. This, he feels, is necessary, and even, in a way, good. He believes that the universe's resources are finite and the needs of the living creatures are infinite. He reasons that by reducing all life by half, plenty of resources will be left for the rest. He succeeds and disaster ensues. So, as we go through this course, remember Thanos's words: "It's a simple calculus. This universe is finite, its resources finite. If life is left unchecked, life will cease to exist. It needs correcting."
- 1.3** At this point, you may wonder, "What does a fictional character from a superhero movie have to do with real life?" The question of the desirability of population growth is not hypothetical or science fiction. It's real, current, important, and still unsettled. Many influential people we will encounter during this course share Thanos's ideas. Paul Ehrlich, he was invited to address a Vatican workshop on biological extinction in 2017. "You can't go on growing forever on a finite planet," he said. "The biggest problem we face is the continued expansion of human enterprise. Perpetual growth is the creed of a cancer cell. It's the aggregate consumption that ruins the environment." In February 2019, Congresswoman Alexandria Ocasio-Cortez told her Instagram followers, "It's basically a scientific consensus that the lives of our children are going to be very difficult, and it does lead young people to have a legitimate question: Is it OK to still have children?"
- 1.4** Countless of other politicians, businessmen, and journalists, and academics expressed similar views over many decades. And in our book, you can find thousands of such references. The impact of this fearmongering is easily discernible. In 2021, researchers from the University of Bath in England polled 10,000 young people between the ages of 16 and 25 in Australia, Brazil, Finland, France, Great Britain, India, Nigeria, Philippines, Portugal, United States. And this study was the latest and the most comprehensive

study of young people's perceptions of the state of the planet. The researchers found that, on average, 83% of respondents thought that people have failed to take care of the planet. 75 % thought that the future is frightening. 56% thought that humanity is doomed. 55 % thought that they would have less opportunity than their parents. And finally, 39% thought that they were hesitant to have children.

- 1.5 Now, the average total fertility rate to keep population stable or constant has be 2.1 children per woman per lifetime. Yet, out of 195 countries and territories, 89 had a fertility of two or less in 2018. The list of countries with lower than replacement total fertility rate is dominated by advanced nations, including France, Sweden, Denmark, United Kingdom, United States on 1.7 instead of 2.1. Germany, Netherlands, Norway, Austria, Canada, Switzerland, Finland, Japan, Portugal, Italy, Spain. South Korea is on one instead of 2.1, one. Lancet, which is a well-known British scientific journal, estimates that the world population will peak in 2064 at around 9.7 billion people and then fall to 8.8 billion in 2100, with 23 countries seeing population shrink by more than 50%, including Japan, Thailand, Italy, and Spain.
- 1.6 Should we welcome or should we bemoan these trends? That, I think, partly depends on answering the following questions: "What is the relationship between population growth and resource abundance?" Does a growing population make resources scarcer? After all, if it is true that resources are finite, the more people there are, the fewer resources every additional person must have. And the second question: "Is humanity on average a net positive or a net negative, regarding the stewardship of the planet's resources?" After all, if all we do is to consume, fewer people would surely be better for the future of our species and the world. The answer to these questions has profound moral and practical implications for all of us. Contrary to the population pessimists, whom we shall call Malthusians, for reasons that will become shortly apparent, Gale and I will argue the opposite. Using logic and evidence, we will argue that humans are not only consumers of resources, they are also creators of new resources, which is why the 8 billion people who live on the planet today have much better living standards than the 1 billion people who were alive in 1800.
- 1.7 200 years ago, people lacked basic medicines and died relatively young. They had no painkillers, and people with ailments spent much of their time, much of their lives in agonizing pain. Entire families lived in bug-infested dwellings that offer neither comfort nor privacy. They worked in the fields from sunrise to sunset, yet hunger and famines were commonplace. Transportation was primitive, and most people never traveled beyond the confines of their native villages or nearest towns. Ignorance and illiteracy were ubiquitous. But that 's not how most of the world lives today. As such, we need to be open to the possibility that growing population does not need to retard human progress. Quite the contrary, population growth enhances it.
- 1.8 How is that possible? In a word, knowledge. Committees don't have ideas. Algorithms don't have ideas. Machines don't have ideas, at least not yet. So far, all the new ideas

have always been a product of human mind. These ideas lead to inventions. In turn, inventions tested in the market to separate good inventions from bad inventions lead to innovations which drive productivity gains, economic growth, and increases in standards of living. The more people we have, the more likely it is that some of them will have ideas which will change the world for the better. They might even help us to protect the environment.

- 1.9** Our book tells the story of population growth, innovation, and human flourishing on an infinitely bountiful planet. Now, implicit in our title is the notion that infinite growth, or, if you will, endlessly expanding living standards for billions of people who are alive today and who will be born in the future, is possible. But how that can be achieved if the resources in the world are really finite? Now, that depends on what your definition of a resource is. The Oxford Dictionary describes a resource as "factors of production, provided by nature." This includes land suitable for agriculture, mineral deposits, water resources useful for power generation, transport and irrigation. It also includes sea resources, such as fish and offshore mineral deposits.
- 1.10** Now, Gale and I find this definition deeply unsatisfying. Why? Land suitable for agriculture can be expanded through the application of artificial fertilizer or genetically modified crops developed especially for arid regions. Water for irrigation can be expanded due to desalination. Fish can be grown for human consumption on farms rather than caught in the wild and so on.
- 1.11** The great American economist Thomas Sowell, it seems to us, was onto something when he observed, "The cavemen had the same natural resources at their disposal as we have today. And the difference between their standard of living and ours is the difference between the knowledge they could bring to bear on those resources and the knowledge used today."
- 1.12** And knowledge, unlike atoms, is potentially infinite. Once you start thinking in terms of our expanding and cumulative knowledge, rather than the finitude of the atoms, the definition of a resource expands as well. So, let's explore some of the ways in which new knowledge can create more resources. First, notice that we constantly increase our understanding of useful atoms that can help us to sustain civilization. Not long ago, hydrocarbons were a flammable nuisance that made the lives of farmers much more difficult. But once we developed the internal combustion engine, hydrocarbons, which is to say oil, became both necessary and valuable.
- 1.13** Second, useful atoms can be made ever more valuable. Consider something as simple as a grain of sand that's been lying around for billions of years. At some point in the past, thousands of years ago, someone figured out that, by heating sand, he or she could turn it into glass. And so people started wearing glass beads, and then they turned glass into glass jars and later into window panes. Now we are using glass to create microchips and fiber-optic cables. So, with every step of the way, the value added in a simple grain of sand has expanded tremendously.

- 1.14** Third, consider increased supply. When the price of a resource increases, people have an incentive to look for more resources. We have only surveyed a small fraction of the planet. We have absolutely no idea what's happening at the bottom of the oceans. So, as surveying and extracting technologies improve, we'll go deeper, we'll go faster, we'll go cheaper, we'll go cleaner to access the untouched minerals that still remain. Fourth, consider efficiency gains. In the late 1950s, when the Coca-Cola can first came onto the market, it weighed three ounces of aluminum. Today, it weighs less than half an ounce.
- 1.15** So, consider all the environmental energy and raw material savings due to efficiency. Market incentives provide people with more opportunities for new knowledge creation to lower the cost of an input so that you can lower the cost of the output. The input here is aluminum, the output here is Coca-Cola can. Why pay more for an input if you don't have to? You can increase your profit margins by decreasing the price of the output. So, as technology will continue to improve, as Andrew McAfee from MIT said, "We can create more with less," or more from less.
- 1.16** Fifth, consider substitution. Long before whales, those beautiful creatures in the oceans, ran out, we used to use them in order to create candles in order to light our homes. Spermaceti, which is a waxy substance in the whales' brains. We used to kill them, extract that spermaceti, and then turn it into candles. But a long time before we killed all of them, we, of course, discovered gas and oil and electricity. Now, perhaps some of you are worried about lithium. As you know, future electric cars will be run on lithium-ion batteries. Are we going to have enough lithium? Well, who says that, in 20 years' time, we are going to be using lithium in order to power our electric vehicles? Maybe it will be sodium-ion, and there is much more sodium on Earth than lithium.
- 1.17** Six, consider dematerialization. Not long ago, every hotel room in America came equipped with a long, thick blue cable filled with copper, and that was a way that you would plug your laptop into the Internet. Today, they're all gone, and all that copper is being used somewhere else. What has replaced it? Wi-Fi. When you go into a hotel, you log on to Wi-Fi. No atoms are necessary. So, all that copper can be used somewhere else. What about your iPhone? Think about the many ways in which it has dematerialized the world. By having a smartphone, you don't need a calendar. You don't need a map. You don't need a radio or a camera or a telephone. You don't need snail mail, that's for sure. You don't need stereos. You don't need an alarm clock. You don't need a dictionary or an encyclopedia. All of that is combined in that little thing called your smartphone.
- 1.18** Seventh, consider recycling. Aside from a minuscule proportion of titanium and aluminum, which we shot out into space in the form of rockets, all the other materials that have ever existed on Earth are still here. Now, vast quantities of steel may have been used up in order to build our buildings, and we may have used a lot of copper in our cables. But all that steel and all that copper could be reused in the future.

- 1.19** If, for some reason, we cannot produce any more steel, we can tear down those skyscrapers and get the steel out of there. This has happened before. Some of you may have seen the movie Oppenheimer. In order to create the atomic bomb, they needed to create these huge electromagnets, and back then, the best way to do it was by using silver. So, the U.S. government gave them 14,000 tons of silver to use these electromagnets in order to produce an atomic bomb in 1945. After the Manhattan Project was over, all that silver was returned back to the American government to prop up the value of the American dollar. But what if, in some distant future, humanity's ever-increasing consumption reaches a level where all the useful atoms are physically incorporated into objects that make our lives enjoyable? At that point, surely, economic growth—the expansion of human enterprise—must come to an end or even go into reverse.
- 1.20** So, consider the ninth way of increasing our resource base—the transmutation of metals. For millennia, transmutation of elements was the domain of alchemists. But then, in 1919, Ernest Rutherford achieved the first artificial transmutation of nitrogen into oxygen. Today, transmutation is all around us. In your smoke detector, in your home, there is some americium, an artificial element produced by transmutation. Nuclear physicists have achieved the transmutation of lead into gold many decades ago, but it still requires far too much energy to be a viable alternative to mining. But cost of energy is bound to fall. After all, we live quite close to a massive nuclear fusion reactor that's converting billions of tons of mass into energy. That huge fusion reactor is, of course, our sun, and in the future, we'll be able to capture as much energy from the sun as we would like.
- 1.21** Or we may find it more convenient to make our own fusion reactors here on Earth. All the elements found on Earth that you will find in the periodic table, other than hydrogen and helium, which were created shortly after the Big Bang, were made by transmutation inside stars. So, the Big Bang creates helium and hydrogen. But then everything else is created by fusion reaction within stars. And when those stars go supernova, when they explode, they populate the universe with all the elements which somehow found their way onto our planet when our planet was forming. In the distant future, we could use artificial fusion not only for energy creation, but also for transmutation of rare elements that we may be running short on.
- 1.22** All that we need in order to achieve that is abundant energy and hydrogen. And 71% of Earth is covered by oceans, which are full of hydrogen. Hydrogen is also the most common element in the universe. Perhaps, at some point in time, in the future, we are going to have the knowledge to be able to bring some of that hydrogen onto our planet. But more plausibly, long before all the useful atoms on Earth and in our oceans are extracted in order to create things that make our lives enjoyable, we will have the knowledge to obtain more atoms and more energy from asteroids and planets and beyond. In that future, just has always been the case in the past and still is the case

today, the only bottleneck will be the rate at which new knowledge can be created. And nothing prevents us from improving that rate, too, by creating relevant knowledge in turn. There are no resources besides knowledge, and no ultimate limits on the creation of knowledge.

- 1.23** So, now that I've explained all that, let's try to look at a little bit of history. I said that this is a very long-held question. People have been debating the question about resources in population growth for thousands of years. In ancient China, in the 6th century B.C., Confucius and many of his followers saw population in a relatively favorable light, but they also believed that population growth needed to be controlled. They theorized that there is an ideal ratio of land to population, and they reasoned that, when that ratio is exceeded, labor productivity and quality of life will diminish, leading to social discord. Therefore, Confucius argued, it was the duty of the government to maintain that ratio of land to people by forcing people to migrate to less populated areas.
- 1.24** The ancient Greeks analyzed population growth from the perspective of the city states. They were mostly concerned with the implications of human fecundity for governments and state security. They thought that population should be large enough to be economically self-sufficient, but not so large as to make democratic governance impossible. Plato, for example, specific ideal of his population that he had in mind was 5,040 people. Now, he advocated for reproductive incentives and immigration—people coming into the city states—if the population was too low, and birth control and colonization—emigration—if population was over his ideal number of 5,040. Similarly, Aristotle worried that since cultivated land could not be increased as quickly as population could grow, it was necessary to abort, expose children, let them to die.
- 1.25** The Romans analyzed population growth from an imperial or expansionist perspective, and they welcomed the contribution of population growth to the increases of military strength of the Roman Empire. And under Caesar Augustus, married and child-rearing couples were given special privileges, while unmarried couples were discriminated against, and celibacy was discouraged.
- 1.26** The Indian philosopher, Chanakya, expressed a similar view, arguing that a large population leads to greater economic and military power. But he also knowledge-checks on rapid growth of populations such as wars, famines, and plague. In 14th century, the Arab scholar Ibn Khaldun developed a fairly comprehensive theory of population. He argued that dense populations were key to rising standards of living, as they allowed for greater division of labor, and he thought that economic expansion and population growth went hand in hand. But he also warned that, in the wake of periods of economic advancements, came luxury, rising taxes, and other changes, which, in several generations, produce political decline, depression, and depopulation. So, he saw population growth in a cyclical way.

- 1.27** In 16th century, mercantilism began to take root, influenced by the ideas of Italian diplomat and philosopher Niccolo Machiavelli and Jean Bodin. Mercantilists believe that the goal of national policy ought to be the maximization of wealth and, consequently, the power of the state. They welcome population growth, believing that it would swell the government's coffers while depressing wages and the cost of labor. So, implicit in the mercantilist thinking was the notion of population growth while strengthening the state would also lead to immiseration of the populace.
- 1.28** The physiocratic school, an economic theory developed by a group of French economists in the 18th century, arose out of opposition to mercantilists. The physiocrats rejected the mercantilist idea that population growth should be welcome, even if overall standards of living fell. The physiocrats believe that agricultural production was key to economic health. They accepted the advantages of population growth as long as it went hand in hand with increased agricultural yields and, consequently, higher standards of living. Now, implicit in physiocratic thinking was that agricultural production could keep pace with population growth given the right economic policies.
- 1.29** As the 18th century progressed, some scholars started to think that population growth makes much greater optimism. They began to see human life as intrinsically valuable and problems concomitant with population growth as eminently solvable. The French economist, Nicolas Baudeau, for example, argued that productiveness of nature and the industriousness of man are without known limits because production can increase infinitely. As such, population numbers and well-being can go on advancing together. Condorcet professed the same faith in the power of science, which would be able to prolong lifespan without any worsening of human condition through the production of food, and also because reason would come to play, to prevent irrational population growth.
- 1.30** Other leading scholars of the day went as far as to argue that good government is the one that leads to the maximization of human population and its well-being. The Scottish philosopher, David Hume, for example, noted that wherever there are most happiness and virtue and the wisest institutions, there will also be most people. Even the French philosopher Jean-Jacques Rousseau held that government under which the citizens increase and multiply the most is infallibly the best. These were, to put in mildly, revolutionary ideas, and the almost inevitable backlash against them soon arrived.
- 1.31** Reverend Thomas Malthus was born in Westcott, England. He studied English, Classics, and Mathematics at Cambridge University. In 1789, he took the Holy Orders in the Church of England. And during his lifetime, Malthus became fascinated with geometric and arithmetic growth rates. A geometrically growing value increases proportionately to its current value, such as always doubling, for example, 1, 2, 4, 8, 16, 32, 64. An arithmetic growth rate, in contrast, increases at a constant rate, for example, 1, 2, 3, 4.

In 1798, Malthus published an essay on the principle of population as it affects the future government and society. He argued that population, when left unchecked, increases at a geometrical ratio—1, 2, 4, 8, 16, 32—whereas sustenance, food production, only increases at an arithmetic ratio—1, 2, 3, 4.

- 1.32** It was that difference between the two rates—the geometric ratio rising very fast and the arithmetic ratio rising very, very slowly—it was this gap that he believed would result in starvation. And so, Malthus warned that, "The proportion of births to deaths for a few years indicates an increase of numbers much beyond the proportional increased or acquired produce—food—of the country. We may be perfectly certain that unless emigration takes place—people leave—the deaths will shortly exceed the births. Were there no other depopulating causes, every country would, without doubt, be subject to periodic pestilences and famine."
- 1.33** Malthus acknowledged natural constraints on population growth, such as war, famine, and plague, but he thought they would cause a lot of misery, and therefore, were to be avoided. Now, Malthus believed that history validated his ideas, which it did. For thousands of years, the world was stuck in a Malthusian trap. The world's population fluctuated, growing during good harvests and collapsing when food got scarce. According to the U.S. Census Bureau, at the time of Caesar Augustus or Jesus Christ, there was anywhere between 200 and 400 million people in the world. 14 centuries later, it was still somewhere between 300 and 400 million. So, you can see that human population didn't grow. But Malthus also insisted that what was true in the past must be true for all eternity, and that was not to be. Malthus lost his main argument even before his book went into print. In the 18th century, the population of England rose from 5 million to over 8 million. Did people become poorer? No, they didn't. In fact, income per capita rose from 12 pounds to 23 pounds. It almost doubled. But then look at the bottom of the slide. The nominal price of a 4-pound loaf of bread increased from 5P to 7P, or 42%.
- 1.34** Now, what does this tell you? It tells you that GDP per person, income per person, increased at twice the speed of the price of a loaf of bread. Over those 100 years, income's increased by almost 100%, a loaf of bread increased in price by 40%. Therefore, relative to income, price of bread actually declined. Incomes were increasing at a faster pace, so the share of your income that you had to pay for bread continued to decrease. It is too bad that Malthus didn't check his theory against facts. Contrary to Malthus, there is no logical reason why human population should grow geometrically, why food production can only grow arithmetically. As Gale will show in subsequent lectures, empirical evidence suggests that the abundance of food can grow geometrically whilst population grows only arithmetically.
- 1.35** Yet, despite his failure to acknowledge the improvements around him, Malthus' ideas spread around the globe like wildfire. Part of this success rests in the human's brain's bias toward a negative, which we will discuss. And another part rests in Malthus' skill as

a mathematician and a writer. He covered his arguments with a patina of mathematical respectability. And he implied that anyone unfamiliar with mathematics could not comprehend the obviousness of his assertions.

- 1.36** Both of those mistakes are common in the writings of his disciples today. Let us turn to the 5 Core Arguments that today's Malthusians believe in. First of all, that population controls are justified and consumption limits are justified because of supply and demand. If planet can supply only a limited quantity of resources, and human demand for those resources continues to expand, then those resources will deplete and the catastrophe will ensue. This concern with the depletion of resources is precisely what drove Thanos to exterminate half the life in the universe.
- 1.37** The second justification for population controls ties to the first. People will enjoy a sustainable existence if the resource demand matches the supply. If the population declines, but everything else remains the same, the demand for resources will decrease and those who remain alive will enjoy access to more resources. Put differently, their standards of living will increase. This is to borrow from Milton Friedman, an American Nobel Prize-winning economist: "This is fixed pie fallacy." Since the size of the resource pie is supposedly fixed, the Malthusians say, fewer people there are, the larger part of the pie they will enjoy.
- 1.38** The third justification for population controls deals with diminishing returns on investment. According to this view, the planet contains finite amounts of high-quality land and easily accessible resources, that we have discussed before. Once those are accessed, it will become more difficult to cultivate more land and mine more resources. Low-quality land will produce less food; deeper mines will produce more expensive resources. Both will result in lower standard of living for an expanded population. Throughout this course, we will argue that the quantity of resources is not fixed. We will show you that human ingenuity allows for our species to expand the stock of planetary resources ad infinitum.
- 1.39** Our view directly contradicts the fourth justification for population controls and consumption limits, which states that technological innovation and scientific products cannot be substituted for existing resources. According to that view, technological solutions to the coming environmental catastrophe can never be as perfect as the ones that nature produces. Synthetic fertilizer, for example, may work to grow plants, but is generally less effective than regular soil nutrients. That would be the kind of argument that the Malthusian will present.
- 1.40** And the fifth justification for population controls states that past successes in overcoming natural limits are irrelevant to present conditions. Put differently, past successes in overcoming population growth and consumption increases are immaterial to the modern world because humanity's problems are much greater than they used to be in the past.

- 1.41** So, let's look at anti-Malthusian arguments, or counterarguments to Malthusian ideas. The first objection to Malthusians and to population controls holds that population growth enables humanity to produce more goods and services, cultivate more land, and access unused resources. Population growth, in other words, enables a greater division of labor, which makes production cheaper and more abundant. The overpopulation alarmists tend to focus only on growing demand, while ignoring the expansion of supply. When demand increases, so does supply. When demand decreases, so does supply. That's why locales with dense populations tend to be richer per person than the locales in sparse populations. Think about Hong Kong, how rich it is compared to Central African Republic, which has barely any people in it, but it's extremely poor.
- 1.42** The second objection to population controls holds that population growth increases rather than decreases, it increases returns on investment. Human beings are the only animal capable of sustained innovation. The greater the population, the more likely a creative mind will emerge to invent a solution to a pressing problem. Increase supply to meet growing demand. Moreover, humans are the only animal capable of building on past inventions. Therefore, as the population grows, information accumulates and innovation accelerates.
- 1.43** The third objection to population controls, or the third objection to Malthusians, holds that human standards of living are not constrained by local resources. And that's because human ingenuity enables people to transcend resource limitations. Humans can and do increase food supply by plowing more fields or increasing yields from the existing fields. By contrast, nonhuman animals must rely on what Mother Nature does, or does not, provide.
- 1.44** The fourth objection to Malthusians and their obsession with population control states that the past achievements are grounds for cautious optimism. Of course, we cannot predict the future. Consequently, there is no guarantee that human prosperity will continue indefinitely. That said, past provides a window into the future. As the British economist Peter Bauer noted, "It is only the past that gives us any insight into the laws of motion of human society, and hence, enables us to predict the future." As we will see throughout this course, looking at the long-term interaction between population growth and resource availability should fill us with cautious optimism. Unfortunately, Malthusian remedies for population growth were not limited to intellectual theorizing. In the mid-1970s, for example, India pursued policies of limited population growth by carrying out enforced sterilizations of both men and women and social ostracism of people with large families.
- 1.45** Nowhere in the world, however, did Malthusianism gain as much purchase as China between 1980 and 2015. It is in that Communist dictatorship that the most horrific abuses of the populace took place. The scale of human suffering caused by government-enforced sterilizations, prenatal and postnatal abortions, and wanton

cruelty is difficult to comprehend. Chinese Premier Zhou Enlai announced a five-year plan to limit population in 1970. The Chinese Communist Party believed that rapidly growing population would hinder China's economic development, and Zhou's plan eventually resulted in a two-child policy that held during much of the 1970s. Although China's fertility rate had plummeted, the government felt that population growth was still too high. And in 1980, the Chinese leadership organized a meeting to discuss further fertility restrictions. One person at the meeting was this man. His name is Song Jian, and he had recently returned from a European trip where he had read two influential books concerning the supposed dangers of population growth.

- 1.46 The first was the Club of Rome's 1972 report, *Limits on Growth*, or *Limits to Growth*. The report looked at the interplay between industrial development, population growth, malnutrition, and the availability of nonrenewable resources, and the quality of environment. It concluded that, and I quote, "If the present growth trends in population, industrialization, pollution, and food production, and resource depletion continue unchanged, the limits to growth on this planet will be reached sometime within the next 100 years. The most probable result will be a rather sudden and uncontrollable population and industrial capacity decline. Given present resource consumption rates and the projected increase in these rates, most currently nonrenewable resources will be extremely expensive in 100 years from now."
- 1.47 The other book that the Chinese saw was the *Blueprint for Survival*, which was originally published in 1972. Basically, in the *Blueprint for Survival*, that also created a huge worry about the future of the planet. The *Blueprint* noted that continued exponential growth of consumption of materials and energy is impossible. Present reserves of all but a few metals will be exhausted within 50 years if consumption rates grow. There will be discoveries and advances in mining technologies, they acknowledged, but these are likely to prove us only with limited stay of execution.
- 1.48 Song—this Chinese intellectual that I pointed out—became convinced that a one-child policy was necessary to curb China's population growth. Not two-child policy, as Zhou Enlai wanted, but one-child policy. And he calculated that the ideal size of Chinese population fell somewhere between 650 and 700 million people, which was roughly two-thirds of the Chinese population in 1980. Song presented his ideas to the Chinese Academy of Sciences and to the Communist Party. The Communist Party bought it and, as a result of which, the Chinese have implemented the one-child policy in 1980.
- 1.49 In a paper from a few years ago, Chelsea Follett from the libertarian Cato Institute in Washington, D.C., outlined some of the measures that the Chinese government took and implemented to achieve the policy goals of the one-child policy. She noted that government incentivized local officials with bonuses to abort pregnancies and sterilize adults, and punished local officials with wage cuts and dismissal for not meeting a quota of one child per couple. To achieve the quota, officials often destroyed property, jailed people, and even threatened to confiscate their children. Legal restrictions went

beyond the one-child limit. The government outlawed single-parent families and required married couples to obtain a permit to have a child. Women were sometimes punished for intimacy out of wedlock. Some were forced to take pregnancy tests every two weeks by urinating in cups in public.

- 1.50** In the 1980s, IUDs became state-mandated for women who had already given birth. Unlike typical IUDs, which last only a decade, the government approved IUDs modified to last until menopause and designed to be extremely difficult to remove. The government in China regularly performed X-rays on women with government-approved IUDs to ensure that contraceptive was still functional. After having two children, women were surgically sterilized. Between 1980 and 2014, the government's official statistics estimate that 324 million Chinese women were fitted with modified IUDs, and 107 million underwent tubal ligation surgery. They got their tubes tied. That averages to 9.5 million modified IUDs and over three million tubal ligation surgeries annually.
- 1.51** Punishment for noncompliance could be ruthless. One official that Follett mentions in her study recalls his part in enforcing a one-child policy, describing how he and his team tore down a family's house after the mother tried to avoid a state-mandated tubal tying. Fees for breaking the one-child policy were extraordinary, usually around 5-10 times of a person's annual disposable income. The Chinese government raked in close to \$3 billion from these social compensation fees in 2013, for example. If the poor families couldn't pay, state officials would send in forces to raid the parents' homes for valuables.
- 1.52** Children conceived illegally were born without "hukou", the legal registration necessary to study, work, marry, and have children. Around 13 million people in China still don't have this document and are forced to either pay huge fines, acquire false identity, or survive without access to mainstream society. One man that Chelsea Follett mentioned spent \$7,000 over three years in a failed attempt to get his adopted daughter registered so that she could go to school. Some Chinese families tried to get around the one-child policy by giving their illegal children to friends, family members, or neighbors. In response to these attempts to circumvent the one-child policy, the Chinese government outlawed adopting healthy children, except by elder childless couples. Children who were adopted illegally were seized by the Chinese government and put up for adoption internationally. Unfortunately, many children were simply aborted, killed, or abandoned by their families.
- 1.53** Given the Chinese cultural preference for sons, females were killed at a higher rate than males, causing a skewed gender ratio that peaked at 121 boys for every 100 girls in 2008. Consequently, many Chinese men fail to find a Chinese wife. Not surprisingly, the shortage of women has resulted in increased instability. One study found that for every percent increase and every 1% increase in gender disparity, there was between 5 and 6% increase in crime.

- 1.54 The state also forced women to abort their children. Often these abortions happened after six months of pregnancy. While these abortions were sometimes criticized in China, the criticism was never directed at the one-child policy. In 2012, for example, photos of a mother and the corpse of her would-be child taken shortly after the woman had been restrained and injected with an abortive agent went viral on Chinese version of Google. The chief editor denounced the forced abortion, but maintained his support for the one-child policy, stating that "the world's resources cannot afford to feed a China with billions of people." An editorial in the Global Times, an English language newspaper controlled by the Chinese Communist Party, argued that forced late-term abortions should be abolished, but the editorial endorsed the one-child policy, claiming that it "freed China from the burden of an extra 400 million people."
- 1.55 So, the bottom line is that Malthusian ideas are not dead. There are a lot of Malthusians around, be they on TV, in the newspapers, or our politicians who keep on claiming that we are going to run out of resources, that having more people in the world is a problem. When these Malthusian ideas got real traction, when they got a brutal government to believe in Malthusian ideas, it has resulted in the destruction of 400 million lives.
- 1.56 In this course, we are going to try to convince you of the opposite. We are trying to convince you that Malthusians are wrong, and hopefully, we will also convince you of the inherent value of human life. Thank you very much.

Lecture 2: Abundance Mindset

At a glance — The abundance mindset.

- 2.1 Welcome to our second lecture in our program, Introduction to Superabundance : The New Economics of Knowledge, Learning, and Time. In this lecture, I want to talk about something that we call "the bet". Now, recall that Dr. Tupy just talked about the one-child policy in China and the issues that were part of the anti-population policy in both China and India. While Limits to Growth and Blueprint for Survival were influential, we cannot ignore The Population Bomb, published by Paul Ehrlich in 1968.
- 2.2 Now, Professor Ehrlich is actually a butterfly scientist. That 's his area of specialty. So, when he approached population, he approaches it from this perspective. He focuses on this natural carrying capacity of insect populations. Now, the mistake that we believe that he made is applying his knowledge in that realm to this realm of human population. And a key difference between insects and human beings is that insects do not innovate. They do not innovate.
- 2.3 Now, this book opens up with this first sentence: "The battle to feed all humanity is over. In the 1970s and 1980s, hundreds of millions of people will starve to death in spite of any crash program embarked upon now." Now, Dr. Ehrlich had visited Delhi, India, and this is what he said about it. "The streets seemed alive with people, people eating, people washing, people sleeping, people visiting, arguing and screaming. People thrust their hands through the taxi window, begging, people defecating and urinating. People clinging to buses. People hurting animals. People, people, people. Since that night, I 've known the feel of overpopulation."
- 2.4 Now, remember that the population in Delhi at that time was 2.4 million people. What do you think the population of Paris was? It was 8 million people. So, one of the economists from India who heard this said, "The issue isn't a population of biology. The issue is economics." There 's an economic reason why all these people here—it was opportunities and government policy that were encouraging Indians to move to Delhi to seek opportunity. It was an economic issue, not a population issue.
- 2.5 We think that Ehrlich made this fatal error, and that fatal error was that he misunderstood how much different human beings are than insects. He made this same error that Malthus made, and really, Thanos made, in misunderstanding what is a resource, what is a resource. But that didn't stop this book from selling over 3 million copies. Ehrlich appeared on numerous TV programs and talked extensively about his ideas and his theories. He was interviewed many times on The Tonight Show. It was the leading evening comedy show in the 1960s and 1970s.
- 2.6 Well, I want to introduce you to another guy, an economics professor. He's got a nice haircut, right? He reads The Population Bomb, and the first thing that he does is—this looks like a pretty reasonable theory, it's like it makes sense—but as a scientist, he says,

"I should really check the theory against the fact." So, he began to go back and look at the prices of resources. The prices of resources. Now, if you're running out of something, what should happen to the price? What should happen to the price if you're starting to run out? Shouldn't the price be going up? Well, what Dr. Simon, Dr. Julian Simon, discovered in his research that all of these prices of these nonrenewable things—copper, metal, wheat, other products—had been doing what? It had been going down. What was the explanation for population is increasing, but prices are going down? What's that explanation? As he researched, everything that he researched, he said he found that it had been going down in price.

2.7 So he publishes his findings in this journal called Science. And of course, when this article got published, Ehrlich read it and just blew up. He couldn't believe that a leading science journal would publish criticisms of his work. He quickly shot back with a scathing counter article asking, "Could the editors have found someone to review Simon's manuscript who had to take off his shoes to count to 20?" Typical Paul Ehrlich—very personal criticisms of people that offered counters to his work. He went on to say that Simon was part of a "space-age cargo cult" and complained about economists' inability or unwillingness to understand the Earth's carrying capacity. He wrote that, "To explain to one of them the inevitability of no growth in the material sector, or that commodities must become expensive, would be like trying to explain odd-day, even-day gas distribution to a cranberry. " So, pretty good at insulting.

2.8 It was around this time that Julian Simon, he said, "Well, here's the facts. Look at these facts. You haven't looked at these facts, or show me other facts that you may have discovered. Because what I'm seeing here is totally contrary to what your claims are. Your theory, your model, your hypothetical model is not matching up with the facts." And so, in frustration, he finally said, "How about?" And Simon and Ehrlich are having this back and forth in the media. They're punching it out with these criticisms and arguments that they're making.

2.9 Well, Simon finally becomes so frustrated that he just says, "Why don't we bet? Why don't we bet?" Because Ehrlich had written in 1970, he said, "If I were a gambler, I would take even money that England will not exist in the year 2000." Simon said, "Well, why don't you just pick any nonrenewable resource? And I'll bet you that, in one year, it's going to be cheaper today in a year, than it is today." So, Ehrlich and two of his friends, two other professors, said, "We'll take that bet." He said, "To accept Simon's astonishing offer before other greedy people jumped in, the lure of easy money can be irresistible." So, Ehrlich was really happy to make this bet. So, here's the terms of the bet: Let 's put \$1,000 on the table. Simon let Ehrlich pick these five metals. He's decided he wanted to pick five of them. They were copper, chromium, nickel, tin, and tungsten. So, let's see what the prices are today, this is September 1980, let's see what the prices are today, and we'll run this bet for 10 years. Ehrlich also picked a 10-year

time frame. So, 1990—let's go back and look at what these prices are in 1990. And we're going to adjust for inflation, but let's see what happened to those prices.

- 2.10** So what do you think happened to those prices? Now, remember, the 1980s, we added 850 million people to the planet. It was the highest growth of human beings in history. So what do you think would happen? If you're going to increase population and you have a limited nonrenewable resource, what should happen to those prices? Well, Ehrlich's model, his theory said, "Those prices should go through the roof." Let's talk about those two for a second. They both believe that there is a relationship between population and resources—a relationship.
- 2.11** Let's just plot a chart here. Let's show resources on the one axis and population on the other axis. What did Ehrlich say? As population increases, what's going to happen to resources? Well, they're going to run out. They're going to become more and more scarce. Simon, on the other hand, made the argument, "No. As you add people, these resources are actually going to become more and more abundant." More and more abundant. So you see the two arguments, the two theories? Population and resource. What's the relationship? I think as population increases, we're going to run out. Simon said, "No, as population increases, we're going to have more stuff." We're going to have more stuff. OK.
- 2.12** So, behold, what are the most important checks ever written in economics? The end of 10 years, the inflation-adjusted price had fallen by 36% on the average. Some of these metals had fallen even more than that. So, Ehrlich writes a check to Simon—\$576.07. What happened? Yeah, what happened? Well, if we can chart this, population is doing what? Population increased by almost 20%. Global population increased by almost 20% over this 10-year period. But the resources, if the price goes down, it means you actually have to spend less time to earn the money to buy it, right? So the abundance increased by 67%. For every 1% increase in population, what was happening to this basket of these five metals? I call them "the five-metal basket." What was happening to those? They were becoming about 3.52% more abundant. Add people, resources increase even faster. How is that possible? Well, a lot of people have criticized that bet and say, "Well, it was just these five metals, and it was a lucky decade. If it was a different decade, it would have been different. They would have picked different metals that have been different."
- 2.13** So we went back actually to 1900 and looked at the prices in 1900 and compared them to the prices in 2019. We extend the period for almost 120 years. What do you think happened to those prices over that 119-year period? Well, first of all, population increased by over 375%. Remember 1900, we're doing this to the population. But what's happening to these resources? They became almost 700% more abundant. We define abundance as "how much time does it take you to earn the money to buy that?" It takes less and less time to buy more and more of this. How did that happen? So, over this longer period, it was like, "Wow."

- 2.14** Now, there are times where we'll have a temporary shortage. There 's a temporary shortage. But typically, what happens when you have a shortage of something? Well, the price goes up. Well, what does the price tell people to do? Tells people to do four things. First of all, it tells them to use less. Second, it tells people to go find some more of it: "Well, I can go make some money if I can go find some of this stuff." The third thing is, it says, "Maybe there's a substitute for this stuff. Maybe there's something else we could use instead of that thing that serves the same purpose." And then, the fourth thing people do is they recycle. All of those four factors combined will do what? Well, you'll ultimately end up with more, so that scarcity actually creates abundance. Scarcity will create abundance if markets and human beings are free to act on that new information that's contained in the price. The price is what's important.
- 2.15** All right. So, we recall our friend, Thanos: "The universe is finite, its resources finite. If life is left unchecked, life will cease to exist. It needs correcting." Now, he was right on this first part. Is the universe finite? Well, we think, well, the Earth seems to be finite. There seems to be a finite number of atoms on the planet. But where did Thanos make his error? The second part of that statement—when he said, "its resources finite"—he equated atoms to resources, and that's the error. Resources are created when you add knowledge to atoms. Knowledge is what makes atoms resources. Look at the facts. What are these facts telling you? You have this idea. You have this hypothetical model that appears to be very clear and logical. But does that model match the facts? What do the facts say about this? That's what Simon did, and that 's what we try to do. But Elon also talks about resources. He said, what—he made it interesting. "Resources will be fine." Resources will be fine.
- 2.16** So we have these two contrasting visions. Too many people, not enough resources. Elon says, "No, we don't have enough people. And the resources will be fine." By the way, his comment about desalination, he says it's cheap. Well, how cheap is it? Well, Israel can desalinate 6.4 gallons of water for less than a penny. So, by all intents and purposes, Israel doesn't have a water problem anymore. They got the Mediterranean. Until the Mediterranean runs out, they're not going to have a problem with water. Because they did what? They applied knowledge to salty water, right? And that did what to the water? You replace salt with knowledge. Suddenly, you got this valuable resource, and you figured how to do it in such a way that it can be done very, very inexpensively. Very, very inexpensively.
- 2.17** So, who are you going to believe? You've got Ehrlich and Thanos and Malthus, who are all making this same argument—the same what I would argue is a hypothetical argument without strong empirical data, evidence, facts. What do the facts say? In contrast to who? Elon, Professor Simon, and Professor Peterson have this opposite view.
- 2.18** All right. You ready for a quiz? When Lego introduced their block, the patent office asked them, How many ways can you put these blocks together? They had a

mathematician that was working for them, and he took six of the blocks, and he came up with a calculation of how many ways you could take six blocks and put them together. Any idea what that number was? It was 102 million ways. Just six blocks. You could put them together in over 102 million ways. Well, this number was reviewed by a Danish mathematician here recently, who said, "You forgot to consider this other possible way that you could put them together." And after he made the calculation, 915 million. Yeah, 915 million different ways. Yeah, it's astonishing, right? With just six blocks, you can configure these blocks in 915 million different ways. I think I actually have seven blocks on this picture, but it was only six—an astonishing number of combinations with just a few blocks. A few atoms can be configured in almost infinite different ways and varieties. First question.

2.19 Second question, "How many keys on a piano?" Got a music major here? I think there are 88 keys. I think there are 88 keys. So, how many songs then are in a piano? It's a trick question. There's actually no songs in the piano, are there? There's no songs in the piano. It's got keys, but there's no—where do the songs reside? Don't they reside in the minds of human beings? And what is that number? Doesn't it approach infinity? Doesn't it approach infinity? So, you've got 88 keys, but how many songs can you create? It's really an infinite number of songs. Now, some songs sound better than other songs. But you have this infinite potential, really, to create songs with this fixed number of keys. Keys are atoms. Music is knowledge. Songs are knowledge. You can think of it as the keys being the hardware, the songs being the software. It really opens up this idea that we've got lots and lots of potential with just a limited number of physical resources.

2.20 Now, if you asked Thanos this question, what would he say? I think he would say, "You got 88 keys? You must only have 88 songs." The Thanos thinking says, "The physical is the resource. I don't consider knowledge." What you can do with that is what counts. And that's what really counts for human beings, because all of us have this potential to discover new valuable knowledge, and then share that knowledge with other people. And by virtue of sharing it, what have we done to the amount of knowledge? If I discover something and I share it with you, we just doubled the amount of knowledge on the planet, right, that I had, and now you have that piece of knowledge. And another really beautiful feature about knowledge is, when you consume it, it grows. It grows.

2.21 So, we're trying to move from this world of scarce atoms to a world of abundant knowledge. Does that make sense? OK. All right and it goes back to, "How do you define a resource?" If it's just atoms, yeah, things are fixed. But if it's knowledge, no, things aren't fixed, aren't fixed of all. Now, Paul Romer, an economist who was awarded a Nobel Prize in economics for his work about knowledge, makes this statement: "We consistently fail to grasp how many ideas remain to be discovered. The

difficulty is the same one we have with compounding. Possibilities do not add up. They multiply." You're multiplying this thing. This knowledge multiplies.

- 2.22** And he uses the example of the periodic table. We've got about 118 elements that we've discovered so far. Of those, about 88 actually exist commonly on the planet. Any idea if you just tried to pick two of them? How many combinations you could have? Well, it starts at 118 times 117, right? That's a pretty big number. Well, what if you take three and put three together? It starts to get bigger, bigger, and bigger. If you put five, you pick five of these, or six of them like the Legos, let's pick six of these, and let's see how many different ways we could put those together. If we just pick six out of this table of 118, it's over 100 billion.
- 2.23** So, we've just barely scratched the surface of how many possible combinations we can come up with to just combine these atoms—just barely scratched the surface. I mean, you think about the time that's elapsed since the Big Bang. It would take that times, times, times to discover all of these potential new ways. So, who are the discoverers of these things? Well, it's you and I, you and I working to experiment and test and see, "What happens if? What happens if we do this? What happens if we take sand and heat it up? I don't know. Let's try it. See what happens. Wow, what did we discover? We can tell somebody else about what we discovered." And suddenly, sand goes to these beads and windows and fiber-optic cables. What are we doing with sand? We're making it smarter and smarter. We're adding knowledge to it. We're, like, we're intelligizing these atoms to make them serve our purposes. And we actually don't even know what they do until we actually start putting these things together. It's like, "Wow, what's this going to do? If I don't blow myself up, maybe I'll discover something that could be really, really valuable." All right.
- 2.24** So, economics is not about physics. It's not about atoms. Economics is about what? It's about knowledge. It's knowledge. That's the key. Now, when we think about that, resources are what? Well, they can be atoms. Resources, we could say, resources are equal to atoms plus knowledge. Well, if you have a fixed number of atoms, what do you want to do to increase your resources? You just add more knowledge and your resources will go up. It's knowledge that's the key. Knowledge that's the key.
- 2.25** Here's our argument. Knowledge transforms atoms into resources. We do not believe, we have not seen, any limit to the discovery of valuable new knowledge. Do you believe that there's a limit to the knowledge that we can discover? Just thinking about the piano keys or the number of atoms, it's like, "Wow, there's lots of possible new discoveries that we could make that could be really valuable." Could be really valuable. And that leads us to this conclusion: "Therefore, we live on what? An infinitely bountiful planet." It's infinitely bountiful—part of the title of our book, part of our argument against Thanos and Ehrlich and Malthus: "Your assumptions were wrong about what a resource is and how they're created."

- 2.26** And that explains perfectly why population could go up during the '90s by 850 million people, and the abundance of these five nonrenewable metals, right, are doing what? They're going up at even a faster rate because you're doing what? You're discovering knowledge, and knowledge is what's creating this abundance. Knowledge creates this abundance. OK, so what do we want to do? We want to move from a world of scarce atoms to a world of what? Really abundant knowledge. OK? All right.
- 2.27** You want to take another quiz? No, we don't need another quiz. Let's just do a quick review. We talked about the bet. What was the bet? Paul Ehrlich is over here. Julian Simon is over here. Simon becomes very frustrated because Ehrlich is not looking at the facts. And he finally says, what? "If the facts won't convince you, how about money? Let's bet." What was the outcome of that bet? Well, it looked like Ehrlich was wrong. At least for that decade with those five, it looked like Simon must have been really, really, really lucky to have offered that bet. But remember, who picked the five metals? Ehrlich. Who picked the time period? Ehrlich. Simon offered him once again, "You want to bet again? Should we bet again? Pick any metals?" He did have one clause on the contract. He said, "If we have a war, then the bet 's off." Because wars tend to do what to prices? They distort everything, throw everything out of whack, so to speak.
- 2.28** If you look at the long-term abundance of most resources, they have this trend line that looks like this. It's going up and down. In the short term, you can find periods where it goes down. They become more and more scarce. But the underlying trend line is doing what? You think about a thing that's going like this. It's like, yeah, in the short run, you're going to see these periods of scarcities. But what happens when you have something that becomes scarce? What happens to the price? Price goes up. What four things happen when the price goes up? People do those four things: I'm going to use less. I'm going to go find some more. I'm going to look for substitutes. I'm going to recycle. What happens? They're really doing what? What are people really doing? And this was also the insight of Simon. He said, "People are going to do what? They're going to take their time and energy and focus on discovering new knowledge." And that new knowledge is ultimately going to do what to that resource? You add more resources to those atoms, it's going to make them cheaper. If you add new knowledge to the atoms, it's going to make them cheaper. And that's, in fact, what these have done in the long run.
- 2.29** So, we got the bet, evidence. The other beauty I like about the bet is, if you're a professor, you get to say whatever you want to, and you're never really held accountable. I can say whatever I want to in class, and I can get away with it. The bet did what? It made a professor accountable for what he said. It was a small little thing, but it was very important because it said, "Look, you make this claim. How much do you really believe that claim?" And the outcome of the bet over this period of time—huge growth in population, never larger increase in the population over a decade in the history of humanity—and what happened to these commodities? Wow. Wow, but what

explains it now? What explains it? Doesn't knowledge explain it? We're discovering knowledge faster than we're making people—much, much faster. And that allows all of these things around us to become less and less expensive, to take less and less time to earn the money to buy.

- 2.30** All right, so we had a quiz. We tried to illustrate this principle by looking at atoms, something that's physical, and then thinking about the potential different ways that we can add knowledge to those physical things and make them more valuable, more interesting, more beautiful. Music, for example, isn't really a physical thing. Well, we use piano keys, but that music, when you come up with a song, what are you consuming? Yeah. So, physical limitation, finite atoms, infinite knowledge can create what? Potentially infinite resources that you and I can value and use.
- 2.31** And then, this conclusion idea—this world of atoms is very important to us. Very important. We live in this material world, but it's really knowledge. Wealth is knowledge. OK? All right. Any thoughts about that? Any questions about that? Any examples you can think about where we have taken something and done what with it? Added knowledge to it, and it becomes what? Really, really valuable. You take this guy. You all have one of these. It's about 6 or 7 ounces of silicon, sand, aluminum, and oxygen. Those are three of the most abundant elements on the surface of the planet. And what are they worth if they're just sitting there? Well, the Neanderthal had those same atoms in his cave. But what was he missing? He was missing all the knowledge that this thing represents. It's really intelligized atoms. It's knowledge that we've been able to take this almost so common elements on the planet and put them together in a new way, organize them in a way, arrange them in a new way, add software to them, have them do functions and things for us that has made this thing so valuable.
- 2.32** What do you think it cost to make the first iPhone? What do you think Apple spent to make the first iPhone? I got to believe it was billions of dollars. All of the research and effort and the testing and the time it took them to develop just the first one. And now we're up to, what, 15? Generation 15. Each one of these seems to likely get better, faster, cheaper. I mean, the price doesn't go down, but the features go up. So, they really are getting cheaper, aren't they? OK. So, let me ask you, what would I have to pay you to never use your smartphone or the Internet again for the rest of your lives? What would I have to pay? It's not what you paid for something that counts, it's how much you value that thing. That's what really counts. So, you paid seven, 800, maybe \$1,000 for this. What's that really worth to you? Well, I have yet to find a student that's willing to take less than \$5 million. Would you give up your phone and access to the Internet for \$5 million today? Yeah, at my age, maybe I'd say, "Yeah, OK, I'll do it."
- 2.33** But look, if you're a young person, it's like, "Not really. My future is really going to be part of this device. It's going to give me access to what? The world's knowledge and resources and the ability to think fast and deep. Fast and deep and really inexpensively—\$35, \$40 a month—and I get that. It's like a dollar a day. I get this \$5 million thing I

get to carry around in my pocket with me. That 's incredible." That's incredible that we have those things with us today that are really the product of what? Yeah, there's atoms there, six ounces of atoms, but there's billions and billions of dollars of knowledge that we get to do what? Because once you discover some new piece of knowledge, what does it cost you to share it with another person? We call it the marginal cost, or the cost to make the next unit. It costs what? Basically zero. Basically zero to make a copy of knowledge and give it to someone.

2.34 Now, what that means is instead of competing over atoms as competitors, this ideology of scarcity has done what to us? The human species. It's made us do this throughout our history. We're fighting over atoms. And if we had recognized that our resources are really knowledge, what does that then turn us into? Collaborators, cooperators. "Hey, why don't you go figure this out? I'll try to figure this out. Let's share with each other. Let's collaborate in this pursuit of knowledge because knowledge can be consumed." There's no rivalry. You and I can consume the same knowledge at the same time. Unlike a Snickers bar where either I get it or you get it, it's like, "No, I give you this piece of knowledge. Now we both have this knowledge." So, it has this potential to do this for us. So, think in knowledge. Think in knowledge. That's what resources really, really are. Any questions before I go on about the first lecture?

2.35 So, how do you think the world is trending now from what's trending—a scarcity mindset or an abundant mindset?

2.36 Excellent question. Where is the world trending in terms of the scarcity mindset versus the abundant mindset? Well, the scarcity mindset's the dominant mindset. It really is because we have relied on that for a couple of hundred years. We're hopeful that our work, our theoretical model—that's what we're going to talk about, how to analyze these things in a quantifiable way, and the evidence—will help move people from this scarcity mindset to this abundant mindset, because I feel so sorry for young people today because they have this idea that we're going to run out of things. It's like, this is the best time we've ever had on the planet.

2.37 There's been no better time to be alive than today with our ability to solve problems and create things for one another. You think about 8 billion people, and 7 billion of them have a smartphone. We're all connected. We could talk to each other for almost what? For a trivial amount of time to earn the money. And we can share knowledge, create, discover. It's like, "Wow. Wow, it's just incredible." So, we 've got work to do. We're hoping to be able to make the arguments, articulate them, and then have an audience that says, "Yeah, I've looked at the facts. I've looked at your theory."

2.38 Once again, what is a theory? A theory explains what the facts are. The facts are the dots. The theories connect the dots. Well, look at these facts. How do they relate to each other? Let's connect the dots with a theory. Or it's some cause-and-effect relationship. I think that first thing, move from atoms to knowledge, and you'll see that we move from scarcity to abundance.

- 2.39 Could this expansion of knowledge and these ideas, could they be stymied or give Ehrlich's theory any weight if this knowledge was to be applied to more creative endeavors that didn't directly assist society, that were more about fictional characters, about just generally amorphous?
- 2.40 Right. Knowledge can be a very dangerous thing, can be a very dangerous thought. The knowledge that we used to create the atomic bomb also created this ability for us to create almost unlimited energy. So, I think that you would say we're rational optimists. We look at the risks involved in giving people the freedom to pursue this new knowledge. We believe that the new knowledge that we can discover is always going to be greater than the knowledge that can be used for not-good things. We know that people can use knowledge, discover things that can be very, very harmful. When the printing press was invented, people used the printing press to print all kinds of things that were evil, bad, not true. But you really want to give up the printing press to not have the benefits of it?
- 2.41 And that's another point that we make that we recognize in economics is, we live in a world of trade -offs. When we want this, it means we have to give up that. We're going to make a trade-off here. Is it worth it? We always ask ourselves, "Is it worth it? Is this trade-off worth it? Yeah, this is going to cost me this much, but I think what I'm going to get is going to be much more valuable than what I have to give up. I'm willing to make that trade-off."
- 2.42 And we all have differences in our valuations. We value things differently. That's why we can trade with each other. Because you don't think it's worth very much. I think it's worth a lot. I have something that you think is worth a lot that I don't think is worth very much. Well, how can we make each other happy? You and I can trade with each other and make each other happier. By trade, trade is the other key element. If people are free to discover new knowledge and then trade that with other people, they do what? We lift each other. We lift each other because we are cooperating, we're trading, we're enriching other people with our knowledge, what we've discovered, and what we then can trade with each other. All right.

Lecture 3: Price Perspective

At a glance — Price perspective — measuring true costs.

- 3.1** All right, welcome to lecture three in our course, Introduction to Superabundance: The New Economics of Knowledge, Learning, and Time. Now, recall from our previous lecture, we had a little clip of Elon Musk and he said, "Look at the numbers." Look at the numbers. Well, that's very important to look at the numbers. But you have to ask two other questions: "What numbers are you talking about? And how do we look at the numbers?" How do we look at those numbers?
- 3.2** So, I want to talk about four things in this lecture: quantities, money prices, time prices, and the change in time prices over time. First of all, if you're an accountant, if you're an engineer, if you're a biologist, if you're an insect biologist, you go out and count things. Let's count up the number of butterflies and let's count up how many resources we have here. And then we watch the butterflies increase and we watch the resource disappear because the size of each butterfly's pie gets smaller. Their resource gets smaller as you increase the population. So, what are the problems with using quantities? Quantities can be very important to us, but we think there are six problems with just using quantities only.
- 3.3** The first problem is we don't know how much we actually have. How much oil is in the planet? How much oil do we have? Now, we'll see publications about, well, we have 10 billion barrels of oil that we've discovered. Well, is all the oil observable on the planet? Have we discovered all the oil that's on the planet? Remember, the earth is 8000 miles thick. We've only gone 5 miles deep. That's the deepest well is 5 miles, 5 miles out of 4000 miles, 8000 miles—so, just to very, very scratch the surface in terms of discovering. So, we don't know how much stuff there is because we really can't observe a lot of these things.
- 3.4** What's the second problem with quantities? There's a difference in quality, right? How much water is on the planet? Well, how much is actually quality that we can drink? So, there's a range of qualities. We really find this with a lot of resources. Yeah, a solid gold nugget is very hard to find, but lots of little fine little flakes of gold—it's much easier to find that. So, the quality of something is also very hard to determine.
- 3.5** The third reason is, it's expensive to hire somebody to go out and search for these things, right? You got to hire an engineer to go out and search for these things. Why would you hire them? Well, you're hoping that you'll find something, but that's a function of the price. So, we don't know how much we have because we really haven't spent the money to go out and look. So, quantities really are limited in that.
- 3.6** We also don't know where to look. The ocean water covers 70% of the planet, that means 70% of the floor of the oceans, we really haven't looked down there. We haven't looked there. We don't know really where to look. Where do you go look for oil? Yeah.

Would you have guessed that you would have found oil in Saudi Arabia in 1920? That's like the last place you would think about. But there it is. There it is. OK? We don't know where to look.

3.7 We also don't know when someone says, "How much quantity do you have?" We don't know about substitutes. We have this much copper. We thought we had this much copper, but what happened to copper? It was replaced by glass. It was replaced by fiber-optic cables. So, we got a substitute for copper. So, counting how much copper you have may not really be the best way to figure out how much you have, because if there's a substitute for that thing, you should consider that, too. You also have to consider, we can recycle this stuff. You know that the typical aluminum can that you drink your sodas out of, it's been recycled over seven times. So, somebody else has drunk out of that can, that aluminum, at least seven times. Aluminum is almost 100% recyclable. It's this beautiful element—light, recyclable. Yeah. So, we don't know when we count how much you can recycle.

3.8 The big thing is, we don't know about future discoveries. What's going to happen tomorrow? Who's going to discover something tomorrow that we don't know is available today? Either a new product, a new quantity of something, or something that could be a substitute. OK, so let me ask you a question. What would be more important—if we had a discovery where we suddenly found twice as much oil on the planet, or we had a discovery of new knowledge that let our cars get twice the mileage? What would be more valuable? Because they both have the same effect, right? You want a car that gets twice the mileage, or do you want your gas to be half-price? What would be better? One is what? One is, you've discovered more atoms. The other is, you've discovered what? New knowledge. It's made these atoms a lot more valuable. I think we're actually going to do both. We're going to continue to actually improve mileage in cars and we're going to continue to discover new energy sources. OK? All right.

3.9 So, those are the problems with counting. When you go into the store to buy a loaf of bread, have you ever gone in? What's more important to you: the quantities, the number of loaves on the shelf, or the price? You ever gone to the store and counted up the loaves on the shelf? Why not? Isn't that important? We're going to run out of bread. There's only, like, ten loaves on the shelf, and I consume a loaf a day. We've only got ten days' worth of supply of bread left, and then we're going to run out and die, right? No, we don't really think in quantities.

3.10 What do we really think in? Don't you think in price? Isn't the price the piece of information that you want to use to be able to make your decision? Isn't it price? So, if you want to think like an economist, quantities are important, but prices are much more important because prices contain this information about relative supply and demand. How bad do people want this thing, and how much is there of that thing? That information is contained in a price. So, when I look at the price of things, it's like, that's

going to tell me whether I'm going to buy it or not. Now, I love Corvettes. Car Corvettes. I've always wanted one. Why don't I have one?

3.11 Price?

3.12 Yeah, price seems to explain a lot of things in life, doesn't it? Price—thinking like an economist, thinking the price. And that will explain people's behavior—why they do the things they do, or why they don't do things they don't do. It's the price that's sending them a signal. That information, the price, will really influence your behavior. If you change the price, you're going to change somebody's behavior. OK? Yeah. So, what if on the way here, I drove by McDonald's and they said, "Now hiring, \$60 an hour." What would you do? I know what I'd do. I'd leave here and I'd go to work for McDonald's. That price can immediately change your mind because people's minds can change as fast as the price can change. People's values change as fast as they can change their mind. And price has this power. So, we move from quantities to prices.

3.13 Now, we also recognize that we buy things with money, but we really pay for them with our time. OK? Ever gone in the store and looked at something and said, "Wow, I'm making, I'm making \$20 an hour. This pizza cost me \$20. I have to work an hour to buy this pizza." OK, you ever do that? I think subconsciously we all do that. So, we have money prices, but we also have time prices. Now, we actually think that time is a better way to measure things than money.

3.14 In fact, we think we've made two mistakes in economics. The first mistake we made is we didn't get the definition of resources right. And the second mistake we made is we try to measure things with money, and that has caused some problems, some confusion about how to really measure things. Money prices, we express those in dollars and cents. Time prices, we express in hours and minutes. How much time did it take you to earn the money to buy that thing? And is that taking more or less time, over time? Well, last year it took me an hour to earn the money to buy that pizza. Today, it only takes 45 minutes. Wow. What does that mean in terms of pizza abundance for you? Doesn't it mean pizzas are getting cheaper for you? Because it takes me less time to buy one. OK?

3.15 So time, prices. Now, time price is simply the money price divided by a person's hourly income. A time price contains more information than a money price because we're comparing a price to your income. It's a ratio of two numbers. When people say, "Well, inflation's gone up 12%," what's that mean? It simply means that things have gotten 12% more expensive. A time price allows you to answer the question, "Has it gotten more or less affordable?" Because if the prices go up by 10%, but if your income goes up by 10%, what's really happened to the price? Isn't it like the same, takes the same amount of time? Well, what if your hourly income went up by 20%? Wow, all the prices have gone up by 10%. Yeah, but your income went up by 20%, which means what? You're better off. It takes you less time to earn the money now to buy things. So, the time price captures that innovation that happens. Innovation—we'll talk about that a

little more—but innovation is this addition of valuable new knowledge. When the time price goes down, it means that we are actually gaining more knowledge, and that shows up in the time price. OK? So the first important thing is that time prices contain more information than money prices, because you're comparing the money price against your hourly income.

- 3.16** The second advantage is that it transcends all of this contention and subjectivity that you have when economists try to figure out the current prices and how much inflation we've had. Now, we've had this experience of inflation over the last couple years. And you hear economists say, "Well, inflation was 8%." Another economist says, "No, it's 12%." "No, it's 3%." It's like, you know, you can transcend all of that. It's very contentious when you get economists together and you try to talk about inflation, because what do we know about money? Money is not a fixed measure. It can be counterfeited. It can be inflated. It can change over time. It's like having a tape. It's like having a ruler that's rubber, that you can move. So, it's very hard to measure things with money because the value of money can change over time. So, if you use time prices, you can transcend that.
- 3.17** So, for example, we can go to any country at any time and look at any product in any currency and figure out the time price. So, I can go back to Paris in 1850, find out what the cost of a loaf of bread was. It's so many francs. And how much were people earning? Well, the bread was 5 francs. People were earning one franc per hour. So, that bread cost me what? Looks like I had to work 5 hours to earn the money to buy that loaf of bread. Well, then I can take that time price, 5 hours, and I can compare it to a time price today of bread, or of oranges, or of any other product. So, it allows me to transcend all of these other issues and go right to the time. What was the time price? That's really what I want to know. What's the time price? What was the time price last year? Ten years ago? 100 years ago? And what is it today? That will tell me if things are getting better or worse for people. All right. Of the 7 measurements that we use in science, these 7 standard measurements, 6 of those go back to time.
- 3.18** So, we should be like the rest of science, and we should use time instead of money to measure things. OK? It's this universal constant. You can't buy time. If you could buy time, rich people would never die. You remember Tony Stark's father, in one of The Avengers, said, "No amount of money ever bought a second of time." We cannot buy time. So, it provides this universal measurement that we all have. It gives us, really, this perfect equality. Each one of us have 24 hours a day to spend. How do you get to spend your time, today versus what was it like for your parents and your grandparents? It gives us ability to measure in a different way besides income. And we think it's a much better way to measure, measure people's time inequality, not their income inequality. So, we think about quantities. What's better than quantities? What's better than the number of loaves on the shelf? The price of the loaf? Yeah.

- 3.19** Prices are better than quantities for economic analysis. And what's better than money prices? Time prices. Time prices are better because we rely on time. This universal standard, it doesn't change, unlike money. It can go up and down. It can go all over the place. And then what do we really want to analyze? What's happening to that time price over time? What's happening to that? So, quantities are counting. Money prices: are we valuing things? What's the time price? It's better than a money price because it's a ratio of the money price divided by your hourly income. So, that's even gotten better information. And what is time prices over time? That's like calculus, because you're measuring the rate of change over time. So, where was Thanos and Malthus and Ehrlich? They're at the counting level. They were counting things and then comparing that count to the population. And we said, "No, what you really need to be is—where, where do you really need to be, to really truthfully understand what's happening? You need to be where? You need to be down here. You need to be looking at the change in time prices over time. That will tell you really what's happening to people's standards of living."
- 3.20** One of the economists that inspired our work is a guy by the name of George Gilder, and he offers three propositions. He says, "Wealth is knowledge, growth is learning, and money is time." Now, from these three propositions, we can derive a theorem. And our theorem is that the growth in knowledge can be measured in time. We can figure out how much knowledge we are discovering by looking at the time price of things. OK?
- 3.21** As you discover more knowledge, what should be happening to the time price? Should be going down because those things are becoming more and more abundant. So, our first equation—we promised you we'd have some equations. The time price is what? It's equal to the money price divided by hourly income. Real simple equation. What's the money price? How much are you earning per hour? That results in a value that is the hours and minutes it's going to take you to buy that item. So, from that, can we calculate the percentage change in time prices? If I calculate the time price in 1980, I calculate the time price today, I have two different time prices. How do you compare those two? Well, we can calculate the percentage change in the time price is simply the end time price—today's time price—divided by what the time price was 1980. And then we just subtract one from that, and that will tell us the percentage change in time prices. If the price is lower today, what do we know is going to happen to that time price? It's going to be negative. Time prices are going to go down.
- 3.22** It's costing you less and less, like everything looks like it's on sale today. Now, the other interesting equation that you can use, because these time prices go down—what happens when the time price, when you have 100% decrease in the time price? What would that mean? It means that it's free, right? Becomes free. That doesn't cost. Is there anything that you use that's free? Well, almost. Lots of things are almost free that we enjoy today. Now, the other interesting way to think about these two time prices is to change the ratio. When we change the ratio, we can calculate the abundance, how

much abundance is increased. Notice this is end time price divided by the start time price? We can calculate what we call the personal resource abundance multiplier. What has happened to your life when we calculate the start time price divided by the end time price? If it took me 5 hours to work to earn something in 1980, and now it only takes me 1 hour, what is my abundance multiplier equal to? Isn't it five? So, for the time it took me to earn the money to buy one of those items, now how much do I get if I devote the same amount of time? Don't I get five?

3.23 So, that's the multiplier, what we call the personal resource abundance multiplier. What's happening? As long as the end time price is smaller than the start time price or less than the start time price, your life is getting better. Your life is getting better. We can calculate the percentage change on that by just subtracting one from that value. All right. You can also just express it that way. You can also look at the percentage change in the time price with this personal resource abundance multiplier. We'll talk about these equations. We'll give you copies of these equations, and you'll get a chance to kind of work through it. They're really pretty simple once you have a chance to work through them. But notice the relationship between this decrease in time price and this increase in abundance. It has this geometric relationship. As the time price goes down, what happens to the abundance? Because it's a reciprocal, it's actually a geometric reciprocal. It not only goes up, it goes up at an exponential rate. So it has this curve. You see the scale here because you can only go to 100% off, right?

3.24 And then what happens to abundance? 100% off means you have infinite abundance. You can have as much as you want. So think about it this way. So there's a percentage change in time price. If something's 10% off, if something's 10% off and you devote the same amount of time, how much do you get for that? I get 1.11 units now. So what's happening to my personal abundance? What if something's 25% off? I go in the stores, 25% off. How much now do I get for the price that it cost yesterday? I now, instead of getting one, I get 1.33. I get 33% more. What if it's 50% off? That's two for the price of one, right? I get two instead of one. My abundance is doubled. It's increased by 100%. What if it's 75% off? That means 4 for the price of one, 80% off 5, 90% off 10. You see how it does this kind of function like this? Yeah.

3.25 And we have little trouble thinking about negative percentage changes. We don't think about that relationship to abundance. We have to actually see it and do the calculation before we realize, wow, if you went into the store today and things were 80% off, you get 5 for the price of one, right? What if they're 90% off tomorrow? Is that better? You go from 80 to 90%. But what do you get? You go from 5 to 10. Now I get 10. OK? So as you start getting above 50, 60%, suddenly things become really, really abundant. All right, so let's do a little example. So, today, let's say bananas cost 60 cents a pound. And I'm earning \$18 an hour, about 30 cents a minute. So what's the time price? What would the time price be? Wouldn't it be 60 divided by 30? So, the time price is, what, 2 minutes, right?

- 3.26** I gotta spend two minutes working to earn the money to buy my pound of bananas today. Now, if 10 years ago, the price was only 50 cents a pound, is that a good, were things cheaper 10 years ago? Well, maybe the nominal price was cheaper. But what about the time price? What do you have to have to calculate the time price of bananas 10 years ago? You got to know what people were earning 10 years ago. And I go look it up. I asked myself, "What was I earning?" Well, I was earning \$10 an hour 10 years ago. I was earning about 16.7 cents a minute. So what was the time price 10 years ago? Well, 50 divided by that 16.7 cents, it looks like it was 3 minutes. Cost me 3 minutes 10 years ago. Only cost me 2 minutes today. Are you better off? How much better off are you?
- 3.27** Well, let's do the calculation. So, remember, the end time price divided by the start time price minus one will tell us the percentage change in the time price. So, we're at 2 today. We were at three 10 years ago, and we subtract one from that. So, 0.67 minus 1 is -0.33 . Convert that to percentage. The time price has done what? It's fallen by a third. It went from 3 down to 2, right? So it's fallen by a third. But the other question is, "How much more abundant is your life if things are 33% off? What do you now get for that price?" Well, we can calculate that. You see how we just flip the start and the end price? Same equation here. I mean, the same variables, but we've flipped the start and end price. So in this case, it's 3 minutes 10 years ago, 2 minutes today, minus one will tell us the percentage change. 3 over 2 is 1.5 . You get 1.5 today for the price it took you to get one. So how much better off are you? You get a half a banana, right? You're actually 50% better off.
- 3.28** So the time price going down is interesting, but what's more interesting is the rate of increase in abundance. And that's what we try to think about. So, here's some other thoughts for you. If you go from one to 1.5 over 10 years, your abundance has increased by 50%. You get 50% more for the same amount of time.
- 3.29** Now we can calculate the compound annual growth rate. This is a term that you'll run into in economics or finance. It's how much is this growing at each year? And we'll show you how to calculate that with Excel or Google Docs or if you have a financial calculator. This indicates that we're doing about 4.14% a year. Your growth is 4.14% a year. How long would it take you to double something in size, if you were growing at 4% a year? It takes about 17 years. Every 17 years, you would double something in size. So you go from 1 to 2, and then where do you go? You go to 4, and then where? To 8. You're having this kind of a growth. It's compounded annual growth rate. So your banana abundance, if you can stay at that 4.14% a year, you're going to double abundance every 17 years. In 100 years—today I can earn, the time it takes me to earn one banana, in 100 years into the future, how many bananas do you get? Well, if you can stay at a 4.14% growth rate for 100 years, you get 57 bananas.
- 3.30** This seems like a lot of bananas, right? Seems like, wow, that seems like a big number. What we found is, there were a number of products that actually were growing at this

rate. So you go back 100 years ago, look at the time price, and you look at the time price today, and it's like, you get 50, you get 100. If it's sugar, for the time it took you to earn the money to buy one pound of sugar in 1850, how many pounds of sugar do you think you can get today? 2 pounds? 4 pounds? A lot? Put a number on that. You get 410 pounds. It kind of explains why. Yeah, yeah. So we've had this huge increase.

- 3.31** Once again, if you get on this compounding growth rate, we don't notice it every day because it seems like not very much, but we discover little bits of knowledge every day. And what does that do? It shows up in higher incomes and lower prices. You put those two together, the time price is going down every day, just a little bit. It goes down. Some days, you have something that really phenomenal happens, it goes down a lot. Other days, you have an accident or something, the time price will go up. But over time, what do you notice? This abundance is doing this. The fundamental trend is more and more abundance, 3-4% a year, every year.
- 3.32** Yeah, if you guys have used Excel, the rate function, the number period function, and the future value function, will calculate all those for you. We'll give you a copy of the spreadsheet that you can use to do your own calculations on these. All right, so, up to this point, we've talked about measuring personal resource abundance. What are these resources doing for you individually?
- 3.33** What are they, what have they done for you? We look at your wage rate, we look at prices. We compare that, we calculate time prices, we see what they've done over time. It's like, wow, looks like your life is getting better and better every year because of these time prices are indicating they're going down, which means my abundance is increasing at an exponential rate, right?
- 3.34** But how do we then go back and think about this question—this question that Ehrlich and Simon had over population and resources. How do you measure resources at a global level? Well, there's a way we can do that, too, and it's simply population resource abundance—not personal resource abundance, but population resource abundance. It's equal to personal abundance multiplied times the size of the population. So how many bananas do you have today, and how many people are there? How many bananas did you have ten years ago? And how many people were there ten years ago? So we use this little simple equation to estimate abundance at the global or population level. OK?
- 3.35** Think about it this way. If you got a pizza, how big is your slice? How big is your slice, and how many slices are in the pizza? Now, how can the pizza get larger? Well, if everybody's slice gets bigger, it's going to get larger. But if you also have more slices, that's going to make it larger as well.
- 3.36** OK, so we're going to try to grab both of these variables, and here's the way I want to do it. Let's think about sandwiches. It's lunchtime. Let's think about sandwiches. So let's do population and personal sandwich abundance. In other words, how many

sandwiches do I get? OK, so we're thinking about sandwiches. I could get one sandwich in 1980, and there was just me. So if we charted that, it would just be one sandwich, and it's just you. Now, in the last 42 years, sandwiches—the time price of sandwich—falls by 75%. That means how many sandwiches do you get today? It means I walk into the store, I get 4. That's 75% off. It means I get 4 for the price of one, right?

3.37 So I get 4 sandwiches for the time it took me to get one sandwich in 1980. So what's happened to my sandwich abundance? My personal sandwich abundance has increased by what percent? Went from 1 to 4. So that's a 300% increase, right? But during this same period, what else has happened? Let's assume that you got twice as many people. If population doubles, now how many sandwiches do you have? Because remember, the time price is true for everybody. You got twice as many people. Now, how many sandwiches do you have relative to 1980? 1980, you just had me and one sandwich. Today I get four sandwiches. My wife gets four sandwiches. We have a total of how many? We have 8 sandwiches. So what's the change from 1 to 8? Isn't that a 700% increase? So our total, or our population level sandwich abundance, is increased by 700%. OK, 300% personally, another 100%.

3.38 How does 300 plus 100, or times 100, equal 700? Remember, population abundance is equal to your personal abundance times the population equals the population abundance. So one sandwich and one person means your population has one sandwich. But in 2022, what did you have? You got 4 sandwiches, but there's 2 people. So now you have 8 sandwiches. Your percentage change going from 1 to 8, you added seven sandwiches. That's a 700% increase in global resource abundance of sandwiches. OK?

3.39 You see how we're going to use that equation to help us estimate what the planet's resource abundance is? Because if the number of people are growing, but the price of something's getting cheaper, what does that mean for the abundance for the whole planet? It means that's growing, growing, growing. Right? I get a larger slice of pizza, and there's more slices in the pizza. So we're growing. Boom, boom. Two dimensions, personally and population-wise. Does that make sense? We'll work through some examples and you'll see this, see this happen.

3.40 A couple of other things that we, equations that we use is—in economics, there's this thing that we call "elasticity", like a rubber band. And in economics, though, when we talk about elasticity, we measure the change in one thing against the change in another thing. So how much did this change by what percent, relative to how much that changed? So the first equation that we use in our framework is called the Time Price Elasticity of Population. And that's simply the percentage change in the time price divided by the percentage change in population.

3.41 So what happened? Well, think about our sandwich. What happened to the personal sandwich price? It did what? Went down by 75%, but population went up by 100%. So

what's the value there? It's actually a -0.75, which means what? For every 1% increase in population, your sandwiches got 0.75% cheaper. Right?

3.42 More people were making personal sandwiches even more and more abundant. Now, we can also measure the elasticity of the abundance of the personal resource, and that is simply percentage change in the abundance of the personal sandwich divided by the population. And what was that? You get 300% more sandwich as population increased by 100%. So for every 1% increase in population, what was happening to your sandwich abundance? It was increasing by 3, or 3%. It was going up by 3%. So are you in favor of more people? Well, if you like more sandwiches, maybe. I'm in favor of more people. Because every time we increase population, sandwiches get cheaper. And why is that? Because more people are doing what? Discovering new knowledge and making these things less time expensive.

3.43 We can also think about the elasticity of population. What happened to elasticity at the population level? Well, it went up by 700% as population increased by 100%. So what would that mean? That means that your sandwiches are actually growing 7 times faster than population. Remember Paul Ehrlich in the bet? Population went up 19%, but the abundance went up, like, 70%? If resources at the personal level are increasing faster than population, you're experiencing Superabundance. That's what Superabundance is. You're getting stuff faster than we're making people. And how is that possible? Because people discover knowledge, and that ends up making us much richer.

3.44 OK, there's the equations. You'll see those. All right, so one last thing I want to show you here. There's population and there's personal sandwich abundance. Let's go to 1980, and let's set population equal to one, and you got one sandwich. OK? So you got this red square, represents the global abundance of sandwiches. Now, let's go out to 2022, and we'll draw the same chart. But what happened to this box? What did it do on the vertical scale? It went up by how much? You could get one. Now, how many can you get? I get four. See how it grew from that dimension? But also, you had an increase in population, so it grew in the horizontal dimension. It went from one to two because you doubled population. So that represents your sandwich abundance in 2022—the red box versus the green box. Now, it's helpful to overlay these. What's that green area? That's all the new, we could say, it's all the new knowledge that you discovered, that's done what to your global sandwich abundance compared to 1980?

3.45 And from that, we can do these calculations and we'll spend some more time. Remember how we can do the compound annual growth rate? We can do the years to double. We can do the elasticity. Now, Jordan Peterson makes this really interesting observation about this relationship between knowledge and time. He said, "If you can do the same amount in half the time, you're twice as smart. You have twice as much knowledge." And isn't that, in fact, what we've been doing?

3.46 Well, we're going to put some more evidence on the table and give ourselves a chance to think about that. So, summary of this lecture is we spent some time on the

equations, and we're going to get more comfortable with these equations, how they work, so we can go out and use this framework of equations to analyze things. One of the equations that we talked about were the time prices, the percentage change in time prices, the personal resource abundance multiplier, the percentage change in that multiplier, the population multiplier. We have about 12 or 13 equations, but once again, they're pretty simple to use. That allows us to quantify and measure abundance.

- 3.47** Remember, scarcity is about what we don't have. You really can't measure scarcity because my wants are infinite. But you can measure abundance. How much did you have last year for an hour of time? And how much did you have this year? I can measure that. I can quantify it, and I can measure it.
- 3.48** And then the last thing we did here in this lecture is we showed you a little chart on how to compare the two abundances between two different years at the population level. The red—we call it the red-green chart—shows the start year, shows the end year, and it shows how that chart changed. Did you add more people? Did it become more abundant at the personal level? What happened?
- 3.49** So our book is filled with all kinds of data. You'll see all of these different charts in those books, and hopefully you'll be able to now better understand and explain to somebody else what they mean. OK, thank you very much. Do we want some questions? OK.
- 3.50** So mine is not really a question. It's just, for me, it's fascinating to be learning this stuff, because since I was young, it was, like, deep in my core, I knew that there was abundance. Even when they told me, "Close the water because you're brushing your teeth, or save the water," it's like my inner child knew that this was a lie, that there was abundance in the world. And I come across the people that are like, "Elon Musk is destroying the world and all of this stuff." And my answer has always been, I never had the facts or the backup that you've given us here, but my answer has always been, "There's no lack of resources, there's just unresourceful people." So it's beautiful to see your data and your knowledge here, because I do believe that knowledge and that creativity, and as we help others grow, we grow, and everything gets better. And it's just beautiful. And I wanted to share that.
- 3.51** I think she should do the next lecture, man. You said "unresourceful people," which is really interesting term—unresourceful person. And typically, what we notice is, remember, until 1800, it was, everybody was poor, and we were just flatlining. Everybody's making, like, \$2 a day. I mean, \$2 a day. There're people are still living on it. And then I said, around 1800, it goes like this. Well, what happened? Well, here's what we believe happened. Lots of things happened.
- 3.52** But one of the key things that happened is people then, people began to have, to treat one another with dignity and respect. They began to realize, you know, that other person there, "Yeah, they have a mouth, but they have two hands, and they have a mind." And that mind, we don't know what that mind can actually create. And if we

give one another this dignity and respect, we could actually become rich. We could all lift one another out of poverty. So, part of it is policy. We have to have this cultural change where people, first of all, recognize that everybody really is a resource. And then we have to have policy that allow people to be able to go pursue their ideas.

3.53 We will talk about that in a future lecture. But part of it is governments, for many years, prevented people from really pursuing knowledge. When government does certain things, they prevent this knowledge discovery process. And it does what? It stops our ability to create new wealth because people are not free to actually pursue these new knowledge. And they also have to be not hungry, right? I can have all kinds of freedom, but if I'm really hungry, it's hard to innovate when I'm really hungry. So our ability to create food for everybody on the planet, that was a key feature. So people no longer hungry, and now you have some more time—what are you going to do now that you're not hungry and you have some more time and you have an idea? You're curious about something, and you have the ability to pursue that and then share that knowledge with somebody else. That starts to happen, and then we start to see ourselves escape poverty.

3.54 So, it is a beautiful story. My mom used to always say to me, "Hey, you're wasting paper." You know, I'd get paper out and I'd draw on it. She'd go, "You're wasting paper." And I figured out how much it costs, and I go, "Mom, if we're running out of paper, why is it so cheap?" Right? Why is the price so low for running out? Because the price is telling us about relative scarcity, right? Stuff that's really expensive, that says, "You got to be careful with that. Don't consume it because it's really expensive." But if things are becoming less and less expensive, it means they're becoming more and more abundant. It means you can use this now, and you don't really need to worry about it, because if you really, truly felt like paper was, we were going to run out of paper, what would you do today? Wouldn't you go buy a bunch of paper and wait for the price to double tomorrow and then sell it? Well, you can do that. You can do that.

3.55 And we kind of had that experience with COVID, right? I go to Costco, and it's like, you got the cart, and it's all full of toilet paper. It's like, "What are you doing with all of that?" "Well, you know, I'm going to go resell it or something. We're going to run out. We'll never get any more of this." It's like, I don't think so. I think people will respond to the higher increase in price and they're going to do something about it. They're going to consume less of it. People will go out and figure out how to make more of it. Now it's like, wow, we're not worried about that anymore because we're able to, as human beings, we're able to react to this new information that's contained in prices. Prices will tell us what to do. It'll tell us what to do.

3.56 So, yeah, key thing. And I think once again, this kind of, we call it the ideology of scarcity because it's an ideology. It's not this empirical, scientific conclusion. It's an assertion that's based on a hypothetical model. And the facts do not support what you're saying there. It's a claim that you're making that doesn't have the facts to

support it. Now, that's not to say that something couldn't happen tomorrow, that a meteor hits the Earth or something that could cause this huge crisis. But human beings, one thing that we know about each other is we are extremely adaptable. We're extremely adaptable. Think about where we live and what we do relative to our parents and grandparents. It's like, it's pretty incredible what we've been able to adapt to and to create and to solve our problems with each other. All right?

3.57 So in this lecture, you talked a lot about evidence as to why to, like, debunk or disprove the population control theory, which I thought was good. But I also had questions. You kind of throughout this series and the other speakers talked about, like, the moral implications to it. And my question is, so, do you think the reason population control caught on so much was because there was a decline in the moral compass and values, as seen in things such as the U.S. Constitution, where the pursuit of happiness, where that was instilled, that everyone has that freedom to pursue happiness? And then it turned to this trying to control people and turn to that scarcity mindset because we didn't believe we had the abundance? This also kind of goes back to, like, my belief in the morals is that God created us to populate and subdue the earth. And I feel like that's a key part of what debunks this whole population control theory, is that we are like not the gods of our own lives, but God is. So I don't know if you want to touch on that moral implication.

3.58 That's a really, really big question, and I'm afraid to touch it. You know, what we've done is we have tried to go back and look at the facts. Start with the facts. What do the facts tell you about things? And then the interpretation of the facts is really what this theory steps up and says, "Well, why did this happen this way? What is the explanation?" And many people have offered explanations about the moral decline or these other issues. We really never touched that in the book because we felt like, "Look, we want to let individuals who read this book maybe use their own theory to try to understand what happened here, and more importantly, what is going to happen in the future?" What's going to happen in the future? Because it looks like we've been able to achieve just astonishing results. And what's going to prevent us from continuing to have this, this kind of growth and abundance?

3.59 Well, one thing that could stop us is, if you don't think you can accomplish something, you don't try. So part of the purpose of the book was to hopefully inspire young people. Look at what we've been able to accomplish. And if someone is telling you that we can't continue to have this kind of growth, it's like, "You got to show me why, because I don't see why out there. I see we've been able to accomplish this thus far. What's now to prevent us?" Because everything I'm seeing—now, we take that with, kind of, consideration. Yeah, bad things do happen, but more good things happen than bad things, right? That's what I think we have to be able to say about the past and we have to say that about the future. It looks like more good things are going to happen than bad things.

3.60 And the moral aspect of that is, the moral aspect is, really, if we can create abundance for everybody on the planet, if we can really do that, if we can lift these people out of their conditions of poverty and want, if we can do that, that's got to be a good thing. OK? All right. Thanks for the lecture. Thanks for coming.

Lecture 4: Personal Abundance

At a glance — Personal abundance.

- 4.1 Welcome back to this lecture. Thank you very much. In this lecture, we're going to apply the framework at the personal level. We'll use the equations that we discussed in the previous lecture, and we'll take that framework, that set of equations, and apply them out into the real world and see what these equations can tell us about abundance.
- 4.2 Before we do that, though, I just want to touch back on this individual, Professor Julian Simon. He wrote this book called *The Ultimate Resource*, and his argument was, the ultimate resource is really people. He makes this argument. He makes a statement that I thought was beautiful. "The ultimate resource is people—especially skilled, spirited, and hopeful young people endowed with liberty—who will exert their wills and imaginations for their own benefits. And so, inevitably, they will benefit the rest of us as well."
- 4.3 Kind of, a couple of key ideas there. It's really, kind of, young people that are full of energy and ideas that, if they have this, if they have liberty, if they have the freedom to be able to act on these ideas, that they can produce these great benefits for the rest of us.
- 4.4 Would also recall back to George Gilder. He makes this statement similar to Thomas Sowell's statement: "The difference between our age and the Stone Age is entirely due to the growth of knowledge." We have the exact same atoms that they did. What's the difference? It's knowledge. Think about your iPhones. You got 6oz of sand, basically, with some aluminum and oxygen atoms mixed in. So if we went back to the Stone Age and somebody says, "Well, make me an iPhone. Well, why can't he make an iPhone?" "Well, sorry, I don't have the knowledge." They have all the materials there. They have all the atoms. What they're missing is the knowledge. So what do they need? They need innovation. They need this idea of innovation, and we got to go find ourselves some knowledge and see if it works or not, and then build from there.
- 4.5 So, economics. The way that we define economics is, it is the study of how human beings create value for one another by discovering and sharing knowledge. OK? How do we, how does an economy work? An economy works when you and I agree to do a trade, that we serve one another, and we serve one another by this search of knowledge.
- 4.6 And we trade. We're really trading knowledge with each other when we go to a market, and we're sharing that knowledge with one another. All right, so let's assume you go out and you buy one of these new cars, and it's \$2 or \$3 million, and you take it off the lot, and in your excitement—and you're noticing that everybody's watching you in the car, right? And you're watching them and—oops. That's a terrible picture, isn't it? But

you go from this to this. What's happened to the value of that car? Now, remember that the atoms, all the atoms, are still there. You haven't lost any atoms. They're all still there. But what's happened to the value? Hasn't the value really disappeared?

- 4.7 So is the value in the atoms or the arrangement of the atoms? It's really the arrangement of the atoms that creates the value. So we have this natural world of atoms. We have this economic world of value, how much we value things. And the bridge between those two worlds is knowledge. Knowledge is what makes atoms valuable to us. Valuable to us.
- 4.8 So, let's go back and do a little history. 1800 to 2022. So we've got, what, 222 years? How many people were on the planet in 1800? Any idea? There were about a billion people on the planet. Today we have, we just hit 8 billion, right?
- 4.9 So eight times more people on the planet today than 1800. What has happened with life expectancy? Life expectancy is simply, if you're born today, how long on the average are you expected to live? How long did people live in 1800, on the average? They lived to be about 30 years old. Yeah. Why was it so low? Well, if you go to a cemetery that's very old, what do you see there? You see lots of little tombstones. They're little babies. And you also see next to them these other tombstones of women. They're, like, 22. It's like, what's the connection? They both had the last name. It was dangerous to be born, and it was dangerous to give birth. And if you got sick, you got sick.
- 4.10 So our ability to discover new knowledge has allowed us to do what? Dramatically reduce infant mortality and dramatically increase life expectancy. Because when you get sick today, you can go find some medicine that will, that will help you to overcome your illness. So we've had this huge increase in life. We get 42 more years. Well, here's a question for you. If the planet is fixed in size, if we have a fixed amount of resources and we have a lot more people, they shouldn't be living longer, they should be living shorter. Right? But we're living longer. So, Malthus Thanos Ehrlich —that model that says we're running out of resources—it doesn't look like it because people's life expectancy continues to increase. Continues to increase.
- 4.11 And then we can talk about income per day. How much money did people on the average earn per day? Back in 1800, it was about \$2 a day. \$2 a day. Imagine what your life would be like at \$2 a day. That's your food, your clothing, your housing, your car, your medical. \$2 a day wouldn't take you very far, but that's generally where we were. 1800. Today, it's about \$30 a day. So we've got what? We've got 8 times more people living 42 years longer, and their per capita income is about 15 times larger. We can illustrate this with one of our charts where we have population and per capita GDP. So let's look at 1800. We had a billion people, and they're earning about \$2 a day. So that red box represents the total amount of wealth that was generated in 1800.

- 4.12 Now what's happened? Well, first of all, we go from a billion people. Now we have 8 billion people. So it's got bigger in that dimension. What else did it do? Well, we went from \$2 a day up to \$30 a day. So there's the GDP in 2022. Let's put back the 1800 GDP. See the green area, how much new wealth that we've created for one another? Right, not only do we have 8 billion more people, but we got people that live a lot longer and are earning a lot more income to be able to not only make their life better, but to make other people's lives better around them.
- 4.13 All right. Economists, we have our own dictionary, and we've got words that we use. And if you've taken an economics class, you'll be introduced to these words that mean something in your econ class, and they mean something completely different over in your psychology or your biology class. One of the first words you'll find in that dictionary is this word—"yabut". Yabut. Anybody know what yabut means? You've all actually said that before. It means, "Yes, but. Yes, but." And what we use that word for is to remind ourselves that we always face this world of trade-offs, always are facing the world of trade-offs. We say, "Yes, that's true"—I like to say "and"—"this other thing is also true."
- 4.14 We have to look at both these trade-offs. We weigh the cost and the benefits. To be rational means to put things on a scale, a ratio, and say, "What are the benefits and what are the costs?" We look at both of those. If you only look at the benefits, you're going to make a bad decision. If you only look at the costs, you're going to make a bad decision. You want to look at both of those things, the trade-offs. So we try to remember to think about trade-offs.
- 4.15 All right, who wants to take a quiz? OK, true or false? Resources equal atoms. Can be true, could be false. How would you make it true? Because isn't resources equal to atoms? Isn't that the mistake Thanos made? Yeah. So, that statement, I would say, is false. How do you make it true? If you add knowledge to it, now it becomes true, right?
- 4.16 Resources are really atoms that are smart, atoms that have been organized, atoms that you've added knowledge to. You've come up with some knowledge that lets you make your atoms much smarter—the way they're organized, the way they're combined with other atoms.
- 4.17 All right, yes or no? Can resources be infinite, and why? Yeah, so if we just said resources equal atoms, we'd have to answer no, because atoms are fixed. We've got a fixed number of atoms. So, if resources are equal to atoms, then resources are limited because atoms are limited. But if we recognize that knowledge is really what drives a resource, then we can say that resources really can be infinite. Because knowledge, our ability to discover new knowledge is, like, we don't know where the limits are there. We seem to be just scratching the surface a little bit on knowledge. So resources really can be infinite. When we make the claim that we live on a planet that's infinitely bountiful, the reason we can make that claim is because we believe that resources are a function of atoms and knowledge. OK?

- 4.18 All right. The price system is a blank system. OK? The price system. When you think about prices, what are prices, really? What do they really contain? They really contain information, right? The price system is an information system. I go to the market, and I look at prices, and I get all this information. And that information allows me to make a decision about what I want to do. "Wow, that's on sale. I'm going to buy two of those. Oh, that's too expensive. I'm not going to do that." That price system is sending information signals to us about how to behave, really, how to change and how to react. If you change the price, you're going to change people's behavior. That information is going to change their behavior.
- 4.19 All right, so last question. What does it mean if population increases: the price doesn't increase? You double the population, and pizzas are still \$20. What's happening? It looks like you're able, your economy is able to continue to produce because the price isn't going up. Your economy is able to continue to produce even though you have more people, right? So we pay attention to prices. What's going on with prices? For economists, that's really our key source of information that we study is prices. What's going on with prices? Because it contains this information that tells us about whether something's becoming more or less scarce. OK? You increase the number of people, and if you have a fixed resource, that price is going to go up. But if you can increase the resource at the same rate or a faster rate than population, then the price isn't going to change. It might even go down. It might even go down.
- 4.20 All right, so recall our first equation—time price is equal to what? It's equal to the money price divided by your hourly income, OK? And equations are simply, they're statements of relationships between things. That equals sign just says there's a relationship between this variable and these other two variables. This ratio is interesting because it's really two variables. That's why time prices are better than money prices, because they have more information there, because you're comparing the money price to hourly income.
- 4.21 So, hourly income—where do we get that number? Because we can go out and look at prices. It's easy to go find prices. Where do you get hourly income? Well, what we try to do is find places that are keeping track of hourly income. And a great source that we use that we found is called MeasuringWorth.com. MeasuringWorth.com. So this website has been sponsored by a group of economists —the Economic History Association, the Economic History Society, the International Economic History Association. The presidents of those associations and former presidents are involved in this website. So they're trying to keep track of all of this historical information.
- 4.22 So it's a great place to go to get hourly income. If you look over here on the left side and you zoom into that, you'll see that one of the data sets is U.S. Wages. It's like, OK, I can go find wages. Let me go find wages because now I can take the wage information and put in my time price equation. So, when you click on that, it'll come up with this little window. And it has two different wage series, or two different lists of wages. It has

unskilled labor, and it has production worker compensation. And note, also, the range. It goes back to 1774. So there's almost 250 years' worth of data that I can go back there and, each year, figure out how much an unskilled worker—that's a person that just kind of starts their job. And also a production worker; a production worker, we also call those blue-collar workers—you might have heard that term, a blue-collar worker—they're working at some kind of manufacturing or retail. They're not managers, but they're blue-collar workers. And we also note that it says blue-collar or production worker compensation, instead of income.

4.23 Now, it's important to remember that you have your income, but you also have these benefits that you get. And we also should account for those. Benefits account for 20%-30% of many people's income. They get insurance, they get pensions, they get other benefits. And you should add those to be fully accounting of what I receive for an hour of my work. OK? One reason that people like benefits over wages is benefits are typically not taxed. So, you want a \$100 raise, or do you want me to give you a benefit that's worth \$100? Well, I'll take the benefit because I don't have to pay taxes on the benefit. So it's important once again to recognize benefits. And that's what we do when we calculate worker, production worker compensation.

4.24 So, if you click on a couple years, I just plug in the initial year, and I plug in 2022, and this is what it comes back to. Got this nice little chart. Here's all the wages. Now let's look at production workers' wages first. That's all the wages. So I can go back to 2000—\$19.36. Last year, it was \$34.76. So we can see that that nominal wage has been going up. This is the cost of unskilled labor. They actually built an index where they put 1860 is equal to \$100. It's been indexed. So we actually converted this index number to an hourly rate, and the hourly rate in 2018 was \$12.78. So that table over on the right has taken this index and converted it to an hourly rate.

4.25 So once you have that, though, what can you do with it? Well, you can go out and start analyzing how expensive or how affordable things were relative to the way—you can start doing time prices. Now, this chart here illustrates the difference between unskilled wages and blue-collar compensation back to 1900. In 1900, there wasn't much of a difference. And why was that? Because nobody really had any skills, right? I mean, there were people that had skills, but the difference began to emerge as people start getting more and more skills and knowledge. You see the gap that starts to form there? So in our analysis, we looked at both unskilled labor and we looked at blue-collar compensation. So we did time prices for both of those categories. So, what was your time price if you're an unskilled worker in 1980? And what's the time price of somebody who's unskilled today? And we compared that from both those perspectives. OK?

4.26 All right, so let's think about time prices. This little table is kind of the little template that I use when I start analyzing time prices. So, got four things down the first column here. The money price—what's the money price? Divided by, what's your hourly

compensation? That's going to be equal to your time price in hours, right? And then we're going to do this personal resource multiplier. That'll be a little calculation, but those will be the things that we'll collect and we'll do it for the start year. We'll pick a start year, and then we'll pick an end year, and we'll then start filling out our matrix here, or our cells. And then once we've got the numbers, we'll do how much it changed by, and then the percentage change. There's the change and the percentage change. OK?

- 4.27 So, if you can set this up in Excel or Google Sheets, just plug in those two values, and Excel will calculate all of this other information for you. It'll do all the calculations for you. OK, so, let's start with Grandpa. Grandpa used to tell me, when he was a kid, Hershey bars—this is my great-grandpa—when I was really small, "They used to only cost 5 cents when I was a kid. And now they're like a dollar." And I'd say, "Yeah, Grandpa, that's true, but how much were you earning an hour?"
- 4.28 So, let's take a look at this. Let's go back to 1900. Now, Hershey's, I mean, chocolate milk was discovered really, by a couple of Swiss candy guys. Nestlé was one of the guys in 1875, and it was really this great candy, but it was really expensive. And along comes this American in 1900, and his name was Hershey. And he figured out how to make chocolate really cheap—I mean, the same chocolate—but he figured out how to make it really inexpensively, so inexpensively that he could sell one of those bars for 5 cents. So, was 5 cents a good deal or not?
- 4.29 Well, what do we say? It depends. The two questions that we always ask in economics, two questions we always ask, is "Compared to what?" Right? That's kind of the "yabut" question. Compared to what? And then, "What happens?" So this one is really compared to what? Well, \$0.05 sounds like a good deal, maybe—compared to what? Well, what were blue-collar, or I think, what were unskilled workers earning in 1900? Well, our data set, we pull that up. 1900 says they're earning \$0.09 an hour. So, what was the time price for a Hershey bar? Well, \$0.05 divided by \$0.09. It looks like its 0.56. It took you a little more than a half an hour to earn the money to buy a Hershey's bar. And we actually just multiply that times 60 to get the minutes. It took you 33 minutes. It took Grandpa, Great-Grandpa, 33 minutes to earn the money to buy a Hershey bar. OK, he got one. That's our resource multiplier. The start year is always one.
- 4.30 So what happens in 2022? Well, 2022, the price has gone up. Now it's \$1.32. If you go to Walmart or Target, you can find a Hershey bar that's \$1.32. But what are people earning, right? The key question compared to what? Well, \$15.72. That's what people are earning today, unskilled workers. So how much time does it take an unskilled worker today? 0.08 hours, or about 5 minutes. So, if we take 33 divided by 5, we get 6.62. If I spent 33 minutes today working, how many Hershey bars would I get? I get 6.62.
- 4.31 So, next column here—how much did things change by? It looks like the price went up. The nominal price went up by \$1.27, but hourly compensation went up by \$15.63. Time

price in hours fell by almost a half an hour. The time price in minutes fell by 28 minutes. The resource multiplier—basically, I went from one to 6.62. I get 5.62 more Hershey bar for the same amount of time. And then we can calculate the percentage change. Note, the price goes up by 2,500%. Wow, the price is a lot higher. And that's what Grandpa thought about. "Wow, the price is a lot more expensive today."

4.32 But what did he forget to think about? Compared to what, Grandpa? What's your wage? Well, it went up by 17,000%. So the time price falls by 85%. In your resource abundance, you get 562% more Hershey bar for your hour today than you did in 1900. Now, one way we like to illustrate this is to show this—1900, you get one. What do you get in 2022? You get 6.6. All right, you kind of see how that works?

4.33 Let's do another one. So, another great resource. There's a company called Sears. They might still be around. They were this retail company and they came up with this innovation. They were really the Amazon of the 1900s. And they would ship these catalogs out to, everybody got these catalogs in their mail a couple of times a year, and these catalogs, they get really thick, and you can find everything in there. You can find appliances. You can even order a house, have a house shipped to you, you could put together. So, it's a great resource to go back and look at prices. So, what's fun in my life? I love to get the old Sears and Roebuck catalog out. It was Sears and Roebuck, two guys that started this. I get their catalog out from, like, 19-whenver. Pull it open, start looking at prices, get my measuring worth number out, start doing time prices on all of these products.

4.34 So, 1910—I pull out the catalog. There's a bicycle in there, \$11.95. Was that a good deal? Huh, well, how do we know if it's a good deal or not? What other piece of information do we need to know to determine whether it was a good deal or not? Yeah, we need to know the hourly compensation. So, 1910, \$11.95. You're making \$0.11 an hour. How many hours did it take you to earn the money to buy a bike? 108 hours. That's, like, over two weeks. 2.5 weeks. I had to work 2.5 weeks to earn the money to buy my bicycle. How about today? 2022. Notice the difference in the bikes? Some people say, "Well, bikes aren't that good today." Well, I don't know. "Walmart, what's in goods got a bike for \$98, and it's got 21 speeds, and it's got a shock, and it has disc brakes." OK?

4.35 What we try to do is find exactly the same thing, but sometimes it's kind of hard to do that. But let's just assume that those are two basic bicycles. So 2022, what's the deal? They're \$98, but you're making \$11.70, or \$15.72. That's what worker? That's an unskilled worker. That's a person that's just, kind of, started working. Been working for a couple years, unskilled. So how much time does it take now? Well, 98 divided by 15, you got to spend about 6 hours. You spend less than a day earning the money so you can go buy a bike.

4.36 So, what's your resource multiplier? If you spent 108 hours working and you took all that money and bought bicycles with it, how many bikes would you get? You'd get

17.43 bikes. So the price goes up, hourly compensation goes up. Time price goes down by 100, over 100 hours. You got 100 hours today that you can do something else with. You get your bike in 100 hours. Grandpa, he got his bike, but he had to spend a lot more time, right? Things used to be really time-expensive. Right? Think about: not the money price—the time price will tell you how really expensive things used to be, because time is the commodity, the thing that you really have. Right? It's time.

4.37 All right, percentage change. Once again, price went up. But look what hourly compensation went up—almost 20 times faster, right? So the time price fell by 94%. So, 1910 to 2022. Look at that. Look at that. So we are really bicycle-abundant today, compared to 1900, aren't we? Yeah. So think about, you know, the thing about it is, you think about, if you start thinking back in time and comparing yourself to yourself or your parents or grandparents, you can be nothing but just profoundly grateful for how abundant our life is today.

4.38 All right, refrigerators. 1956, you could buy a new refrigerator for \$469. I actually grew up in a house with one of those refrigerators. I remember what that thing looked like. It's \$469. You go to Home Depot today, and it looks like refrigerators are a little more money—\$549. OK, tell me what we're going to do to analyze this. Do you have the information there to really draw a conclusion? Not really. What do you need? How much were people earning, OK?

4.39 How much were people earning? Let's shift gears and think about someone that, say, blue-collar worker. 1956, \$2.16 an hour, which means how much time do you spend? You spend over 200 hours, almost 218 hours, earning the money to buy your refrigerator. 2022, you're now making almost \$35 an hour. You're going to spend about 15 hours. You spend 2 days instead of what? Instead of 5 weeks, you spend 2 days working. So you get how many refrigerators? You get almost 14 refrigerators. So percentage changed. Yeah, the money price went up a little bit, but look at the income rate. It went up a lot more.

4.40 As long as your income is going up faster than the money price, your time price is going down, right? That's what you got to think about. Yeah, prices are going up, but is my income going up as fast or faster? It should be going up faster because we should be getting smarter every day, if we're free to be able to discover new knowledge and then share that knowledge. OK? All right. One in 1956. They're almost 14 in 2022.

4.41 OK, let's talk about air conditioning—the magical, the magical innovation that makes this place livable, right? Can you imagine what life would be like in Miami without air conditioning? I mean, the whole southern United States, lots and lots of different parts of the world. It's like, you couldn't live. I mean, you could live there, but you couldn't really do anything there in the summertime. I used to—I'm from Idaho—and I would see these movies about people in the south, and they're, like, just sitting around in the sun. I go, "Wow, those people seem lazy. How come they're like that?"

- 4.42 And then I realized that, the first time I came here, it's like, "Wow, how does anybody get anything done around here?" And this was, like, in February. And then, it's like you go into air-conditioned places, like, "Wow, this is such a blessing—innovation that we can live in a place where we can control the temperature." We can control our environment. We get a little bit hot, let's cool it down. We get a little bit cold, we can warm it up, and we just touch a button, and it happens magically almost.
- 4.43 So, 1952, you could buy an air conditioner for \$350. OK, is that a good deal? Well, that depends, right? It depends. And it depends on what? Yeah. Blue-collar guys making \$1.83 an hour. So it takes them 191 hours to buy an air conditioner. Today, actually, you can go to Walmart today, and you can get this, actually, a better air conditioner. It's got a remote control on it, has more BTUs—that's how you rank air conditioners—for \$154. So not only did the price go down, but what else happened? Wages. Wages went up by almost 18 times. Went up 18 times. So what's the time price now? A little more than half a day. I'm going to work till noon. I'll take my money, and I'll go buy an air conditioner. OK? So how many air conditioners do I get today? Well, if I spend 191.3 hours working, I could get 43 air conditioners. My parents, they got one. I get 43. Right? I'm kind of embarrassed when I think about it. It's like, "you had one, I get 43. Is that fair?" Yeah, it doesn't seem like it's fair—one versus 43. OK.
- 4.44 All right. Anybody ever had a cordless drill? This is the first cordless drill that you had. You can still buy one of these. If you go to eBay, you can find one of these that's still for sale. Now, Black and Decker, this company makes these little tools. They came out with their first cordless drill in the 1960s, early 1960s, and their claim to fame is one of their drills actually went to the Moon, and Black and Decker had one of the drills that they took core samples from the Moon from.
- 4.45 OK, so 1961, they're selling this drill for—and we'll just go straight to the table here—it was \$100. And the president of the company said, "It's really a good innovation, but we're not selling very many. Why weren't you selling very many?" The price, compared to the wage—\$2.60 an hour. Blue-collar workers making \$2.60 an hour. I'm a carpenter. I'd like to get one of these drills because I could see how it could make my life really easy. I don't have to worry about the extension cord if I get this drill, but it's going to take me almost a week of work to earn the money to buy a drill—38.5 hours.
- 4.46 Today you can buy one of these drills for about \$30. Nice little cordless Black and Decker drill at Home Depot for about \$30. You're making \$30, you're making \$35, so less than an hour. I work less than an hour. So, how many drills do I get? I get 44 drills. In '61, they get one drill. Once again, 98% drop in that time price, which leads to almost a 44x factor in my drill abundance. OK, you get one, I get 44. That seems pretty good. That seems pretty good. Not only for me.
- 4.47 Now, if Elon Musk wants a drill, what price does he pay? He pays the same price you and I pay, right? They don't give him a discount because he's rich. You and I get the same drill. Everybody else gets the same drill, right? We all get these things that used

to be these luxuries, very expensive luxuries that have become what? Very inexpensive, abundant commodities. Like, everybody can get one of those. Everybody can have one now, rich or poor, middle or wherever you want. It's like everybody can kind of have one of these.

- 4.48** OK, you had breakfast this morning. Let's think about eggs. Eggs. 1919. So, the Bureau of Labor Statistics, they do these nice surveys of the prices of things. So you get out their book, and you go back to 1919, and they had surveyed all the prices of commodities—basically food, basic food commodities. They look at eggs. And the eggs, 1919—what do you think a dozen eggs cost in 1919? The price. Today they're what? Today they're like \$1.50. For \$1.50, you can buy a dozen eggs. Remember here last, during COVID, eggs went up to, like, \$4. It's like, "Wow, eggs are, like, \$4."
- 4.49** And what do we say as economists? Are they going to stay at \$4? They were \$1, they went to \$4. Are they going to stay at \$4? Probably not, because what's going to happen at a \$4 price? People are going to use less. People are going to try to make some more eggs. People are going to look for substitutes. It's hard to recycle an egg, but you're actually going to see what? You're going to see this response to the price.
- 4.50** The price is sending a signal to everybody, "You got to change your behavior." And once people changed their behavior, made adjustments to it, what happened to the price of eggs? It comes right back down. It ends up, actually, it might be lower because people who are focused on trying to make eggs smarter, chickens smarter, you end up with more intelligence, more knowledge in the egg business, which yields a lower price. You can now sell eggs lower because you figure out how to make eggs faster, better. OK?
- 4.51** So here's the table on eggs. 1919, eggs are \$0.61 a dozen. Yeah. \$0.50. \$0.61, 50/60 cents a dozen. But once again, how much am I making? I'm a blue-collar, I'm making \$0.43 an hour. So it takes me, what, almost an hour and a half to earn the money to buy a dozen eggs—85 minutes. 2022, today I can get a dozen for \$1.54. I'm making \$35 an hour. So it takes me about 3 minutes to earn the money to buy one egg. One dozen eggs, right? It's a dozen eggs. So, how many eggs do you get? 1919, you get one egg for your time. Today you get 32 eggs. So you get 32 times more eggs. We're enjoying this huge egg abundance. Once again, we've had these periods where the price has gone up and down. But what's the overall trend been? More and more and more abundance because eggs are getting smarter. We're getting smarter. We're adding knowledge to chickens so they make eggs faster and cheaper for us. OK?
- 4.52** All right. Sugar, the lovely substance. 1850. For the time it took you to earn the money to buy one pound of sugar, how many pounds of sugar would you get today? I've got you kind of set up here. You're a little scared, aren't you? You get 410 pounds of sugar. Now that's the world price of sugar. Now, the United States actually has these tariffs that they put on sugar, and they restrict the price of sugar, or they restrict the supply of sugar that makes the price go up, so the people who make sugar actually can make

more money. So they get this little benefit, this special benefit from the government because we actually pay this little tariff tax when we buy sugar in the U.S.

4.53 But if you go to Brazil or Mexico or any place else, sugar is a lot cheaper. That's why some companies actually go and make their candy bars in Mexico instead of the U.S., because they can buy the sugar a lot cheaper. If you drink your soda, you notice that some soda doesn't have sugar in it. It's got, like, corn syrup, right? Why is that? Because the price of sugar got really high relative to corn syrup. So the point is as though sugar's done what? And we can talk about whether it's good or bad. That's a different subject. What we can talk about today is the price —has done what? It's become dramatically more abundant.

4.54 So, one fun exercise we had is we pulled out a 1980 Sears catalog, looked it over, and said, "OK, there's probably stuff in that catalog that we still kind of have today." So we picked out six items that you find in a typical kitchen—a coffee maker, toaster, a blender, a mixer, a can opener and a food processor. And we tried to find the same thing at Walmart today, right? Sears used to be Walmart. Now you got, now you got Walmart. And we said, "Let's look at these prices. Go to Walmart's website, see if we can find a coffee maker that looks like the same thing, has the same features." And what did we find? This slide's a little bit, lot of things on here, but we show the price for an unskilled worker, right? \$4 an hour in 1980.

4.55 That's kind of when I started working. I remember my first job. I'm making \$4 an hour in 1980. Well, how long did it take me to buy all these things? It's going to take me a day to buy the coffee maker. It's going to take me 5 hours for the toaster, another 5 hours for the blender. I'm going to be spending about 55 hours to buy these six items. In Walmart today, how much time do I spend? I spend less than 7 hours. For the time it took me to buy one in 1980, what does an unskilled worker get today? They get almost 8. So they're 8 times better off in terms of kitchen appliances.

4.56 OK, so do you see where we're going here? What do you have to do to really compare things? Compare it against people's income, and that will tell you what the true price is, because the time price really is the true price that we have to pay. I get 24 hours a day. How am I going to spend that 24 hours? OK? I'd like to spend less time doing some things. It gives me more time to do other things. If we have an economy that's creating more and more abundance, it means we all have more and more time each day. More and more, it's really more and more choices. We don't get more time. We get more choices with what we can do with our time.

4.57 All right, let's talk about gasoline. Is gasoline expensive? Well, if you look at the price of gasoline, looks like, "Yeah, it's expensive." The nominal price. We have really good data on gasoline, so we can go and look at the gasoline prices every year, historically. Looks like the price of gasoline went up, and just recently, it really shot up, right? It seems like it's coming back down. But if we do what to the nominal price, what should we do? Compare it to the blue-collar hourly compensation—and what does that tell us? This is

the price of gasoline. Used to be really expensive. Go back to 1929, 1930, it was really expensive. It took me 25 minutes to earn the money to buy a gallon of gasoline. Today, it takes me less than 5.

- 4.58** Now, if we think about how many gallons I can purchase with an hour of labor. I'm going to work for an hour to buy my gas today—how many gallons could I get? Well, in 1900, I'm going to work an hour. I'm going to get about 3 gallons. Today, I can get somewhere around 10 gallons. Now, notice that we've got spikes happening there, right? Events happen. Things happen in the world that cause these prices to go up and down. Earthquakes. Natural disasters. We also have economic things that happen that cause the price to go up and down, but what generally happens over time? That price comes back, comes back down, once people react to that price. OK.
- 4.59** All right, how about cars? What do you buy a car for? Well, it's nice to have a nice car, a comfortable car, but I really buy the car that allows me to actually transport myself from one place to another place. So, that suggests that what's really important with cars is the mileage of the car, right? So, 1980, the most popular car was a Cutlass. I think it was Oldsmobile Cutlass. Beautiful car. One of my friends, actually, his dad had one of these cars. I remember him in high school. He had this car. Got 20 miles a gallon, though. The most popular car today is a Honda CRV. I actually have that car. It's 31 miles to the gallon. So I get 50% more miles for one gallon, right?
- 4.60** So what should you do? Should you just compare the price of gasoline? No, you should compare what it costs you to get your car to go for a mile. How much do you have to spend in terms of your time to get a mile's worth of car driving? And when you do that, what do you discover? Well, 1980—a gallon of gas was \$1.19. Today, it's around \$4. So the price of gasoline, the nominal price, has gone up. But what happened to blue-collar salary? It also went up. Look at the two numbers there, the percentage change—241 versus 281. It looks like gas has gotten a little bit more expensive, right?
- 4.61** The time price looks like it's very similar between 1980 and today. But what's the big change? Car mileage. And really, what are we doing? We can go out and try to drill more oil, but what else could we do? We could use our time and effort to try to figure out how to get this engine to be more efficient. We could try to take knowledge and use our knowledge to make a car get better mileage. And that, actually, is going to also be the benefit. Lower-priced gas would be a benefit, but a car that gets much better mileage would also be a benefit. And so, we've got that happening. So when we look at the miles per gallon, and the time price in seconds it takes to go a mile, it was about 25 seconds of your work to earn the money to buy fuel for a car in 1980. Today, it's about 13 seconds. So, the time price of driving your car has actually fallen about 42%.
- 4.62** Now, some people say, "Yeah, but the cars are a lot more expensive today." That's true. But what else is also true? You make more today. So, car in 1980, the Cutlass, sold retail price \$6,735. It took you about 738 hours of work to buy your car. Today, you can buy a CR V for about \$28,000. It takes you about 850 hours. So you spend about 10% more

time to earn the money to buy a car today than you did in 1980. Now, who would rather have a 2022 CRV than a 1980 Cutlass? Would you really pay 10% more? Yeah. Yeah, I've driven both those cars. It's worth 10%+ to have those. All right, we OK with that? So what are we trying to do? We're trying to move from measuring things with money to measuring things in time—because time, once again, gives us a much better picture, a sense of what things are truly and really costing us.

- 4.63 OK, I'm going to introduce you to this guy. His name is William Nordhaus. He was awarded the Nobel Prize in economics, and he did lots of, he's done lots of research. One of the papers that he wrote that really inspired us is this paper about the cost of light —the time it costs you to earn the money to buy light. We think about the light in here. Now, light's measured in something called lumens. And about, you need about 1,000 lumens to be able to have enough light to read. OK?
- 4.64 So, he went back in time and started to look at how much time people had to work to earn the money to buy that much light. I want to buy 1,000 lumens for 1 hour. Think about that. I have a light bulb I want to turn on for 1 hour. Well, what would you do in 1820 or so to get that? Well, you'd have to buy a candle and you'd have to spend about 3 hours working to earn the money to buy enough candles so you could have candle light at night for 1 hour of that 1,000-lumen hours —enough where you could read comfortably. So I worked 3 hours to get 1 hour of light.
- 4.65 So, what's happened to light? Well, when he studied this, he discovered that light —the time price of light—went from 3 hours down to what? One-sixth of a second. Yeah, you can buy 1 hour of light today for one-sixth of a second of your time. And these were the numbers. So, cost you \$0.18 to buy these candles. You're earning \$0.06 an hour back in 1820, 1830—\$0.06 an hour. And that was both unskilled and skilled labor because there really wasn't skilled labor. Everybody's making \$0.06 an hour.
- 4.66 So it took you 3 hours, about 180 minutes, or about 10,800 seconds. Today, the money price is that number. It's a tiny little number. You're making \$35 an hour. There's the time price in hours, minutes, and seconds. So what's the bottom line number there? What's the key number? It's this one here, compared to this one here. For the time it took your ancestors to earn the money to get 1 hour, how many hours do you get? You get 64,000. You get 64,000, almost 65,000 hours. So you've had this huge increase in light abundance.
- 4.67 Light has really burst forth in this astonishing abundance. And that's one of the things that we think might explain why we were able to escape poverty. We escaped the darkness. We discovered how to make light really, really cheap. The knowledge that we were able to discover that allowed us to now create light for everybody—it kind of defeated the whole darkness problem. It used to be the sun goes down, everybody, you go to bed, and in the wintertime, you were in bed a lot. It was cold and it was dark. But now we can have light and we can be warm. OK?

- 4.68 All right, you guys ready for a quiz? Let's do a little quiz on pizza. So, pizza costs \$20 today, and you're earning \$30 an hour. 20 years ago, pizzas were \$10, and you were earning \$8 an hour. So, you got to answer these 6 questions. First of all, what was the time price 20 years ago? What's the time price today? And how much has the time price decreased or increased? How much has pizza abundance increased? What's the compound annual greater, the compound annual growth rate in pizza abundance? What's it growing on a compound basis? Then, how many years will it take to double pizza abundance? OK, let's see if we can work through this.
- 4.69 Before I do that, I want to tell you about something that we call the Rule of 70s. Some of you may have heard of it before. It's a little rule of thumb that we use in economics to quickly determine how long it takes for something to double in size. How long is it going to take something to double in size? If it's growing at this rate, how long is it going to be before it doubles? It's a real simple little equation that's based on logarithms, if you remember those from high school, maybe.
- 4.70 It's just 70 divided by the compound annual growth rate. Will give you approximately—remember, it's not equal. That little sign there is the approximate sign. Squiggles means approximately. That's how many years it's going to take to double something in size. So if you're growing at 10% a year, how long will it take you to double something in size? It'll just be 70 divided by 10. It's like, every 7 years, you're going to double, right? OK.
- 4.71 Why isn't it 10 years, 10%? Why wouldn't it take 10 years? Because you're doing what? It's not linear. You're doing this. You're growing on growing. You're having this exponential growth rate. So if I'm going to grow at 7% a year, how long would it take to double something in size? 70 divided by 7, it's got to be 10 years. 10 years, you're going to double in size. If you're growing at 5%, about every 14 years. If you're growing at 3%, every 23 years. So, if you tell me, "Hey, I can grow at this rate," we can quickly calculate how long it takes you to double in size. Now, there's more precise equations we can use, but this is a pretty simple one that we use—the Rule of 70s. Everybody got that?
- 4.72 OK, so let's go back to our pizzas. So, 20 years ago, pizzas are \$10. I'm making \$8 an hour. What's it cost me? About 1.25 hours, or about 75 minutes, to get one pizza. What's your time price today? What would that equation be? Price is 20. I'm making how much an hour? 30. So it looks like it only takes me about 0.67 hours, or about 40 minutes of time, right? So, what happened to the price? Well, the price went up, but hourly compensation did what? Went up much faster. Therefore, the time price has fallen by how much? 70 down to 40 is a 46% decrease.
- 4.73 All right, remember how to do percentage changes? What's the difference between the two? It looks like it's -35, right? 70 down to 40 is -35. And then you divide that from where you started, so 35, -35, divided by 75 is going to be -46.7% . So the time price of pizzas have done what? Fallen by almost half, which means if I spend the same amount of time today working to earn the money to buy one pizza —excuse me, if I spent the

same amount of time today that I spent 20 years ago, how many pizzas would I get today? Almost double. Excellent. 1.875 pizzas. I get 87% more pizza today than I did 20 years ago.

- 4.74 Now, we can use a calculator. We can use Excel to calculate this compounded annual growth rate. It looks like it's 3.19%. Tell me, how many years will it take to double something in size if you're growing at 3.19%? What equation could you use to tell you that answer?
- 4.75 That Rule of 70s, right? 70 divided by 3.19 should tell me somewhere around 22 years. Every 22 years, I get twice as many pizzas. So I get one pizza today, wait 22 years, now I get two for the same. 22 more years, what do I get? I don't get three—four, because it's doing this. 1, 2, 4, 8, 16—this is doubling. It has this doubling effect. And, you know, we don't really think about that. We think in linear, we don't think in compound. We don't think about that. But that's really the way the world works, is in these compound growth rates. Not linear. Linear, we like linear because it's real simple. Looks nice and clean. It's a straight line. But it's really doing these kind of things.
- 4.76 OK, so how would we answer this? What's the time price 20 years ago? Looks like it was 1.25 hours, or about 75 minutes. Time price today? 0.667 hours, about 40 minutes. So how much did it decrease? Went from 75 down to 40, so it fell by about 46%. How much is abundance increased? You remember how we calculate that one? Just in the first case, it's, yeah, it's 75 divided by 40. That's how much abundance, and -1 will give us the percentage change, and then the compound annual growth rate, or we shorten that. We call it CAGR. C-A-G-R, CAGR, OK? About 3.19%. You're going to double, how many years? Take you to double at 3.19? Somewhere around 22 years. OK?
- 4.77 All right. So, here's what we would also say. Jordan Peterson, 12 Rules—his first book—rule number 4 is this really important rule. He makes a statement. He says, "You should compare yourself to who you were yesterday, not to who someone else is today." What's he mean by that? Well, you know, if we compare ourselves to other people today, what do you feel like? Look, there's always somebody that's smarter, younger, richer, more attractive. I'm going to be a loser, right? Because you can always find somebody that makes you feel like a loser if you're comparing yourself to other people today. And it can also lead to envy and resentment and all these other things he talks about, if you're comparing yourself to other people. He says you should not do that.
- 4.78 You should do what? Compare yourself to who you were yesterday. In other words, think about where you were yesterday. And what he really means by this is, compare life over time, not across space—over time. Go back in time and think about what life was like yesterday or 10 years ago. My parents. My grandparents. If you do that, you can be nothing but profoundly grateful, because in almost every dimension we think about, your life is, is much, much more abundant. Much, much more abundant. OK?

- 4.79 So there's wisdom there in using time prices over this comparison across other people—their incomes, their heights, their weights, their attractiveness. No, no, no. Don't do, don't compare that way. Compare what yesterday was like, and you'll have a completely different perspective. It's your perspective that will lead you to this conclusion. You change your perspective, you're going to change your conclusion. And it can also change, kind of, your attitude about life and the way you feel about life, depending on how you compare, because we can't help but compare ourselves to things, right? So the proper dimension for comparison is time, not space. Does that make sense? OK.
- 4.80 All right, so our lecture summary, first of all, what we do? We took the equations that we learned in the previous lecture, and we went out and we looked at the world and said, "OK, let's take these equations and apply them at the personal level." Let's see what's happened to what? Hershey bars, air conditioners, bicycles, food. We just barely touched on some of these things. In the book, we go in and we look at hundreds of different commodity items and products and services. And what we concluded is, it was really, really hard to find something that has become less abundant—very, very hard to do that.
- 4.81 All right, so we took our tools—our framework, our set of equations—we went out into the world. We got some prices from Sears' catalog or from published reports of prices, what prices used to be. We then pulled up our wage rates, and we did time prices on all these things. And then we had the time price back at some beginning date, some start year. We compared it to a current year, and then we calculated the percentage differences—got the change in the time price and then got the change in the abundance.
- 4.82 And then it's like, OK, we've got some facts. Now we can talk about things, right? It's one thing to say, "Well, things have become more or less abundant." Isn't it much more powerful to be able to take a fact and lay it on the table and say, "Here's the evidence that I have found that leads me to this conclusion," right? Nothing worse than someone making a statement. It's like, "Where did you come up with that? What evidence do you have to support that?" It's not a conclusion, it's just an assertion.
- 4.83 We want to make conclusions. In science, in economics, we advance through this process of evidence—equations that help us understand relationships—and then we can draw conclusions about what has happened. And then from what has happened, we can do what? Make predictions into the future.
- 4.84 What is a theory? Clayton Christensen, Harvard professor, said, "A theory is simply a statement of causation." "I think A causes B." Why are theories really important to us? Because if you have a really good theory, you can predict the future. You know what's going to happen tomorrow, because this theory says, "If this happens, then that's going to happen." OK?

- 4.85 So when we do our evidence research and we start finding all of this evidence about things, what does it help us do then, in terms of our theory, to then say, "Let's connect all these dots and make a forecast about tomorrow?" It allows us to predict the future. And this is why Julian Simon was able to bet—because he had done all of this research on all of these prices, and he knew what history said about these prices. And that's why his confidence in making this bet—he knew that there was a chance he could be wrong because something could have happened, right? I mean, we could have had some kind of event and things could have went bad. Yeah, war. He did write a little clause in that contract. If we have a war, this bet's off because things all go kind of chaotic during a war. But he was confident because he had done the research and he had the data and he'd done the analysis and he figured out what was happening.
- 4.86 If I can figure out what's happened for the last 100 years, what can I do about tomorrow? Well, I can probably make a prediction about tomorrow, assuming things aren't going to change dramatically, something else isn't going to change. We should be able to continue this path, this path of what Julian discovered, this path of abundance, and what we then reconfirmed—that this is still true. This is still true.
- 4.87 So, we talked about hourly compensation sources. Where can you go to get good hourly data, on the U.S. Market? MeasuringWorth.com. That place has become like one of my best friends because they've got all this really good data, and I'm really confident that it's probably some of the best data out there. There's other places you can look to find data, but that's a pretty good source. And in many cases, you want to look at multiple places to see who else is collecting data and see what they have to say. OK? So, MeasuringWorth.com, good place to get U.S. data. They also have U.K. wage data, and so you can go find that, too, if you want to go and compare some prices in the U.K.
- 4.88 So we have some good hourly compensation sources. And that's a beautiful thing, because now we have the denominators we need to then go out and look at the numerators—the Sears catalogs, the old publications that we pull up when we find these prices. We go and we find newspapers in Paris in 1850, and we can figure out, "Oh, this is the price of bread in 1850. Well, how much were people earning? Well, we found another source, and this is how much people were earning." Bingo. We've got our two numbers. We can now do the time price.
- 4.89 So we can do it in any country, with any commodity, with any product, with any currency. We can do it in dollars, we can do it in euros, we can do it in francs and marks, in any currency. And that lets us transcend all of these issues with differences in currencies because now we're expressing everything in hours and minutes, not dollars or francs or marks or euros. It's hours and minutes. Now, what this also allows us to do is figure out what the time price of something is in the U.S. and go compare that time price to the same product in China. Because we know what people earn in China, what the price is in China. We can then start making those comparisons. OK? So it's a

beautiful and elegant and very simple way to be able to analyze, kind of, our standards of living.

- 4.90** OK, we went through a bunch of little examples. The book—we spend a lot more time going through lots more examples. You'll be able to read about those. And I would encourage you once again to take these tools and see if you can't do a time price on something. Pick something that's of interest to you and say, "I'm going to go see if I can calculate what the time price of this thing is." Now, one challenge you have is products typically change over time, right? They typically get better. So it might be kind of hard to compare the bicycle in 1910 to the bicycle today because it looks like it's a different bicycle. And that's certainly true if you're talking about iPhones or even cars. Things change over time. They typically get better over time. So you have to maybe make an adjustment, but fundamentally, you can calculate the time price between those two different periods of time with that commodity.
- 4.91** So the first thing that we did is, we actually analyzed a bunch of basic commodities that really don't change. Wheat. A bushel of wheat is kind of like the same bushel of wheat today as it was 100 years ago. So, wheat was easy to compare. Crude oil. Yeah, we had crude oil sales and, 100 years ago, we knew what the prices were. Crude oil really hasn't changed. So, we looked at all these different commodities that really haven't changed. We didn't have to make an adjustment for changes in quality, and we compared those time prices. That was kind of our first exercise. And we discuss that in the book, and we'll talk about that in a later lecture—what we found when we did that first analysis.
- 4.92** And then we spent some time doing a quiz to see if we could actually apply this. We practiced a little bit, we looked at the theory, which leads us to—here's the framework with the equations. It's the relationships between things. Now we have those equations. Let's go out and try them on something. We can then say, "Here's the facts. Here's what we concluded. Here's the numbers we were able to quantify and then measure things." Science is based on being able to quantify something and measure it. We make advances when we are able to assign numbers to things because that allows us to begin to measure. So time prices give us this way to really, really measure things across time. OK?
- 4.93** Then we had our little pizza quiz, and you all got an A on the quiz. OK. All right. Thank you for attending this lecture.

Lecture 5: Resource Abundance

At a glance — Resource abundance.

- 5.1** All right, welcome back to our next lecture. In this lecture, we're going to talk about applying our framework at the population level. Remember, our framework allows us to analyze what's happening at the personal or individual level, but it also lets us analyze what's happening to resources at the population or global level. So that's what we would like to do in this lecture. So, personal level—that's really looking at what individuals are experiencing in terms of their personal change in abundance, that they're having in their own lives. The population level looks at what's happening to the planet. OK? Or what's happening to, maybe, a country. We can then analyze this change in abundance at that level. OK?
- 5.2** So, our equation, if we recall—population resource abundance—is equal to simply the personal resource abundance multiplied times a population. So, think about it this way: How big is your slice, and how many slices are there? How many people are there? Here's the size of the pie. Here's the slices. How big is your slice, and how many slices are there? So what we're trying to do is figure out the size of the pie now—not just the size of the individual slice, whether it's getting larger or smaller—but the size of the whole pie. And you can make the pie bigger two ways. Individuals—their slices can get larger—and you can also get more slices. You can have more people. OK?
- 5.3** So, once again, if population increases, but the time price doesn't increase, what does that mean? It means you're growing the pie. You're growing the pie. OK?
- 5.4** And that's what we've been able to do, right—is we've had more and more people. The time price tends to go down, which means the abundance is increasing. And what we observed is, the abundance is increasing at a faster rate than the population is growing. That's what superabundance is. It really goes back to this initial argument made by Malthus, this relationship between population and resources. What is that relationship? He offered a hypothetical model and said population—excuse me—that food increases at this linear rate, but population grows at this exponential or compounding rate. And that gap between the two, it means your society, your civilization, is going to collapse because you're making people faster than you can make food. And everybody looked at it and said, "Wow, the math looks good. That looks right. Must be true." And what have we learned? What do we know as economists? Well, you've got to question those theories. What did the facts actually say? Do the facts match your theory? Did the facts match this model or not? And that's where we get in lots and lots of trouble, is instead of relying on data and facts, we rely on hypothetical theories and models. We can get ourselves into lots of trouble if we do that. OK, so there's our equation.

- 5.5 So, think about our little table. We did this first part here. When we did the personal level, we stopped right here. Remember, it was money price. Time divided by hourly compensation gave us the time price, and then we could calculate the multiplier there. What we're going to do is add two more rows to this, and we're going to take this personal resource multiplier. We're going to multiply it times the size of the population. That'll tell us what the population resource level is—what the resource level is at the population level. OK?
- 5.6 So let's go back and talk about our Hershey bars. Remember, we got to that bottom line. We had the resource multiplier. That's the row that we need to be able to go to the next level. So we're going to take that multiplier. Remember, 1900, one Hershey bar? 2022, you get 6.62? So we have our two values that we need for the personal resource abundance. Now we're going to plug that into the population. So, we're now going to take those personal, and what do we do? Multiply it times population. We're going to use U.S. population because this was really a product that was sold in the U.S. And so we're going to look at the U.S. population. 1900, we had about 68 million people. Today, we have over 330, about 336 million people today.
- 5.7 So, what's happened? The resource multiplier increased by 562%, but population was also growing. So the population resource multiplier—that times that gave us this number—now you get 6.62 bars. But how many people are there? 336 million. So you have this resource factor of 2,223. It's the change between those two numbers that we're looking at. So what's happened there? Yeah, Hershey bars have increased by over 3,000%. Our abundance of Hershey bars has increased by over 3,000%. And once again, how did we get there? Because every individual now gets 6 for one. And we have how many more people? We got 268 million more people. So it's like Hershey bars are—become really, really abundant, as you can see.
- 5.8 All right, so there's all of the data, shows us the population level, starts with the money price divided by hourly income, gets the time price. Then it shows us the personal level. Then we multiply that times the population, gets the population level. We can actually index that back to one. And we note that we end up with what? About 32 times more Hershey bar abundance today than we had in 1900.
- 5.9 We can actually also chart this. Remember the chart that we used where we looked at the size of the resource in a start year and compared it to the size of the total resource in the ending year? So, if you go back to 1900, there was population. Let's just make that little red box the Hershey bar abundance in 1900. So, what happened? Well, population grew by 394%, but personal abundance grew by 562%. I know I'm throwing these numbers out here, but we can actually go back and look at the calculation. Just look at the chart, and the chart will say, "Wow, every individual is getting more, and you have a lot more individuals." So that green area represents the percentage change—how much growth, really, how much abundance you've had with Hershey bars.

- 5.10** Now, is this a good thing? Well, it's a really interesting thing. Let's compare it to some other things and see what happens. Once we have that population level abundance, we can calculate the CAGR, the compound annual growth rate. Looks like Hershey bar abundance at the population level was growing about 3% a year. If it grows at 3% a year, how long does it take to double? Somewhere around 25 years or so. Yeah.
- 5.11** You remember our equations with elasticity, which had to do—if you increase population by a percent, what happens to the Hershey bar abundance? Does it go up by a percent or down, or does it go up at a different rate? Well, at the personal level, it looks like, for every 1% the population increased—so we take this number—every 1% the population increased, you get 562. That ratio says that every 1% increase in population, everybody gets 1.42% more Hershey bars. So you're making Hershey bars faster than you're making people. And then at the population, it's even much greater. Every 1% increase in population corresponded to about an 8% increase in Hershey bars for everybody. OK?
- 5.12** You OK with that? Let's think about bicycles. 1910, we had this resource multiplier. You got one bike. How many bikes do you get today? You get 17. So there's at the personal level. Look again at the population. In this case, we're going to look at the whole planet. Had about 1.75 billion people on the planet in 1910. Today you've got 8 billion. So the population resource multiplier goes from basically one, if we index it, from one up to almost 80. So what do we have today? We got 80 times more bicycle abundance than we did in 1910.
- 5.13** And when we chart that, what's the chart going to look like? Let's see if we can think about that chart. So if we make the red box 1910, what's that green box going to look like? It's going to be taller. How much taller is it going to be? Is it going to be 17 times taller? And how much wider is it going to be? It's going to be about 3.5 times wider, right, for a population? And that's kind of what it looks like. OK, so there's 1910 right here. We have this population increase, and we got this personal abundance. So we got this huge increase. Bicycles are running at about 4% compounded annual growth rate. So every 18 years, we double the abundance of bicycles on the planet. Bicycles—interesting thing, right?
- 5.14** Air conditioning. Look at air conditioning. 43 times more personal abundance with air conditioners compared to 1952. U.S. population increased, almost doubled. So our abundance multiplier about 9,000 times. What's that red-green chart going to look like? Start with a little red box. How far is it going to go up? Like, 43 times, right? And then it's going to go wide, about, it's going to double almost, little more than double. So when you look at that, you can just—you don't have to look at the numbers, just look at the chart. There's where we were, it's like we went, we increased population, but we really, everybody got air conditioners. Everybody got air conditioners for, like, almost free compared to what they were in 1952.

- 5.15** OK, you guys look like you're ready for a quiz. OK, pick up where our pizza example was. Remember that, in the last 20 years, pizza abundance increased by 87.5%. During this population—let's assume that during this population, or during this period, population increased by 50%. Let's see if we can answer these 5 questions. How much has population of pizza abundance increased? What's the CAGR, the compound annual growth rate? How many years will it take to double? And what's the personal resource abundance elasticity of population? I know that's a long term, but it's a pretty simple meaning. It says, "Well, what's that divided by that? What percent there, compared to what?" We're going to compare those two rates of change and see what they say about the rates of change.
- 5.16** So how would we set this up? Well, there's our personal resource abundance for pizza. We went from one 20 years ago to 1.875 today. What do we do with population? We made the assumption that population increased by 50%, went from 1 to 1.5, so what's the population resource multiplier? One times one is going to be one. 1.875 times 1.5 is going to be 2.8125, which means what? It looks like you've increased what with pizza? How much more pizza abundance do you have? I get 3. Right? We had 1, now we have 3. OK?
- 5.17** Once again, are we counting up pizzas? No, we're not counting up pizzas. We're looking at the time price. What's more important to economists—the quantity of something or the price of something? Isn't it the price? And what price should we look at—the money price or the time price? Time price. So we're looking at the time price of all these things and comparing them over time, and it's like, "Wow, it looks like things have gotten really cheap for a lot of people." You keep adding population, and things get cheaper and cheaper.
- 5.18** How could that be possible if you add more people and things get cheaper? Because resources are not just atoms. Resources are what? Atoms plus knowledge. It looks like you're adding lots and lots of knowledge. You're figuring out how to make pizzas that don't cost very much. Now, we could argue about how they taste, but you can certainly say something about the time price of pizza. It looks like it's going down. Takes less and less and less time to earn the money to buy pizza today than it ever really has. OK?
- 5.19** All right. So, can we do the compound annual growth rate? You're going from 1—maybe I'll use my finger—which is going from 1 up to 1.875 in 20 years. So if you plug that into a financial calculator, it will tell you the compound annual growth rate. So, I go from 1 to 1.875. That's the growth over how long? 20 years. Plug in the 20 years and ask it to tell me what the CAGR is, and it comes back with 3.19. OK? Does that make sense? OK, that's at the personal level.
- 5.20** What two numbers would I use for the population level? I'd use these two numbers here. I'd use this number and this number. Yeah. All right. So if I go from 1 to 2.8125 in 20 years, what is my increase going to be? Excuse me, let's go there. Looks like it's 5%, right? Pizzas are growing about 5% a year. So how long does it take for you—how big is

your slice growing each year? It's growing at about 3.2%. It means every 22 years—remember our Rule of 70s?—70 divided by that, every 22 years, your slice is twice as big. But in terms of the population level, if you're going at 5%, what does that mean? It means their slice, every 13 years, the pie doubles in size. The pie is doubling in size.

- 5.21** OK, elasticity of population. What's that a function of? Elasticity of population? Let me see if I can do this again. That elasticity of population. How about, this is—let me just point at this. Elasticity of population is what? It's that number divided by this number, right? Elasticity—personal abundance resource, elasticity of population—is how much pizza you're getting. Right? And how fast is your population growing? 87.5 divided by 50 says, every 1% growth in population, everybody's pizza grows by 1.75%. In terms of the whole population, it's this one, right? 181.25 divided by 50 gives you that number. So the pie is doing what? It's growing about 3.6% every year for the whole population. OK?
- 5.22** All right. We OK with that? So, can we answer these questions? How much has it grown? Population level, 81.25. Our compound annual growth rate, about 5%. 5%, how long is it going to take to double? A little under 14 years. Our elasticity of population, our personal elasticity—every 1% growth in population means each one of us get 1.75% more pizza. The whole planet is getting what? 3.63% more pizza. Once again, you got more people, but the time price is going down, which has to be what? You're enjoying this pizza abundance. OK? All right. We OK with that? Everybody got 100 on the quiz? OK.
- 5.23** All right, so I want to talk about something a little bit different. Remember when we, remember when we were talking about the benefits of using time instead of money to measure things? And one of the benefits is that we have this equality of time. We have this equality of time. So maybe time is a better way to measure differences in inequality than money is. OK? What do you get to do with your 24 hours a day, and what do I get to do with my 24 hours a day? Because we wake up and each one of us gets 24 hours. What do you get to do with your time relative to what everybody else gets to do with their time? OK?
- 5.24** So, one way to think about this is, let's go back to 1960, and let's go to India and China and the U.S., and we look at these basic commodity prices. We looked at the price of a bushel of wheat, a bushel of rice, and a bushel of corn—the three basic commodities that people eat on the planet, those three things. And we said, "OK, what's the time price for those three things?" And we looked at what the time price was in China, the time price in India, and the time price in the U.S. Do you think there was a difference? Yeah. The price of all these commodities—the same, right? But what was the denominator? It's much different in the U.S. than it was in India and China, right?
- 5.25** So, what was the time price in China? It's a lot higher. In India, a lot higher than the U.S. In fact, if you look at the time price in India—there's India. The blue dot's the U.S. It took about an hour to earn equivalent of a bushel of those three commodities. India

and China, on the other hand, it took what? It took about 8 hours. So when you talk about somebody, who is a subsistence, living on subsistence, it means, "I have to work all day today to earn the money so I can buy the food to eat today. I'm not earning enough money to buy extra food, so I don't have to work till tomorrow. I've got to work every day." So what was the case in 1960 in India and China? Both those places, on the average, people spent their whole day working just to earn the money to buy their food for that day. The U.S., we were a lot richer. Only took us an hour. India and China, it took 8 hours. You see the difference there between the inequality? Big difference there, big difference in the time inequality.

- 5.26** But what happened over time? Well, India—because what did India do? Their incomes increased dramatically. Their incomes actually increased much faster than incomes in the U.S. Now, they started at a much lower income, but their income went up really fast. So now they only spend about 90 minutes a day, 1.5 hours today, to earn the money to buy their food in India. In China, it's even more dramatic because their income was actually going much faster. Then China, so they spend about 18 minutes. Now, are things better in the U.S. today than 1960? Yeah, things are even better for us. But now, look at the difference between those two. Look at this gap versus this gap. OK?
- 5.27** You see what we're thinking about there? Now, these Indians and Chinese, they've got a lot more time to be able to do something else. They get to go from being consumers to being creators. Isn't that where we want everybody to be? Move from being just a consumer only to being able to create? And if you have more time, that allows you to do that. So it's a really, really important variable, we think—time inequality. And as things become more and more abundant, it means people's time—we all, our 24 hours starts to look very similar. Our days start looking very similar in terms of what we can do there. OK?
- 5.28** India, they save 6.5 hours. China, over 7, almost 8 hours. U.S., we didn't save much because what? We weren't, we didn't start with very much. We didn't spend very much time to begin with. We were down only an hour a day. Now we spend less than 13 minutes a day to buy that commodity basket. So, 1960, there was a 7-hour spread. Today, India, it's about a 1-hour, a little more than 1 hour. China, it's about 4.5, 5 minutes. OK?
- 5.29** All right. Time inequality versus income inequality. OK? I've always thought about income inequality. It's like, "Yeah, you have that income, but what are you really able to do with it?" Time, on the other hand, you get to make choices. What would you rather have—more time or more money? When you're young, you say, "I want more money." When you're old, you say, "I want more time." But if you think about it deeper, you realize that time is actually more valuable than money, because time gives you the time to earn money. You can't buy time with money, but you can use your time to earn money.

- 5.30 So it's really, what is your time going to give you today? And you're hoping that you're going to be able to spend it in a way that's going to give you the highest value per minute in your life. And that's a function as if you're creating knowledge and everybody else is, because we all then do what with our time? Discover new knowledge that allows us to create value for one another. OK? All right.
- 5.31 So, second thing we want to talk about is how to build an index. Index is important because if I'm comparing two things and they're really different, like a barrel of oil and a bushel of wheat, how do I compare those two? You know, do I compare it by weight? Do I compare it by volume? How do I compare the two? So, one thing we can do is, we can go from quantities—physical quantities—to prices, right? And then we can start comparing prices and changes in prices.
- 5.32 So let me show you some data here. I had 1980. Hourly income is \$3.24. Crude oil—you could buy a barrel of crude oil for 36, about \$37. Wheat, for a metric ton, was about \$172. An aluminum metric ton was about \$1,775. So what's the first thing that we need to do with these money prices? Convert them to time prices. So we have the hourly income there, and we have the money prices. So let's just take these, these numbers, and divide them all by that number, right? And that will give us the time prices for each one of those. OK, there's the time price.
- 5.33 Now the next thing that we can do is, we can actually index these things. So we're going to set each one of these values equal to 100 for each one of these commodities. And we'll see how that helps us to start to doing this analysis. So let's set each one of those equal to 100. OK? Now the next step is, let's go to 1990. 1990, the data that we find—income's \$5.41. This is what's happened to those prices. We then calculate the time price for those. Now we can compare these two time prices, right? These two time prices. If we've indexed that to 100, what's that allow us to do? We can then compare that percentage change on each one of those.
- 5.34 All right, do you see what crude oil's done? If crude oil, we index it to 100 in 1980, 1990 it had fallen to 37. OK? Is crude oil becoming more abundant? Yeah, because the time price is falling relative to these other indexes. See, we can look at these products. Which one of these products now is falling at a faster rate? Which one is becoming more and more abundant? Yeah, it looks like the crude oil. See how crude oil is becoming a lot cheaper? Think about it. Cost me 100 to buy crude oil. It cost me 100 to buy wheat. It cost me 100 to buy aluminum. 1990, ten years later, it looks like they're all on sale. Now from here, I can take the average. On the average, it looks like everything is on sale for—used to cost me 100, now it's only 46.
- 5.35 And we can extend this and we can do—let's do 2000. We got the hourly income, there's the prices. Convert those to time prices, index those, and now we get that number. See how we're taking specific commodities? Because sometimes it's hard to just look at one commodity because someone says, "Well, there was something that happened to that commodity that that's not true for everything else." Well, let's put

everything else into one basket, and look at the whole basket, and see what's happening to the whole basket. This is how we do that. We convert it to time prices and then index everything, so the start year is 100. Then we can see the relative change with each one of those commodities over time.

- 5.36** Then we can chart that. So this is what the chart would look like. OK? They all start at 100—index value. But look what's happening. Crude oil looks like it's gone lower. Looks like wheat actually ended up at the bottom. And the index value, all three of those—there's the average of all three of them—now we can start seeing it.
- 5.37** So, in your research and when you're presenting your findings, what thing do you want to show people first? Don't you want to show them this first? And then, when they ask you about it, you go back and show the data where you got this—because this really communicates kind of what's been happening to these, to these individual commodities, as well as that basket or that group of commodities. And then you can, once again, go back and show the facts that support your analysis, OK? You're trying to be able to communicate as fast and as simple as possible what the facts say. That's why we like charts. That's why we like charts.
- 5.38** OK, we OK with that? How to index a group of commodities? OK, once again, in our book, we go on and explain this, we give you some more examples how to do this thing, because we're really—once again, we want you to become, we want you to become time price economists. So you're able to, kind of, build these indexes and build these, do this analysis yourself.
- 5.39** OK, so let's talk about another thing. It's kind of, kind of important. It's how do we average percentage changes? Now, the problem with percentage changes is they can't be added up to get the average. If you think about this for a minute, you'll see why this is the case. So let's say we have two commodities, and wheat goes up by, wheat goes down by 50%, but corn goes up by 50%. If we just took the mathematical normal average, it looks like the average is 0%, right? But that's actually not the case.
- 5.40** To figure out the geometric—we have something that's called the geometric average—it's different than the arithmetic average. And the way we calculate that, is we just determine what the resource multiplier is. So if wheat's 50% off, how much wheat do you get for the same amount of time now? Don't you get two, instead of one? And if the price of corn goes up by 50%, how much corn do you get? You only get two-thirds of the corn that you got before. OK, so what happened to wheat? What did it change by? You get 100% more wheat, but how much corn do you get? You get 33% less. If you add those two together, what do you end up with on the average? Yeah, you get 33% more overall—which means what? Means that, on the average, the geometric average, the price fell by 25%. Didn't fall by 0%, it fell by 25%.
- 5.41** To think about this again, look at another example. What if you have wheat and corn? One goes down by 10%, the other one goes down by 90%. If you just average the two

numbers, you say, "Well, it looks like, on the average, it went down by 50%." And that would not be correct, because if something's on sale for 10% off, what do you get? You get 1.1 for the price that you used to get. But if it's 90% off, what do you get? Something's 90% off, you walk in there, you get 10 for the price of one. So now you get 11.11 units. On the average, you got two. 11.11 divided by two means you get 5.56. So you get 456% more—more when those prices fall like that. So the average here, the calculated average going from here back to here, is about 82% off. OK?

- 5.42** So what you have to just remember is that, when you are trying to average percentage changes, you cannot just take the average. You've got to do the geometric average. So it's an additional step that you have to go through to get the actual number that represents the whole basket. OK?
- 5.43** All right, so recall that we have this. We started off with this bet between Simon and Ehrlich. And they, Ehrlich picked 5 commodities, and over a 10-year period, and he lost the bet because the prices all fell. And he had to write Simon this check. So, our initial research started when we asked, "Would Simon win this bet today?" What would happen today? So there were two criticisms of the bet. The first criticism was, "Hey, there was only 5 commodities. Your basket wasn't large enough to really represent an economy."
- 5.44** So, to address that objection, we expanded our data set from 5 commodities to 50. We not only looked at those original 5, we looked at 45 other commodities, including energy. So we looked at oil, coal, natural gas. We looked at food, so we looked at wheat, bananas, oranges. We looked at materials like lumber, cotton, rubber. We also looked at other minerals like aluminum and iron ore. And the beauty of this is that the World Bank keeps track of all these prices. Every month they produce a report. So we're able to go to their website, find all of these nominal prices. So we have all the price data, and then we divided by income factor that we calculated for the global GDP per hour worked.
- 5.45** Once again, it isn't just how much money you make, it's how much time it takes you to earn that money. If you have this kind of income, but you worked fewer—last year I worked 40 hours a week, but this year I only worked 35 hours a week—my hourly income is actually going up, right? My total income is the same, but my hourly income is going up because I'm working less and less. So we analyzed how much people were working each year, and there's an organization that keeps track of that. So we got this number that we could use for the denominator in our price time analysis for these 50 commodities.
- 5.46** So we said, the other objection, "It was only ten years." So we said, "Well, let's start in 1980 and we'll go to 38 years." We'll look at every year for the past 38 years and see what we find out. And we called this index that we built, the Simon Abundance Index. And there's our commodities. Oil. Natural gas. We have natural gas in the U.S. and natural gas in Europe. There's two different markets there. We have coal. We have tea

and coffee. And the reason we picked these commodities is we had good data on the prices. World Bank, once again, is keeping track of all these prices of things. It's like, "Wow, you've got this really good data there." We'll use your data, because everybody trusts the World Bank.

- 5.47** When you do research, you want to be able, if you can, use a data source that everybody else believes is credible and trustworthy. So then there's no dispute over the data. So World Bank helped us out a lot, had all of this data for us. They had data on soybeans and wheat and corn. They had it on beef and chicken and shrimp. They had it on wool and 2x4s, and plywood, iron, tin, gold, silver, and platinum. So we take all of these commodities and we figure out what the time price was back in 1980. And then each year—we have the time price each year—and we also then calculated the change in that time price over time. So one of the first charts that you'll find in the book—I know you can't see this from there—but there's all 50 commodities. And then we actually make this thing we call a chartable, which is a chart plus a table we put together. I'm going to zoom in here so you can kind of see one of these. That would be better.
- 5.48** OK, so sugar, 1980 to 2020. Remember, the first thing we looked at was a percentage change in the time price? The sugar price fell by 86%, which means how much—what happened on the abundance factor? Means for 1 hour of time in 1980, how much sugar did you get in 2020? 7.25 units. What's the percentage change? You get 624% more sugar. What's the CAGR? Sugar is growing about 5% a year at the same rate that I'm growing. And then, years to double is what? The Rule of 70 tells us that. So we can look at this, whole chart's got this, all these 50 commodities and all these different ways to look at the commodity. What's happened? Time price goes down. That means abundance goes up. Well, how fast is abundance going up? Well, here's the compounded annual growth rate. Here's the rate that it's growing at. Well, that's interesting. Well, how long is it going to take to double in size? Oh, well, this is how long it takes to double that commodity's abundance.
- 5.49** So we have all of these commodities, and here's what our conclusion was. From 1980 to 2020, what did the average time price do? Fell by 75.2%. Now, if the time price falls by 75%, it means what? And you walk into the store at 75% off, I get 4 for the price of one. I go from one to 4, it's about a 300% increase, right? So I get 300% more for the same time it took me to earn the money to buy that back in 1980. What happened to population? Global population increased by 75.8%. So what's happening? Every 1% increase in population corresponded to a 1% decrease in the time price. What explains that? Why could you have population growing and the price going down? What explains that gap? What's in that gap? The K-word. Knowledge.
- 5.50** We're growing knowledge faster than we're growing people, because we grew people—we grew 75.8% more people—but what did we grow in abundance? 300% on the individual personal level. At the population level, the global level—remember, that's

this abundance level times the population—it's 600%. So, our compound annual growth rate at the population level is about 5%, which means what? Excuse me, at the compound annual growth rate, it's about 5%. Which means what? Look at this number. Every time you increase this by a percent, you increase by 75% here, but you increase by 600% here. So that divided by that is where you get 8%. Every 1% increase in population corresponded to what? An 8% increase in this commodity basket.

- 5.51** Now, some of those commodities were getting more abundant than others. When Marian and I first did this study, we thought, "Oh, man, we're going to find something that's gotten more scarce." We thought for sure something had become more scarce since 1980. We got to be kind of running out of something because we hadn't fully understand what was happening there. So it really astonished us. It really shocked us when it's like, there's nothing here that's becoming more scarce. There were times where you would have a scarcity issue, but then it would go away. You'd have a war, and then things kind of go chaotic during a war. But after the war, it seemed like things kind of got back to normal, and things continued to grow. Abundance continued to grow. So we were quite surprised.
- 5.52** And then we checked the data, and we rechecked it, and we had other people check the data, because we were hoping to find an error. Right? True science hopes that someone finds an error in the data or your logic that you can correct yourself from. So we had all these people looking at it saying, "Can you find something that we're missing here? What are we missing here?" And we just really couldn't find anything because the facts tell a story. The facts tell a story. They don't have an agenda, right?
- 5.53** So we chart all of this, and we put all this on one chart. We start with this as a percentage change of everything. 1980, we start at zero. What happened to the time prices? They fall by 75%. That black line is just the zero line. What happened to population? It's growing. It grew at 75%. What happened to personal resource abundance? You notice there were periods here where it collapses and it comes back up? Yeah. What do you think—this is down to 2020—what do you think happened in 2021 and 2022? Yeah, we had COVID, which was like a war, and we're going to have this event. And then it came, it's now turned around. It's coming back up. But we have these events that occur. But look at that underlying trend line.
- 5.54** If you were to plot a trend through there, what would you say? It looks like you could forecast out into the future. And if Julian Simon, once again, he'd done this on all these prices, he says, "If I'm going to forecast 10 years out, I'm probably pretty confident 10 years out. We might have some issues, but we're probably going to be more abundant 10 years out, because I have all this data that's predicting, and I also have a theory that explains the data. I understand why things become more abundant." Because more people make things more abundant because, well, we're going to talk about that in the next lecture—the explanation for how that happens.

- 5.55** OK, so, to summarize this lecture, we applied the framework at the population level. Remember, we can use the framework at the personal level, and we can also use it at the population level. We can look what's happening to individuals, and we can look at what happens, what's happening to the resources at the population level. So that's our framework that lets us do that. We looked at some examples, kind of saw how it looked. We looked at our little red-green charts to see how the change happened, what it looked like to us.
- 5.56** We could see a lot of information in that little chart. Then we talked about time inequality. Why is time inequality maybe as good or better than income inequality? Because everybody wakes up with 24 hours a day. How do you get to spend your time today? OK? What do we notice about time inequality over the planet? It seems to be doing what? Seems to be collapsing. Everybody seems to be getting more and more choices with what they can do with their time every day, because we're doing what for each other? We're lifting each other out because we're creating, discovering, and sharing knowledge.
- 5.57** We talked a little bit about how to build an index. The book goes into more detail on that. We also talked about averaging percentage changes. Those are kind of some math things that we had to deal with, but we had to get that straight in order to make sure that our indexes were working right. And then we talked about the Simon Abundance Index. We named this index in honor of Julian Simon because he inspired both of us, you know, to look at the world from a new perspective, to think about—to think about things based on what the facts were telling us.
- 5.58** And that's what I'd probably leave you with on this lecture is, look at the facts and see what the facts are trying to tell you. And then think about your perspective that you look at those facts from. And maybe your perspective, you need to change it a little bit. And suddenly the facts might reveal something that you didn't see at all. If you change your perspective, it's going to change what you see—and the perspective of thinking in time, it's a new perspective. It's a new economics of knowledge, learning, and time. How do we think about knowledge instead of atoms? And how do we think about how we learn, how we acquire this knowledge, and then how do we measure? Do we use money to measure, or should we use time to measure? OK?
- 5.59** All right. Thank you very much for coming to this lecture. Okay.
- 5.60** So just a couple of questions for you. But in your research, did you look into the change of housing prices? I was curious about if that has, over time, gotten more affordable relative to the time price?
- 5.61** All right, really good question. We actually published an article in Forbes magazine about the time price of housing, and I'll give you a reference to that. And our conclusion is housing has actually fallen by about 60%. And it's really a function of a couple of things that you have to think about housing. It's like, "I can't compare a house

today with a house in 1980 for one reason. Houses today are about 50% larger." So you have to go to the per-square-foot rate. How much did it cost you to buy a square foot of house today? And then the other big factor was, "What are interest rates?" 1980, they're like 14%. You know, last couple years ago, they're down to 3%. They're back up again. But how much time do you have to spend to earn the money to make your mortgage payment? And then how many square feet do you have in your house? That's the way that we think is the proper way to analyze housing costs. And when you do it from that perspective, it looks like houses are getting a lot cheaper.

- 5.62** We know that building costs really haven't gone up. The time price cost to go buy lumber and nails and concrete—it really hasn't gone up in terms of the time price. What's changed in many cases with housing is the land. The land costs more. And why would the land cost more? Because we actually have government policies that limit what you can build, and that's an artificial restriction on the supply of land that really leads to this higher price. Who would want to restrict the price? Who would want to restrict the supply of land and make prices go up? Who would benefit from that? That's the other thing that we always ask in economics. When you see a policy, we say, "Who benefits from this policy?" Well, if I have a house now and I want it to go up in value, how could I make sure it goes up in value? Not letting anybody else build a house. Try to restrict the supply, and that lets the price go up. So you've got that political thing that's happening in the housing market that you don't really have in the car market or the pizza market.
- 5.63** So you got some things happening there with it. But fundamentally, the time price is, you're getting more for your hour of housing abundance today than, really, you've gotten—well, we went back to, I think it was 1970, 1980, I think is where we started the analysis, 1980 to today.
- 5.64** I know houses have gotten much, much bigger since the '50s. I mean, if you look at older houses, they're very small.
- 5.65** Yeah. I don't know where you grew up, but I was 1,800 square feet and one bathroom with one light bulb and the bathroom, too. It's like, well, I have 13 light bulbs in my bathroom. What's the deal? Because light's so cheap today, right?
- 5.66** Another quick question too, though. So with knowledge increasing with time and life has become easier and more prosperous, what do you think AI's impact will be on production—the price of goods, general prosperity—while controlling for the loss of certain jobs? And you mentioned earlier, consumers versus creators. I'm interested if we want to expound on that.
- 5.67** OK, so the question—the fundamental question we ask is—what is wealth? Wealth is knowledge. So anything you do to increase the ability to discover new knowledge faster and cheaper is going to create more wealth. The one view is, look, AI is going to help lots and lots of people discover knowledge faster and really cheap. On the other hand

is, what are we going to do with that knowledge? Now, one of the big concerns is, "Wow, I'm going to lose my job. AI is going to take, I'm not going to be a professor. I'm going to have an AI guy that's going to be a professor. AI is going to take over all of these occupations."

- 5.68** Historically, what do you notice that people do? AI ends up—all of these new technologies, all this new knowledge—does what for most people, most of the time? It enhances their ability. They become more productive because now they have this tool. I remember the first time I saw Excel. This was in college a long, long time ago, and it's like, "Wow, I can't believe how much better my life is going to be now that I can do my accounting class with this program." So, yeah, there are going to be some disruptions, some changes, because AI is going to take over a lot of these things, but it's going to be people that use AI tools. It's not going to be AI versus humans. It's going to be humans that use AI versus humans that don't—because it's a tool of learning and knowledge and coming to discover new things really cheap and really fast.
- 5.69** So, yeah, we're going to maybe have to change the way we do lots of things, but overall, it's got to be good because that production of new knowledge is going to translate to new wealth, and that wealth tends to do what? It tends to benefit all of us. All of us now get that benefit, because if I have new knowledge and I share it with you, I don't lose it. You get it. So, I tend to be pretty, kind of, rationally optimistic about what these new learning tools can do for us. Does that help? OK.
- 5.70** So mine is kind of a two-part question, but who do we have to thank for the the time price, how things have become more affordable? Do we have the economist to thank? Have they helped businesses market their products and lower the price and think about the cost of living and the hourly wage? Or do we have to thank industries and businesses for a boom in success? Also, how did economists come about? Did they come about after the fact to research how the time price has worked, or did they come out to help come out of a Great Depression or something?
- 5.71** Yeah. You know, we think about the Great Depression, and it's kind of the anomaly, it's the exception to the American history, because we were doing this, and all of a sudden, we stopped. We did this, we got stuck down there, and then we came back out. That's the general trend. Lots of other things are going on there. But economists are still having disagreements about what caused that, what caused it to start, and how we got out of that. And we had lots of things happening in the world. We had lots of political things that were happening—wars and other regulatory things that were happening that maybe have, maybe could explain why that happened. So that would be the answer on the Great Depression issue, maybe.
- 5.72** But the idea of going forward, and how do economists maybe predict the future? Look, if you could predict the future, it wouldn't be the future, right? Because you would know it. Innovation is surprise, right? It's surprise. Information is really surprise. If I tell you something you already know, it's not information. You already know that. It's

surprise that's really information. And that information then can be converted, can actually become knowledge, new knowledge.

- 5.73** So, economists will talk about this, and we're really kind of moving this profession, you know. We're trying to move the economics profession. There's been a number of economists that have preceded us that have tried to do this or are doing this—move it away from counting atoms, weigh the atoms to do what? To think in knowledge. It's knowledge that we have to really think about. How do you measure knowledge? And how do you focus your policies on figuring out how to get everybody on the planet to get into this discovery game, these learning curves of discovering new knowledge, and then sharing it with the rest of us? That's really what economics is kind of interested in.
- 5.74** And, you know, Elon Musk and those guys, it's like, yeah, they live at this level. But what we're really concerned about is, we're thinking about people at the bottom. How do we lift those people? Those are the people that we're really interested in. How do we help them escape poverty and then become creators? And become creators? And we think that this, this time price methodology allows us to do that. And the theory explains how this, how you can continue to have this growth.
- 5.75** Because knowledge isn't just a function of time. We plot it over time and say, "Over time, it looks like we're doing this," but it's not time that's doing it. It's a growth in knowledge that's doing it over time. Right? So it's, we've been able to figure out how to grow knowledge at a faster and a faster rate. How do you do that? Well, you get a bunch more people, and you give the freedom and the tools to go out and discover and share with other people. That's what we have to think about. Because then, you and I, we're not competing over atoms. We're collaborating and cooperating with each other to discover and share knowledge with each other. So it changes people's approach to other people.
- 5.76** It also says, "We got to treat each other with dignity and respect because we don't know what ideas that other person might come up with. It could be really, really good." They have to have the freedom to act on their ideas. And some places on the planet are much better than other places. And we'll talk about that in one of our later lectures about that cultural capital that's different. And it creates more opportunity for people to create more knowledge. OK?

Lecture 6: Capitals of Innovation

At a glance — The capitals of innovation.

- 6.1** Welcome back to this lecture. Thank you very much for coming today. In this lecture, we're going to talk about the Capitals of Innovation. Before we do that, though, I'd like to talk a little bit about one of the most important food products in colleges across America. They provide about 200 calories of energy, 33 grams of carbohydrates, and 7 grams of protein. But they didn't really become popular until the innovation of the microwave oven.
- 6.2** Now, the microwave was accidentally discovered by someone who was a guy by the name of Percy Spencer in 1945. He was working on this vacuum tube called the magnetron, and he noticed that he had a Hershey bar in his pocket and it completely melted. So he continued to work on this thing, and he put some popcorn in there—and that's popcorn. So, we had this beautiful invention that he developed. So, the first microwave oven was born. Eleven years later, in 1956, Duane Roberts invented the frozen burrito. Now, we have the two key innovations that could change college cuisine. Let's think a little bit about our time price of methodology and how it relates to microwave ovens.
- 6.3** In 1979, actually, when I started college, microwave ovens were around \$400. But unskilled labor at that time, I was an unskilled worker, making about \$3.70 an hour. What was the time price for a microwave? When I started college, it looked like it was about 108 hours. So, I had to work over 2.5 weeks—actually, longer than that, 108 hours. Yeah, it was about 2.5 weeks to be able to earn the money to buy a microwave. And when I saw this the first time, I thought, "I gotta have one of these." I tried to convince my wife that it was really a good deal. And she goes, "Well, if you want one, I guess you got to go earn the money to get one." And I said, "The reason I want one is look at how quick I could make a burrito! "
- 6.4** So, look at that time price. Today, you can buy a microwave at Walmart for about \$74. And people—unskilled laborers earning about \$15.72 an hour—so it takes a little less than 5 hours to work to earn your microwave. So, a little more than half a day of working gets your microwave. So, what's the resource multiplier there? Yeah, for the time it took me to earn the money to get one microwave, you get 23 microwaves a day. So, you get 23 microwaves today.
- 6.5** But that's just one thing that we can think about. The other thing, of course, was how much time it saved you. So, here 's off of a wrapper of a burrito. One minute and 15 seconds in a microwave, you put it in an oven, it's 28 minutes. So, I convinced myself, "Look at all the time I could save if I could have this microwave. I go from 28 minutes down to 1 minute, 15 seconds." My time price had fallen by over 95%. But recall that a microwave oven uses a lot less energy than a conventional oven for a lot less time. So,

the electricity that you use also fell dramatically. So, the overall burrito-nomics: microwave ovens fall by 95.7%. The resource multiplier there is 23 microwaves. The heating time, I can do 22 burritos with a microwave in the time it takes to do one in a regular oven, and electricity fell by almost 99%. So, you have these huge savings. We basically made, we figured out how to make this energy much smarter when we innovated a microwave oven.

- 6.6** So, what we want to talk about in this lecture is something that we call a model that we use, and a model is just a way that we try to understand how the world works. We're going to talk about this thing we call the Capitals of Innovation. Now, what we're seeing here, we're going to go through and look at each one of these pieces in this puzzle and put it back together. But this is where we're going to end up at the end of this lecture, what this looks like. We're going to start with each one of these pieces of this puzzle.
- 6.7** Before we do that, we're going to ask this question, "What is wealth? What is wealth?" Is it money? Is it oil? Is it gold? We have one definition that wealth is knowledge. We understand that. We're going to broaden that definition and say, "Wealth is anything that we value." Anything that you value is wealth. So, what is economics, then? Economics is the study of how human beings create wealth or how human beings create value for one another. How do you and I create value for each other? That's what we study in economics. How does this process happen?
- 6.8** Another term we use is "capital". You may have heard of this term. "I need some capital. We need capital." Well, capital—the way that we're going to define it—is anything that we can use to create something of value. Anything that we can use to create something of value is capital. Our model is the Capitals of Innovation. What do we use and how do we use those things to create things of value for one another?
- 6.9** Now, we begin this model, fundamentally, with human beings. Human beings are at the foundation of our model. Why do we put human beings there? Because human beings are the only thing that we've been able to discover that can discover new knowledge, that can learn and can innovate. Now, we've developed lots of tools that help human beings to discover new knowledge, but it really begins with human beings. So, they are at the foundation of our model—human beings. Now, human beings.
- 6.10** The second piece of our model—and we draw this as a big blue space—is culture, cultural capital. Everybody lives someplace on this planet. And depending on where you live, you're going to have a different culture. The culture is going to be different. Well, what is cultural capital? Your culture really comes down to, "How do people treat one another?" In different places on the planet, people treat each other differently. Their cultures are different. How we treat one another is your cultural capital. I notice that there's a difference between the way people treat each other here in Florida and the way they treat each other in Hawaii and in Saudi Arabia. There's a difference in cultures, and that culture can be capital. It can be something we can use to be able to create value from one another.

- 6.11** So, cultural capital is really determined by the worldview of the leaders of that country. Now, Dr. Tupy talked a little bit about Korea. This is the Korean peninsula. You fly over this at night, and this is what you see. When I first saw this picture, I thought, "Oh, that looks like Hawaii, and it looks like there's volcanoes going off or something there." No, it's actually the Korean Peninsula, and it's very clear what's North Korea and what's South Korea, isn't it? By the manifestation of the light that you see at night, right? You can see right where the border is.
- 6.12** Now, recall that that peninsula is the same geography, same language, very similar climate, very similar resources. It's very similar in all these other respects. But what's different between the two? They have a different culture. In the early 1950s, this country gets split in two, and we run this experiment and we say, "Well, what happens if you have a leader that actually has a worldview, and then you have a different leader in this part of this country that has a different worldview? Let's run the experiment."
- 6.13** What do we observe today? Well, the light will tell you. The light will tell you how much freedom to innovate you have in that part of the world. You don't even have to land your airplane and get out and look. You can just fly over it at night and you can see. The light tells the story. Dramatically different cultures, and it shows up in what people are able to generate in their standards of living—22 to 25 times greater standards of living in South Korea versus North Korea.
- 6.14** All right. So, what do we think about as we drill deeper into the culture? First thing that we think about as economists is property rights. Can people own things? Can you own things? Can you own a car? Can you own land? Can you own a house? Or is it owned by the state or the government? What do you think the property rights are in North Korea versus South Korea? Well, they're much different. Much different. That's the first thing. What we note is when people can own things, the lights come on. You have prosperity that begins with this ability to own things. So, one thing that we could look to is where we see innovation, where we see people lifting themselves out of poverty, is when they can begin to own things.
- 6.15** The second thing that we think about under culture is the rule of law. We hear that term: the rule of law. And what it simply means is this idea that we're going to have a set of laws that everyone is going to be expected to obey. We're not going to make a distinction between your race, your gender, your religion. Everyone is going to be expected to obey that law, including the king.
- 6.16** Now, something symbolizes that, you may have seen a statue like this. It's called Lady Justice, and it's a symbol of what the rule of law means. You look at it, first of all, and it's like, "What does this symbolize?" Well, first of all, it's a lady. There was something in Roman culture that said that women had this innocence and justice about them that maybe men didn't possess. So, it was characterized in this statue of a woman. But she's holding this scale. You notice the scale has these two different sides that you would put

weights on each one of these dishes on the scale. And then you would watch if it was balanced or not. She's holding this scale up there to weigh the evidence.

6.17 Now, you also note that she's wearing a blindfold. What's the symbol of the blindfold mean? It means that she is not looking at the status of the parties. She's not looking at who's putting the evidence in the scale. She's making a decision based on the weight of the evidence, not the status of the parties. "Are you my cousin? Are you in my tribe? Are you in my religion?" No, we're going to just strictly look at the weight of the evidence, and we're going to ignore the status of the parties. That idea that I could go to justice, and get a fair hearing based on the evidence, that 's part of rule of law. We move away from people being able to have an influence over the judge because there's a relationship to say, "No, we're just going to wait the evidence." Key element.

6.18 What else does she have? She's got a sword. What's that symbolize? That justice is quick and painful. Then she also, if you noticed in the corner, her foot is standing on a book. What does the book symbolize? The book symbolized that rules are written down so everyone can have access to this rule book. Everybody should know what the laws are. They're written down. This symbolizes this idea of rule of law and justice. There's really no country that is perfect in terms of applying this sense of justice. But as countries strive to get closer to that ideal of justice, we note that they also tend to rise out of poverty. If you have a sense that things are going to be fair, you tend to devote time and energy to be able to create and share your creativity.

6.19 That's the culture. We've got human beings. We have cultural capital. Once again, there's places on the planet that have lots of cultural capital, and other places, not so much. That's going to affect how human beings are going to be able to react and respond and treat one another.

6.20 The next element that we're going to put here in the puzzle are ideas. Ideas are really only the product of human imagination. It really begins with ideas. Human beings have ideas, and that's why we value human beings—because they have ideas, and these ideas have this great potential.

6.21 And what is an idea? We define an idea as a thought about how to potentially create new value. "I've got this idea. I think if I did this, I could create something of value." That 's an idea. Do you have an idea? Human beings, we have ideas. You've probably had a couple of ideas already this morning about things, right? So, human beings have ideas. The culture will determine what that person is going to be able to do with that idea. OK? All right.

6.22 The next capital we want to talk about is physical capital. Physical capital. What's that? Well, it's all the atoms and the energy around us. It's the physical material world. It's things that, we call them natural resources. We understand we have this material world, and we add knowledge to it, and these things become resources to us. But it's all the things that we have to work around here that naturally occur in our planet.

- 6.23** Time is a physical capital. The buildings that we occupy, the equipment that we use, the things that we, the physical things that we can touch that help us make things—that's physical capital. Physical capital. And some parts of the planet are really blessed with lots of physical capital. Saudi Arabia has got all of this oil. It's like, "You guys have a lot of physical capital. What can you do with it?" Well, we'll see that you've got to be able to be in a culture that allows you to combine these capitals with each other to be able to produce these products. It 's just one little piece in the puzzle. All right.
- 6.24** So, next element, human capital. Human capital—what is that? Well, the first thing that we think, well, it's you and I. Are we capital? Can we create something of value? Do you know somebody that can help you create something of value? Well, if you do, that person must possess human capital.
- 6.25** The first thing that we put under human capital is trust. Can you be trusted? Cultures that encourage one another to trust each other, to build trust relationships, tend to be able to grow. A key human capital that we possess is our ability to be worthy of trust. Human capital. Second one is health. How do you do physically? Are you hungry? Can you function? Can you perform? Then we can think about skills. What skills do you have? What do you know how to do with your hands? Can you make something with your hands? Knowledge. What knowledge do you possess that can actually be activated in a way that you can understand things?
- 6.26** The other element that we put under human capital is personality. What is your personality like? Can you change your personality? Are there people that you like to be around and people that you might not like to be around? Your personality is part of your human capital, part of your human capital, and your relationships with other people. When I meet you, I get to meet potentially all the people that you know. So, part of your human capital is your network of relationships with other people. So all of these things, really, are things that you and I can develop that allow us to be able to create value for other people and trade that with the things that they hold. Now, what are we doing in class today? I think we're trying to develop our human capital right now, right? We're trying to figure out how we can grow in skills and knowledge and relationships that allow us to be able to serve one another, to serve one another in an economy.
- 6.27** Now, I put trust there. There's two elements of trust that we talk about—integrity trust and competence trust. Integrity trust—do you do what you say you can do? Competence trust is, "Can I trust that you're competent to perform this function?" Now, do you know people that have integrity trust, but they're not competent in certain things? Or people that might be very competent, but they don't have integrity? Now, why are we so troubled when someone cheats on an exam? Because they 're lying about their competence. So they're violating both of these trust elements. So we don't like it when people do that.

- 6.28** Warren Buffett makes this interesting observation. He says, "I look for three things in hiring people. The first is personal integrity, the second is intelligence, and the third is a high energy level. But if you don't have the first, the other two will kill you." Right? Yeah. So, this integrity-trust thing is very, very key in your human capital, and it's very key in a culture. Are the people around you—can you trust them? And how large is that circle of trust? The larger that gets, the greater prosperity that culture is going to enjoy because I can begin to trust more and more people. Because of that, can I be trusted, and can I trust you? If that's the case, then you and I have the basis for a relationship that we can trade with each other. Do you really want to trade with somebody you can't trust? I don't want to. Yeah. That 's a key element in a culture. How can we trust one another? All right, so that's human capital.
- 6.29** The next one we put in the puzzles is intellectual capital. Intellectual capital is the knowledge that we have access to, the technology, the methods, the patents. A patent is a special license that a government will issue to you when you come up with a new invention, and you get to be the person who owns that invention for a period of 20 years. After 20 years, everybody else gets to use your invention. But the idea is, we want to incentivize people to be creative and let them own this invention for a period of time so they can enjoy the benefits, but all of the rest of us get that benefit as well.
- 6.30** Here's the patent for the light bulb that was issued to Thomas Edison. Each patent has a number on it. If you look up here, this is patent number 265 '311, issued in 1882. How many patents do we have today? Well, you can go to the Internet, to the Patent Office, and they'll show you all of these patents. We're up to about 10 million patents today. 10 million patents. If you have a culture that creates some kind of a system that gives people the incentive to be able to be creative, to be able to keep at least a portion of the benefit of their effort, you've given somebody an incentive to go be creative.
- 6.31** So, patents in the United States—it was a really interesting thing because our Constitution actually had a clause that's created a patent office. Yeah, the first couple of patents were actually signed by George Washington. Yeah, so, pretty interesting. All right, so are we OK with intellectual capital? It's all these ideas that we have out there that we have a knowledge that we can have access to.
- 6.32** Now, in many cases, someone will come up with a patent, and they'll actually give a license or they'll sell the patent to another company. I've developed this idea, and I can go to a company and say, "Would you like to buy or license my patent?" We can separate the inventor from the company that actually activates the patent and uses the patent. This system of cooperation, of creativity, and then sharing our creativity with one another with these property rights. A patent is a property right. All right.
- 6.33** Then the next element we want to talk about is financial capital. Whenever you use the word capital, you always want to qualify it. What kind of capital are you talking about? Financial capital. You hear of venture capitalists: "We need to get some capital." In

many cases, we're talking about this idea of financial capital. Financial capital is really money put at risk over time. Finance is about money put at risk over time.

- 6.34** It really consists, you've got to have a system where you have people that are willing to save that allows borrowers to have access to this financial capital. If people don't want to save, you're not going to have the financial capital. "I've got this great idea I need, but I don't have any money." And what's interesting is, a lot of times you'll find people that have great ideas with no money, and you have people over here that have money but don't have ideas. So they've got to figure out how to cooperate with each other in order to be able to make this creation real.
- 6.35** So, people willing to save—they earn the money, they're willing to save some of that money. Why would you want to save it? "I don't need it right now, but I'm also not worried about it losing its value." That's why one of the important things about an economy is the currency. Can you trust the currency, that it's going to hold its purchasing power? That's one reason economists tend to not like inflation, because if you have inflation, people don't want to save. They don't want to save. You don't have the capital there with which to be able to finance new ideas. You've got borrowers and savers. Is your country, are people incentivized to save? That allows people to be able to borrow.
- 6.36** We've got this little fundamental equation—one plus the rate of interest raised to the power of N . That's the fundamental basic equation of all finance. If you take a finance class, you'll learn about this equation and how it works. Beautiful equation.
- 6.37** Other elements under financial capital are insurance. Insurance. You want to take this risk, but if something goes bad, you could lose everything. Insurance is an innovation that allows us to share the risk with a bunch of other people. "I have this idea. I want to buy this boat, and I'm going to take this boat from London, and I'm going to go to India, and I'm going to get some spices. If I can bring them back to London, I can take these spices and I can charge 1,000 times the price it cost me to go get them." So I have this opportunity to go get these spices.
- 6.38** But the problem is, I got to get a boat, and the boat has to be on the ocean, and it could sink. So what do I do? I know there's a really small probability that it could sink, but if it sinks, I'm gone. My investment's gone. Insurance allowed us to find other people that were willing to take that potential risk and put a pool of money aside that, if our boat sunk, that this insurance would actually pay for our losses. We couldn't get the boat back, but we could get the loss in the money that was put at risk. Being able to buy insurance was a phenomenal step forward in terms of our ability to begin innovating because now we could take a really big risk. Because it's not me taking the risk—I'm going to spread it across a bunch of other people who have similar risks. All right.

- 6.39 Another innovation was a limited liability corporation. That's essentially a business form that says, "We're going to allow you to create this business, and we're going to let you look for stockholders in the business, but their losses are limited."
- 6.40 The only thing I lose is what I paid for that share of stock ownership in that corporation. What that allows us to do? It allowed us to raise a whole bunch of money. If I needed to borrow a million dollars, I could go find one of my uncles, maybe. One of my relatives. I could go to a bank. My uncle didn't want to loan it to me. The bank says, "No, I'm not going to take a risk with you." But if I could find a million people that had a dollar, what have I done? I've raised the million dollars, and if my idea fails, what have I lost? What have my stockholders lost? They've only lost \$1. So are you willing to put up \$1 with a chance that maybe you could double your money? Yeah, maybe I would take that risk.
- 6.41 See how we've been able to innovate this idea, this concept of a corporation—a limited liability corporation—that allowed us now to go out and take a big risk and pursue a big project? By the way, I need to raise a million dollars to buy the boat to take to India. Can I raise the money to buy the boat? Can I also buy the insurance on the boat to potentially go out, take this big risk? But if this risk proves, if this is successful, all of us would actually be much better off.
- 6.42 Those two things allow us to take much more risk. But why we were able to do it is, we were able to diversify the risk across a bunch of people. Countries that have insurance systems and limited liability corporation laws tended to do what? They started to grow, and grow much faster. Why? Because people in those countries could take a lot more risk, potentially creating lots of new wealth.
- 6.43 Then the last thing we put there is double entry accounting. Double entry accounting was another innovation. The reason that was important is because it allows us to standardize the way we analyze businesses. We have the standard language that we use to analyze businesses. It's their accounting statements. All these businesses are going to use a standard way of accounting, of reporting their income and their assets and liabilities.
- 6.44 That allows me to compare one company to another company. Now I have this common language between businesses that now allow me to do this analysis with this standardized language. It really is this language of business—accounting systems. It was this great innovation that this Italian accountant came up with. We put that in the financial capital category, as well. We're OK with financial capital? How important it can be? How necessary it is for us to take risk? How we use it to be able to advance ourselves? You take all these capitals, and each one of these little items is really a capital, isn't it? Things that we need to create a new product.
- 6.45 Now, the next element that we want to put in the model are inventions. You notice inventions, they're different than ideas. Here's how we're going to define an invention.

An invention is a proof of an idea. "Hey, I have this great idea. " "OK, can you actually show me how it works?" The invention is the "show me." I've proven the idea. I've taken my thought and I've made it real. I move from thought to reality. That's an invention. That's how we define an invention. Who's an inventor? Lots and lots of people are inventors. Lots and lots of us can invent.

- 6.46** The next thing that we've got to put there—you notice that we haven't got to innovation yet. What's the difference between an innovation and an invention? Well, the difference, what we're going to realize, is an innovation is a successful invention. An innovation is a successful invention. And how do we determine whether this invention is successful or not? Well, we go test the invention. And where does it get tested? It gets tested—actually, the people that go and take these inventions, we call them entrepreneurs. They're the individuals that are taking these inventions to a market.
- 6.47** An entrepreneur is someone who's taking the risk in hopes that there's something that you've done here that's created value. When we say created value, what we mean is, you can sell it for more than it cost you to make it. It's a profitable invention, a market-successful invention. That's what an innovation is. They're trying to take an invention and move it to an innovation. In order to do that, they have to have this freedom to discover whether or not this happened or not.
- 6.48** I don't know about you, but I would come up with these little inventions, and I'd go show them to my mom, and she'd say, "You're a genius." It's like, "But nobody else thinks I'm a genius, Mom, because they won't pay me for what it cost to make these things."
- 6.49** I made lots and lots of inventions, but I never really did an innovation because I wasn't able to convince somebody to pay me more for this thing than it cost me to make it. It wasn't a successful invention. Where does that happen? It happens in a market where people can learn. I'm going to go out and figure out whether or not this invention has created value or not. I've got to take it to a market where I can learn about it.
- 6.50** A key element that we have to have in those markets is whether or not you can allow this thing to fail. Failure is a key part of learning. I've got to determine whether or not what I've learned here has really created value or not. How do I know that? I "hope" that. My mom has told me that it's a great invention, but I don't really know. I've got to take it to a market. I've got to either be an entrepreneur, or have an entrepreneur take it to the market and see if that's happened.
- 6.51** All right, anybody know who these two guys are? Both of their first names are Steve. Yeah, Steve Wozniak is the guy on the left, and Steve Jobs is the guy on the right. What's Wozniak looking at? He's looking at his creation, and he was really this creative inventor guy. But who is Steve Jobs looking at? He's looking at you and I, thinking, "I wonder if this invention is an innovation. I'm hoping it is. I think it is." So who's the inventor here? The inventor is really Wozniak, but the entrepreneur was really Jobs.

You see the difference between the two? Yeah, one guy does the invention—and that's interesting, too—one guy does the invention, but he's really not the entrepreneur, and the other guy is the entrepreneur, but he's not an invention, which means most people, most of the time, have to do what? They got to go find somebody they can cooperate with, somebody they can trust that allows them to be able to actually have this relationship that lets them create value.

6.52 Now, there's some very rare people on the planet that can do both those things at the same time. One of those guys was this guy, Thomas Edison. He happened to be a pretty good inventor, and he was also a pretty good entrepreneur, but that's really rare. In most companies, most of the time, it comes down to these two guys—two people that got together and used their different talents, combined those in a way, one had entrepreneurial talent, the other one had inventive talent, and let's get together and see if we can create something, the value between the two of us. Somebody that does both those things is very rare. You're probably an inventor or an entrepreneur, either one of those two. Most human beings fall into—they're entrepreneurs, or fall into one of those two categories.

6.53 All right, so the next piece of this puzzle we want to put in here is free markets. Now, we talk a lot about markets. We qualify that with the word "free." And by free, what we mean, a market, first of all, is a place where buyers and sellers go to discover value. And what do we mean by "they go to discover value"? Well, it's a place, a market is really a place where I go to find somebody that's better at solving my problem than I am.

6.54 When you got up this morning, did you have your own oats that you grew in the backyard to make your own breakfast? Somebody else figured out what you might want for breakfast and put that product on the shelf, and you bought it. Somebody else is better than you at solving your problems. Isn't that true? And what we really do in a market is, we go find other people that can help us, that we can discover a price that allows us to be able to help one another.

6.55 The market has to be free. And when we use the word "free" in this context, what we mean is a market has to be free so buyers and sellers can come and go. I'm free to go into that market and try to sell something. I'm free to go in that market and try to buy something. But it also has to be free in the sense that the price has to be free to move up and down, that the price can go up and down. How are prices typically negotiated? Well, we'll have a price tag, but in many cases, it's you and I talking back and forth. We're trying to discover a price that will make both of us happy. We're trying to discover a price that makes both of us happy.

6.56 A price system is really what kind of a system? A price system is really an information system, and it's contained in that price because that price will create incentives. Have you ever walked into the store and said, "Wow, that's a really good deal. That's a really good price. Why isn't anybody else buying it?" Because they have a different value than you do. OK? All right.

- 6.57** So, entrepreneurs and markets. It's like this gauntlet that the invention has to go through to see if it can actually get to the innovation stage or not. You need entrepreneurs that are willing to take the risk with it. You also need to have markets where it gets tested to see if this thing has actually created value or not.
- 6.58** How do we define an innovation? An innovation, we can say, is just a market-successful invention, or it's a profitable invention. Now, all the patents that we talk about, we talked about the 10 million patents. Of all the patents that have been filed, what percent do you think actually went from the invention stage to the innovation stage? About 3%. It's like, "Wow, I got a 97% chance of being a loser? Why should I ever want to get involved in invention?" Well, the 3% that actually win—you can be phenomenally successful with your invention. We rely on everyone having this interest in trying to put their idea into an invention, and then bring that invention to the market and let us look at it.
- 6.59** It might be that the invention, great idea, great invention, but the price isn't right yet—right? Like the cordless drill. 1961, the price wasn't right. Great idea, but wow, price is too high. So something has to change in order for this thing to now move into the successful innovation category. Are we OK? Human beings have ideas. Those ideas can become inventions, and those inventions can become innovations. Ideas to inventions to innovations. That process, see that? We're using all of these different capitals to come up with these ideas.
- 6.60** Back to a trade. Whenever there's a trade that occurs, remember, there's one price that gets negotiated, but there's two values. The seller would prefer the money over the thing they're selling, and the buyer would prefer the thing over the money. The reason that you and I can trade with each other is we have a difference in values.
- 6.61** It's the difference in values that allow us to create value for each other. Because we have a difference in the values that we assign to these things, it allows us to trade with each other. "Look, I have something that you would really like, and you have something that I would really like. I like the thing that you have more than you do and vice versa. Why don't we trade with each other?" Then we both walk away what? We both walk away happier because we've been able to trade up to something we value more. All trade is based on this, isn't it?
- 6.62** Whenever you go and buy something, isn't it worth it to you? I'd rather have this thing than the \$3. I'd rather have this thing than \$3. Right now, at this moment of time, I'd rather do that. And what does the store do? The store says, "I'd rather have the \$3 more than this thing." And they're willing to trade. What's the price? Well, let's negotiate a price. And we find a price, we discover a price that does what? Makes both of us happy. Both of us walk away now. Because if it doesn't make both of us happy, does the trade happen? No, it doesn't happen. It doesn't happen. OK? You're OK with that?

- 6.63** All right, so let's go back and think about our model again—the Capitals of Innovation. What is wealth? Wealth is anything that we value. What is capital? Anything we can use to create something of value. What is economics? The study of how people create value with each other, for each other, in these trades that we make. Where does the model begin? Begins with human beings. Everybody lives someplace, and that place is going to have a culture, so human beings all live in a culture. Some of those cultures are better than other cultures. Human beings have ideas. Those ideas entail really putting capital together in new ways. "Hey, I have this idea. If I take this, put this together, boom, I got this new—I've got this potential new invention I could come up with."
- 6.64** Steve Jobs, when he introduced the iPhone, remember? You probably don't, you guys are all too young, but I remember it. He showed a cell phone and he showed the iPod. The iPod was this innovation that Apple had done. It had all this music on it. He goes, "We're going to put these two together. Our idea is to put the phone and the music player together. We'll have one device. And oh, by the way, we're going to put a browser on there so you can access the Internet, too." So he's putting those two things together.
- 6.65** And all of these inventions really entail putting our capitals together in new ways. How do we do that? So, first thing, physical capital. Human capital, intellectual capital, financial capital. It takes all of these things, in many cases, for your product to actually launch and become real, to become the invention.
- 6.66** What do you have to do once you've done the invention, though? You got to go test it to see if it's a really valuable invention, which is an innovation. What are the two key elements there? You've got to have what? You have entrepreneurs, people that are free and willing to go take these risks and have the talent for that, but you also have to have a market that you can go and let other people vote on whether or not you've done something worthwhile or valuable. If you make it through that, this thing ends up potentially as an innovation.
- 6.67** Then what happens? We got one more piece in the puzzle here. These innovations feed back into this system, and human beings say, "Wow, did you see what I saw at Best Buy? Wow, I think I could actually take that thing and add it to this thing." Or, "Wow, I went to this restaurant and I tasted this. And if I mix that with this other thing, I might be able to create something that's really good." We went to a restaurant last night and they had a coffee pizza. I really wanted to get it, but nobody else wanted it. I thought, and the waitress said, "This is like one of the most popular things that's on our menu." It's like, who would ever think putting coffee and pizza together would be a good idea? But evidently, it was a good idea. It became what? It evidently became an innovation because it was on their menu and it was one of the most popular items. See how this feedback loop works?
- 6.68** Look, be observant, be curious about things. It's like, "Wow, I can come up with this new thing. I'm going to put these things together in this new way." If you have a culture

that really encourages individual human beings to have ideas and then act on those ideas, and you've got these other pieces of capital out there, and it's pretty easy to actually create the invention. Then you have markets that can be tested. You're going to have an economy that's going to allow people to escape poverty.

- 6.69** Now, do you think about places on the planet that—and really, the culture capital, the way I think about it is, the cultural capital makes it easy for these other capitals to work with each other. Okay? In some places, they don't have these capitals. Japan doesn't have any oil. Well, how did Japan become rich if it doesn't have any oil? Hong Kong doesn't have, like, anything. How come they're so rich in Hong Kong? Because they've been able to figure out how to cooperate with other people and trade with other people to get the capitals that they need. They've been able to extend trust to other people and trade for the things that they need. "I don't have this, but I have that. Maybe I go find somebody that trades me." Is it easy to do that or not? Is it easy for the people in that country to do that thing or not?
- 6.70** All right, so what do we do? We create value by continuously innovating our capitals. We create value for each other by this continuous innovating of our capitals. Does that make sense? Yeah, it's a key principle here. How do we escape poverty? We innovate. How do you innovate? Well, you got to have capitals. What do you do, if you don't have capitals? You could find somebody that does have capitals, who you trade if you can.
- 6.71** All right, so what's an iPhone? It's about 6 oz, 7 oz of sand, primarily silicon, silicon sand. You got some aluminum. You've got, like, I read somewhere that you had 14 different elements, little trace, little pieces of elements that go into this device. We've been able to do what? Make our sand really smart by adding knowledge to it. Now we have this device that we can use.
- 6.72** Where is the iPhone actually made? Where is it made? If you look at the back of the iPhone, it says it's designed by Apple in California. OK, that's actually not true either, but it's on the back of the phone. But it says over here, it says it's assembled in China. A lot of people say it's made in China. Well, where is it made? Well, truth be known, there are over 200 different companies that Apple works with to make parts and pieces for this device. And all these companies are making these parts and pieces. They ship them all to China, Foxconn factory in China, 100,000 plus employees, and they're putting all these parts and pieces together. But look at where these companies are from.
- 6.73** And people look at Apple and say, "Well, Apple started as a computer company, and then they became, like, a phone company." And now it's like, what is the iPhone? Is it a phone? I think it's like a camera. It's got this really good camera now. It's a video camera, and it does all of these other things as well. But look at the participation of all of these people on the planet to make this thing work.

- 6.74** If you really put where it was made on the back of it, you would have to put it 's made on the planet Earth. It's an Earth product, right? If you had an alien come to the Earth today, and they said, "What have you created?", that's what I would show them. I 'd show them the iPhone and say, "This is what we've created on this planet." It's required that we figure out how to cooperate with everybody else on the planet to make this thing work. It really requires billions of people cooperating with each other to make this work.
- 6.75** That was the genius of Steve Jobs when he created Apple. He really created an innovation company. "Let's come up with an idea. Hopefully this idea, we can make it work into an invention and we can take it to the market and it becomes a profitable invention. We take the profits, at least a portion of them, and we do what with them? We put it back in to research and development for the next innovation. We're hoping to do what? Continually innovate these ideas into successful inventions. That's how we're going to do business." That's how Apple became the most valued company on the planet—because they've got billions of people cooperating with them, trusting Apple to provide these services and products. It's only possible because there are billions of people that have learned to cooperate with each other, discover new knowledge, and share that knowledge. It's really people that have learned to serve one another.
- 6.76** Now, there's a market involved where we have a price system, but it fundamentally is, "I'm going to create and discover knowledge, I'm going to share it with you, and you're going to share it back with me—your knowledge—in the form of money." Money allows us to do this. So, spent some time in Saudi Arabia. I'd ask my Saudi students, "What does Saudi Arabia have to do with the iPhone?" And they go, "Well, we really don't have anything to do with the iPhone." I'd say, "No, you actually do." You actually do because you remember all those parts and pieces get flown to China? And then when the iPhone gets assembled, they get flown to the rest of the world.
- 6.77** What's it cost to ship an iPhone from China to the U.S.? What do you think it costs? Well, you can take one of these 737s, take all the seats out, and you can put 450,000 iPhones in one of these things. They'll charge you about \$232,000 to fly it from China to the US. So \$232,000 divided by \$450,000, it's about 54 cents per iPhone. Isn't that astonishing? 54 cents, you can fly from China to the U.S. in what, 14 hours? Yeah. So, what Saudi Arabia have to do with the iPhone? Oh, about half the cost of that flight is jet fuel. And jet fuel is a product of crude oil. Saudi Arabia has 20% of the crude oil on the planet. So, in every iPhone, there's a little bit of Saudi Arabia.
- 6.78** There's a little bit of every one of us in these devices because we're participating. We're either producing—we're a producer of something that's used to make this system work. This whole Internet-system iPhone world works because we cooperate with each other as consumers, as developers. Some of you are writing code, some of you are going to be using your iPhone to do all kinds of interesting things. The iPhone is this

fundamental part of your capital, isn't it? It's a fundamental capital today because you're going to create wealth with this device.

- 6.79** All right, you ready for a little quiz? OK, first question, what is wealth? How do we define wealth? It's anything that we value. That's wealth. I have wealth because I have something that you value that I can potentially trade with you. What is capital? Anything you can use to create something of value is capital. Do you possess any capital? You say, "I don't have any money." Well, money is just one kind of capital. There's like 11 kinds of capital, right? You got human capital for sure, and you can grow that every day.
- 6.80** Anything we can use to create value is capital, which means all of us are capital-ists. We're capitalists because we believe in creating value for one another, and we use capital to do that. We discover it, we create it. We're looking for new capital because it allows us to create value. In a trade, there's always one blank. There's always one blank and two blanks. Yeah, there's one price, right? You see the price—we can all see the price—but how many values are there? There's two values, and we don't know what those values are.
- 6.81** We just know that that person who paid the \$5 for the Ben and Jerry's little ice cream, they thought it was worth more than \$5. At least it was worth \$5. It could have been worth \$10. Sometimes I go in there, it's like, "I would give you \$50 for that." Fortunately, the price was only 5. Now, what do we know about the value on the seller side? The seller thinks it's worth less than \$5. Otherwise, they wouldn't sell it, right? So both parties walk away after the trade happier because they've traded up values. They've traded up values. You see how the value gets created for both parties? I walk away happier, you walk away happier. We've created value for one another by trading with each other.
- 6.82** What is an invention? The proof of an idea. I got these ideas. Everybody has ideas. When I was in college, everybody thought about, "Wow, wouldn't it be cool if you could get movies over the phone? Use your phone and you could hook it up and watch movies." Everybody had that idea, but there was just a few guys that actually took the idea to the invention level. We ended up with Netflix, right? Wow, you can watch movies on your TV using your phone, using your phone system. An invention is the idea made real. You've proven your idea. It's a proof of a concept. What's an innovation then? It's different than an invention, right? An innovation is a profitable invention.
- 6.83** So, last question, how does an economy grow? How do we grow an economy? How do we lift one another out of poverty? We continuously innovate our capitals. We're trying to figure out how to take these capitals that are all around us, put them together in new ways, that become this innovation, that suddenly now we have this new knowledge that does what to the time price? What does innovation do to the time price? It makes it go down because either the price is going to go down or my hourly income is going to go up, and that is going to make that time price go down. When

we're looking at time prices, that's how we measure innovation and new knowledge is, "Wow, time price went down by 50%. It means I get 2 for the price of 1. It means I'm twice as smart. " We've figured out how to put twice as much knowledge into that particular product than we did last year. Because look at the time price. See the connection between time and knowledge there, and innovation, how they connect together?

6.84 So we've got this planet, beautiful planet, and we've got atoms on this planet that allow us to, really to create value for each other—if we're free, if we have a culture that's free, that allows people to innovate. As Dr. Tupy discussed, how did that happen? Well, it didn't happen until fairly recently in our history, where we began to treat one another with respect and dignity and allow each other to get on these learning curves and become innovators. OK? All right.

6.85 I could probably call this—I call my class, really, "Apple Economics" because I love to talk about Apple because I think that they made a lot of mistakes, but they've also achieved these, created these brilliant products that have made my life, and I think everybody's life, much better. So, Steve Jobs—remember, he was adopted. His father was actually from Syria. So, what would our life be like if Steve Jobs had been born and raised in Syria instead of raised in California, in Silicon Valley? Do you think our life would have been—you think his life would have been different? I think his life would have been a lot different. He probably would have been dead, knowing his personality in Syria. It's not a very nice place. But I know that our lives would have been different because he would not have been able to manifest himself and act on his ideas. He was fortunate that he was in a place where he was able to flourish.

6.86 Now, this parable of the seeds that we find, that we have seeds and some get planted in a hard place on a rock, and they don't do anything. In some places, some seeds get planted in a place, and then the weeds grow up and take over them. But these other seeds get planted in a place, and they get the water, they get the sunshine, and they flourish and blossom and bloom and bear fruit. Well, he was very fortunate that he was able to grow up in a place, in a culture, that had all of these capitals. All of these capitals kind of work together.

6.87 Think about what the iPhone really is. Isn't it a combination of all these capitals? You have physical capital. You've got human capital that put it together. You have intellectual capital, all these ideas and patents. You had financial capital. Steve Jobs and Steve Wozniak, they didn't have any money. They had an idea, they had an entrepreneur. They had to go find financial capital. They had to find somebody that trusted them, that was willing to loan them the money initially, so they could actually prove that their idea could actually work. So they had all of these capitals, and they had a cultural—they had cultural capital. Let them all work together.

6.88 How many seeds are in an apple? Well, it's pretty easy. You open up the apple and you count the seeds. How many apples are in a seed? You don't really know, do we? We

can guess, but the potential of every seed is the potential of every human being that's born on the planet. Because we can count the number of babies that are born today, but we really can't count how many ideas that they could have—what their potential is to be able to create and discover and share and do what for the rest of us. We all do this for each other if we have all of these capitals working together. So that's part of what we're hoping that we're able to come to understand. What do we do that we can create an environment where people can really achieve their potential? They can achieve their potential.

- 6.89** So let's summarize our lecture. We talked about the Capitals of Innovation. We got this model. It's like this puzzle with all these little pieces we put together. We arrange them, we talk about what they are, the relationships to each other, how they work together. We get to this end point where this is how innovations occur. Now, this model is based on lots and lots of different economists that have done research and have explored these different ideas, and we've tried to take all of those guys and put them together and come up with this explanation. We're hopeful that you use these ideas to come up with your explanation for how things work—your own theory about how innovation works.
- 6.90** We talked about innovation. What is wealth? We talked about wealth. What's the difference between wealth and capital? Then we go down to this fundamental idea—human beings have ideas. Those ideas can potentially become inventions if you have the capitals. Then if you have markets and entrepreneurs, those inventions can then become innovations. Ultimately, innovations are what we need to lift us out. Lift us out.
- 6.91** All right, so I would just conclude with this statement. Humphry Davy was a British chemist. He said, "Nothing tends so much to the advancement of knowledge as the application of a new instrument." We're hopeful that you look at our time pricing framework as a new instrument that you can use to look at the world. A new way to look at the world—a new additional perspective that you can use to look at the world, in addition to the perspectives that you already have.
- 6.92** When Galileo first saw a telescope, he ran home as fast as he could and he made one himself. Then he was able to use it to look at the world, to look at the world out there. And he saw all of these things suddenly. All these things suddenly appeared in the heavens to him because he had this instrument that he could use. And we're hoping that what you've been able to learn in this course allows you to look at the world and see the world in this way that, we believe, is the true perspective that lets us appreciate one another and the abundance, the superabundance, that we can create for each other. All right. Thank you.

Lecture 7: Intellectual Revolution

At a glance — The intellectual revolution behind growth.

- 7.1** In the previous lecture, Gale explained how innovation happens. In this lecture, I want to explain why innovation happens. Now, that may seem like a silly question. Humans have always innovated. That's what makes us different from other animals. But the speed of innovation used to be very slow.
- 7.2** For example, the Oldowan stone tools, which our ancestors used to kill and skin game, are some 3 million years old. And the Oldowan tools remained in use for the next 1.5 million years. They remained in use without any improvement until 1.5 million years ago. They were replaced by the Acheulean technology. In contrast, today, innovation is a daily occurrence.
- 7.3** In fact, some people are complaining that there is too much innovation. The world is moving very, very fast, right? So, it is very important, in my view at least, to distinguish between the era of innovation—which has always been there—and the era of sustained innovation and increasingly rapid innovation, which is only the hallmark of the last 200 years or so. And one of the reasons why I believe that we are currently in the age of sustained innovation is the importance of human institutions, or formal and informal rules that organize social, political, and economic relations between humans. So let's start there, with human institutions—or formal and informal rules which organize our life.
- 7.4** In our distant past, humans were quite egalitarian, so equality was, in a way, a human institution. As the economic historian Deirdre McCloskey wrote, "The scientific consensus is that a core characteristic of documented nomadic foragers is their political egalitarianism. Nomadic foragers have no hierarchical social stratification"—right? Politically, everybody is at an equal level—"Leaders, if they exist at all, have very little authority over group members. Rotation of roles and functions occurs regularly. People come and go as they please, in and out of the group, and no person can command or subject group members to act according to one's political aspirations."
- 7.5** Equality, alas, did not mean prosperity. Equality did not mean prosperity. When food was not found, death followed. Then, some 12,000 years ago, humans discovered that a good way of mitigating the risk of starvation was to plant seeds in advance of the next season. Or, as Matt Ridley notes, "True, a life of farming proved often to be one of drudgery and malnutrition for the poorest. But this was because the poorest were not dead. In hunter-gathering societies, those at the margins of society, or unfit because of injury or disease, simply died. Farming kept people alive long enough to raise offspring, even if they were poor."
- 7.6** Now, eventually, this technique, known as agriculture, became a primary means of subsistence. This transition meant that we had to settle down. Sedentary farming—the

hallmark of the Neolithic age—allowed us to grow unprecedented amounts of food on a relatively small land area and store food for emergencies. But agriculture, whilst it was a huge step in the history of our species, included or involved some trade-offs.

- 7.7** No matter how skillful an individual hunter was, hunting was always a bit of a gamble. In some days, the hunter caught nothing. Moreover, the foragers had limited means of preventing food spoliation. As a result, some food, and especially game, was widely shared. So here again, we come across this idea of greater egalitarianism in the hunter-gathering societies.
- 7.8** By contrast, farmers didn't share quite as well. In foraging group, it is easy to tell who is not working. Just keep an eye out for the one who never brings home a carcass of a dead animal. A farmer's harvest, on the other hand, is a culmination of months of labor, which makes it difficult to measure how much work everybody puts in. People are tempted to slack off, knowing that the hard work of others will ensure a good harvest.
- 7.9** This free rider problem was amplified by larger size of farming communities. Foragers lived in small bands consisting of their own family and closest friends, a much more trusting and cooperative group than a large farming community full of mere acquaintances. As such, agriculture resulted in a cultural shift from collective ownership to private property, to mitigate the problem of free riding. Farmers who owned their land, tools, and harvests could work knowing that they would reap the benefit of their labor.
- 7.10** In a world of private ownership, differences in ability, effort, and opportunity inevitably led to income gaps, wealth gaps, and status gaps. Of course, inequality is not exclusively human. Other species exhibit inequality through territoriality, for example. The biggest and the most dominant lions have the largest hunting groups, featuring the plumpest buffalo, and so on.
- 7.11** So, we are already seeing here an emergence of private property and inequality. And after accepting private property and inequality, which were not present in the hunter-gathering days, but were present during agriculture, humans ran into a psychological conflict. How does one justify being relatively rich while those of you around were relatively poor?
- 7.12** How come that you have emerged from poverty, but others still continue to live there? We have evolved as social and cooperative animals who are hardwired to help the needy. At the same time, the economics of agriculture dictated that sharing was a bad survival strategy. And so, wrote William von Hippel, a well-known psychologist, "New patterns of thinking were required to mitigate the dissonance that this wealth disparity aroused. First and foremost in these psychological changes was the idea that some humans are better and more deserving than others. Once you accept that as a fact of nature, why stop at rich and poor? Why not king and peasant? Or a master and a slave?"

Of course, that's exactly what happened. Worldwide, egalitarianism was kicked to the curb, and newfound beliefs in innate superiority led to all sorts of human suffering."

- 7.13** So, to justify these new hierarchical beliefs which emerged in the agricultural era, people began to argue that certain bloodlines were better than others—that when, you know, when one lazy son was just an aberration in the line of worthy kings, right? We have all seen movies like the Lord of the Rings and so forth. And the logic of hierarchy has been used to justify conquest and oppression throughout history. One great example is imperialism, which was often defended as a way of bestowing the gifts of civilization on the so-called savages.
- 7.14** Interestingly, when colonialists enforced unequal systems on indigenous people in Latin America, they were fiercely resisted by the hunter-gatherers, who were more accustomed to the egalitarian way of doing things, but were much more accepted by the agriculturalists, because the farming communities in Latin America were already taxed and oppressed by local lords. So, ordinary people didn't care whether the tyrant was a Spaniard or an Inca. Right? So this is very important because, due to the arrival of agriculture, we have this emergence of hierarchies and ruling elites. In the next section, I want to explain how hierarchies and ruling elites held back innovation.
- 7.15** OK, there is a reason why I kept talking about the agricultural revolution and how we have created these new norms of human beings living together and working together—because the emergence of the ruling elites has a profound effect on innovation. So let's turn to that section.
- 7.16** So, prior to the modern times—let's say the last 200 years or so—the vast majority of governments that has existed consisted of elites exploiting the rest of the population, while defending the latter from predation by other elites. So, the implicit bargain here was that the local lord will protect you, the peasant, from being raided by a lord from another jurisdiction. But in exchange for that, you are basically their chattel, their property, in a way. Now, today, government predation is something that accompanies hierarchy and lawlessness. Think about Venezuela and the—Chávez and Maduro.
- 7.17** But social structures that emerged in the agricultural era were highly hierarchical and relied on brutal enforcement of often unjust laws. Different sets of laws imposed different sets of rights and obligations on different people, depending on their status within the hierarchy. And that's still the case in places like Cuba and North Korea.
- 7.18** What distinguishes the modern world from the past, therefore, is not the rule of law. People always had rule of law—"You must do that, you must know that." What distinguishes the present moment from the past is equality before the law, that it doesn't matter whether you are man or woman, whether you were born to money or born to poverty. Everybody is equal before the law.
- 7.19** And the best way to ensure an equal set of rules is to pit one elite against another. When the elite is homogeneous, as was the case with the plantation owners in the

American South, they will create rules that benefit the plantation owners. When the elite are diverse and include, for example, farmers and industrialists, bankers, they will create a political and economic system that is much more fair. So, in a heterogeneous society, other groups always outnumber one's own. So, equality under the law is in everybody's best interest. And it is to that transition that I want to now turn.

- 7.20** So, once in place, the new institutions of private property, inequality, and hierarchy help to perpetuate the interests of the governing elite—the aristocracy, the clergy, and the soldiers. Now, this elite accounted for only for a tiny sliver of population, maybe 5-10%, while the vast majority of the populace were peasants, compelled to exchange their labor for a modicum of greater security in this life and a promise of better life after death. In a word, the political and economic institutions of the agrarian era were extractive. Extractive.
- 7.21** Now, over time, political and economic institutions became much more inclusive—from extractive to inclusive. Or to put it differently, they became much more reflective of the needs, wants, and desires of ever-larger parts of the populace. Now, this process of institutional change differed across the globe, spanning different lengths of time. But most scholars agree on a general, long-term trend toward greater inclusivity of human institutions, political and economic. This change started in Western Europe and North America, and it contributed to the age of sustained innovation.
- 7.22** It then spread to other parts of the world and continues to this day. As political and economic institutions have changed, our understanding of private property inequality and hierarchy has also changed. In the agrarian era, property rights were widespread but unequal. They were also arbitrary, and they were also often arbitrarily violated by the powerful, like the Egyptian pharaohs, and Roman and Chinese emperors, or the Russian tsars. Today, all people in advanced countries enjoy substantially the same property rights.
- 7.23** Moreover, they can usually expect to have their property rights protected with equal vigor. Expropriation and taxation still take place, of course, but they tend not to be arbitrary, and they follow due process under the law. It is, in part, the security of private property that distinguishes today's rich countries, where people have an incentive to save, invest, and innovate, from today's poor countries, where people's private property continues to be insecure.
- 7.24** Inequality in the agrarian era meant unequal rights and freedoms enjoyed by different groups of people. Think of different rights and freedoms of the nobility and peasants during the Ancien Régime in France. Today, unequal treatment before the law, be it sexist legislation in Saudi Arabia or anti-Muslim discrimination in China, still persist in some places, but it is universally condemned. That said, not all types of inequality are detrimental to human well-being. Think, for example, of the mathematical genius of Albert Einstein or the business acumen of Jeff Bezos. Such people demonstrate that

inequality of ability can benefit us all, whereas inequality under the law is generally harmful.

- 7.25** And finally, the meaning of hierarchy has also changed. Hierarchy used to mean an inherent right to rule, be it by individuals like Montezuma or the French king Louis XIV, or by nations, such as the feeling of some Europeans during the colonial period that they had to, that they had the right to rule other nations. So, those kinds of notions of hierarchy have disappeared.
- 7.26** But other forms of hierarchy are unavoidable and, on the whole, beneficial. The hierarchy of competence, for example, often elevates smart people into positions of power and influence. Think about Thomas Jefferson or James Madison. Think about Steve Jobs or great surgeons, like the South African heart surgeon Christiaan Barnard. These people are recognized for the hierarchy of their competence.
- 7.27** So, in their 2012 book, *Why Nations Fail: the Origins of Power, Prosperity, and Poverty*, Daron Acemoglu from the MIT and James Robinson, formerly at Harvard and now at the University of Chicago, explore the distinction and the interaction between inclusive political and economic institutions, on the one hand, and extractive political and economic institutions, on the other hand. Their thesis provides one useful prism through which we can view the differences in economic outcomes between countries as these are manifested in the world today.
- 7.28** Consider North and South Korea. North Korea's extractive institutions restrict the autonomy of the individual by setting stringent rules on how people may earn and spend their money. The government uses state education as a propaganda tool. It requires everyone to enlist in the military for 10 years and sets limits on property and business ownership. To the extent that wealth is still created by the citizenry, it is diverted to members of the elite, such as the members of the ruling Communist Party and the top echelons of the military.
- 7.29** By contrast, South Korea's economic institutions are inclusive. People are largely free to earn and spend their money any way they like. The citizenry has access to an open educational system, information, and technology. The property rights of ordinary citizens are protected, and entrepreneurial activity is encouraged. South Korea's market is open and, consequently, innovative. It produces wealth. That goes a long way toward explaining why South Koreans are at least 22 times richer than their cousins in the north. Ethnicity, culture, history, geography, and climate, those two countries largely share. So, ethnicity, history, geography, and climate cannot explain the difference between North and South Korea. It is the different institutions which explain the difference between the two.
- 7.30** The problem is that, as things stand, economic inclusion is determined by actions of the state. Market activity requires, among other things, the legal enforcement of contracts, infrastructure for exchange, and protection against fraud. Since government and

economic activity are interwoven, economic institutions tend to conform to the objectives of the ruling class.

- 7.31** Inclusive economic institutions rely on the existence of political institutions characterized by power-sharing and the wide distribution of decision-making amongst elites, businesses, civil society, and ultimately, individuals. The elite, in other words, is constrained. Unconstrained decision-making, by contrast, centralizes the power in the hands of a small elite or, in extreme cases, of that of one individual, like the Chinese leader Xi. The former—the inclusive institutions—foster competition, coalition-building, accountability, and innovation, whilst the latter, the latter, fosters elite predation.
- 7.32** Finally, economic and political institutions, according to Acemoglu and Robinson, tend to reinforce each other. Here, the feedback loop between the governing elite and the rest of the citizenry becomes salient. In extractive political systems, there's a lack of appropriate feedback mechanism between the citizenry and the government. And as such, the elite can use their power to further entrench an extractive economic system that benefits them at the expense of the population as a whole. In contrast, in an inclusive political system, the elite is constantly exposed to the feedback of the citizenry, be it in the form of opinion polls or elections. And as such, the elite has incentives to improve the functioning of the state—and that includes the creation of more inclusive economic institutions.
- 7.33** If prosperity is so good, why doesn't everybody embrace institutions that promote economic growth? And here we come to the essence, the role of the elite. If prosperity is so good, why doesn't everybody embrace institutions which create more wealth and prosperity? The Austrian economist Joseph Schumpeter noted that, "Vibrant and innovative societies create wealth through the process of creative destruction, or the incessant product and process innovation mechanism, by which new production units displace or replace outdated ones." That process creates losers and resentment. Those threatened with losses, therefore, try to prevent innovation from happening. Innovation undermines the status quo. Elite has an interest in the status quo. Therefore, innovation threatens the elite, and therefore, elite tries generally to squash innovation. OK?
- 7.34** During the industrial revolution in Britain, new business entities, such as textile factories, replaced individual clothing producers. The latter attempted to thwart the spread of industrialization by mounting a Luddite rebellion and smashing stocking frames or machines that were used to knit fabrics. In addition to economic changes in the industry, industrial revolution produced a dramatic shift in power relations. As peasants left the British countryside to work in factories, the power of the landed gentry, the aristocracy, collapsed. You can see why the aristocrats, who derived all their wealth from land, were not terribly keen on industrialization —because they were the losers in the process of creative destruction that Schumpeter talks about.

- 7.35 In places where aristocracy had more power than it had in Britain—think about Austria, Hungary, and Russia—the elite initially blocked industrialization rather than lose control. Such actions can delay the emergence of widespread prosperity by decades. Political and economic changes tend to be heavily influenced by the attitudes of the elite, whom we define as a relatively small group of people who hold a disproportionate amount of wealth and power in society. Elite emerges in the agricultural era, and then they get a hold on human institutions for the next 12,000 years. OK? It's the hierarchy, it's the elite.
- 7.36 And the elite has the power to resist and block changes, and it will wish to do so as long as they perceive those changes to be contrary to their long-term interests. Political and economic institutions that dominated human affairs during the discovery of agriculture 12,000 years ago tended to be the extractive ones. Then, a few centuries ago, 250-300 years ago, politics and economics of Western Europe started to change.
- 7.37 In 2012 essay, a well-known economic historian, Gregory Clark, asked, "Why was the transition to this modern regime of sustained innovation delayed at least 10,000 years from the development of settled agriculture? Why did it occur in a small country on the fringe of Europe, and not in the great center of world population, such as China? England had a population in 1760 of 6 million, compared to 270 million in Qing China, 31 million in Japan, at least 100 million in India. Why did it not occur 2,000 years earlier, in the classical civilizations of Europe, in Greece and Italy, or in the already developed economy of China?"
- 7.38 So, my goal for this lecture is to explore some of the possible reasons for the emergence of the age of sustained innovation in Western Europe. In his 2019 book, *The Wealth Explosion: The Nature and Origins of Modernity*, Stephen Davies noted—as we have discussed before—the emergence of two elites, or rather, two classes of people, the elite and the rest. We've already discussed that. And for a long time, he observed that the elites were satisfied with political and economic arrangements which they had imposed onto the rest of the people. But beginning in the 14th century, the incentives that the Western European elites faced began to change. Around the year 1300 A.D., there was suddenly an onset of the little Ice Age, a 500-year period of longer winters, short summers, lot of rain. And as a result of that, agricultural yields declined and harvest failures decreased by about 20%. There was a population decline, a Malthusian trap.
- 7.39 Everybody was worse off. And on top of that, you have the arrival of Black Death, which kills anywhere between one-third and half—and one-half of all Europeans. And the ruling elite begins to be very desperate, because it can increase revenue only in two ways. First, they can tax more heavily and more widely. And second, they can wage war in order to control larger populations, in order to get taxes from them, right? So you can either maintain your population, but tax them more heavily, or you can expand

your population that lives under your rule. And that way you get—you don't have to tax them more heavily, it's just more people, right?

- 7.40** So this increased taxation and the onset of war produced terrible effects. Disease, malnutrition, uprisings in Europe, China, and India. Across the world, dynasties and empires fell and crumbled. And out of the ashes of the old system, a new balkanized and highly competitive political order emerged.
- 7.41** Now, starting in 1400, so about 100 years later, political competition between states caused warfare to go through a number of changes. And the chief among them was the invention of the gunpowder. Heavy artillery was originally used only in prolonged sieges. But the improvements in artillery meant the development of lighter and more mobile guns that could be used on the battlefield. And larger armies needed a more structured hierarchical chain of command, resulting in a more permanent and elaborate military establishment.
- 7.42** The ruling elite had to provide full-time soldiers with food and shelter, which meant that they needed to build additional infrastructure and secure more funding. Armies became a constant expenditure, paid for with current revenue, not just accumulated treasure. Put differently, a large and expensive, expensive military needed a large and efficient tax administration. This military revolution resulted in a huge political shift.
- 7.43** Since innovation before then was generally slow, powers that adopted some of these military changes first had a big first-mover advantage. That was especially true for artillery. It was this increase in militarization and increase in the efficiency of artillery that leads to the emergence of what we call "gunpowder empires," such as China, Russia, Mughal India, Ottoman Empire, Safavid Iran. These empires became more and more powerful with each conquest and only stopped expanding when they encountered an empire of similar proportion on its borders.
- 7.44** These new empires controlled vast majority of Eurasia and significant proportion of the world's productive base. They were also very large, and internal stability was, therefore, the ruling elite's primary concern. The gunpowder empires reduced predation and exploitation within their borders, leading to some economic growth. But ideas and innovations that threatened the traditional order were suppressed in the name of stability, hindering long-term growth.
- 7.45** So, you have the emergence of these vast empires, but they are so large that they cannot be conquered from outside. They are simply too big, too powerful. If they are going to be destroyed, the enemy must come from within. And so all of these empires have one most important obsession—to destroy domestic economic and political innovations that could upset the ruling order, the ruling elite. It's the internal stability which is most important to them. They are not worried about the outside world. They are too strong. "Nobody is going to conquer us." They are worried about domestic stability, and they clamp down on innovation.

- 7.46 There is one major exception to this rule in the entire world, and that's Europe. Europe, of course, had empires. Spaniards had their empires in South America. The British had their empire in North America and, later, in Africa. The French had their empire in Africa and many other parts of the world. But Europe was never one country. It was never an empire. Nobody ever succeeded in turning European countries into one large country with one large government that could then exercise its power in order to clamp down on innovation, in order to promote domestic stability.
- 7.47 So, it was because Europe was dismembered, because there was constant competition between these small European states, because the elite constantly feared that another country would defeat it and destroy it, that the European elites had to make a choice. "Do we allow for greater innovation in our little country—Switzerland or France or Britain or Holland—therefore, generate more economic growth, more technological innovation, so that we can defend ourselves from our enemies from abroad? Or do we emphasize domestic tranquility and stability by destroying, eliminating innovation? But then we'll be poorer and, consequently, subject to being consumed by our competitors from abroad." OK?
- 7.48 The big empires in China, France—sorry—China and Russia and Iran didn't have to face those worries. But European countries were tiny, and so the elite figured out, "We would rather risk some domestic instability, some challenge to our elite rule, from new ideas, new technologies, more wealth. But in return for that compromise, we will generate more growth, and therefore, we will be able to protect ourselves from foreign predation." OK?
- 7.49 So, without a single hegemon, Europe remained in a state of constant war and, most importantly, competition. The military revolution thus continued. European armies became the most advanced in the world. Their navies became most sophisticated in the world. Expanding military needed to be paid for, and economic and scientific innovations were allowed to take place, along with military reforms. So, in other words, economic and scientific innovation was allowed because it generated more economic growth, which created more revenue, with which you could then pay for greater defense against foreign invasion.
- 7.50 And in addition to that, European countries also incentivized greater technological—not only were they not opposed to innovation, like the rulers in Peking, or the rulers of India, or the rulers of Russia, they actively encouraged innovation through the creation of, for example, the Royal Society in England, or the Académie de France in 1666. So, here we have a mental switch of the elite. The elite, under normal circumstances, hates innovation, wants to preserve stability. But in Europe, it's the geopolitical competition between states which actually creates an elite which promotes innovation. And that, according to Stephen Davis, is one of the reasons why the era of sustained innovation starts in Western Europe about 200, 250 years ago.

- 7.51** Now, along with the geopolitical competition also comes trade with the New World. Very important. Very important because you have the so-called Columbian Exchange, which has enormous consequences for the world, and especially for Europe. Produce from the New World, including tobacco, potatoes, chile peppers, tomatoes, and maize, entered the world system. The potato, in particular, was consequential, for it allowed more people to be supported on a smaller area of land, leading to population growth. In Europe, introduction of potato is incredibly important. It has another advantage. Unlike wheat, which is stored in big containers that bandits or passing armies can rob, potatoes are in the ground, and they are protected from predation.
- 7.52** Another far-reaching consequence of the European trade with the Americas was greater access to quantities of silver. And once you have, basically, the world system monetized—once you have a global currency in silver—then you can start trading. You take stuff from America, you bring it to Europe, or maybe you ship it over to China, and you bring stuff from China, because now everybody suddenly has access to silver. We have, for the first time in the world, a global currency, which is silver coin. And that is enabled by the European conquest of the New World, specifically the silver mines of South America. So, the adoption of silver as a currency means that, for the first time, we have a universal means of exchange. Before then, what would you trade with in China? China was much more economically developed than Europe. They didn't want any goods that Europeans could produce, but they wanted silver. So now, whatever Europeans needed, they could pay for with silver. OK?
- 7.53** So, all of these economic changes that I described, and the introduction of New World crops into the Old World, meant that population recovered from famines, from plagues, and started to grow again. Urban trade hubs rebounded, and more types of goods were traded, allowing for greater specialization amongst regions.
- 7.54** The next on our list is a revolution in thought. And here, I'm borrowing from work by Joel Mokyr, a well-known American economic historian, and his 2004 book, *The Gifts of Athena*. Joel Mokyr separated human knowledge into two categories. Propositional knowledge refers to what humans believe to be true about the world. Today's propositional knowledge consists of our understanding of concepts like gravity and evolution. These ideas have no direct effect on day-to-day lives of most people, but they do frame our perceptions of the world around us. Right? We know why the magic carpet doesn't hover in the air —because of gravity. So it has no profound impact on day-to-day life, but we understand it, sort of. We understand how gravity works and how it affects us.
- 7.55** Prescription knowledge, by contrast, refers to instructions on how to do things like grow crops, cast metal, and navigate the seas. In the premodern world, let's say 250, maybe even 300 years ago, these two types of knowledge—the propositional and the prescriptive knowledge—barely intersected. Barely intersected. As such, people had little or no theoretical understanding of why prescriptive knowledge worked. So, when

Edward Jenner, for example, starts to vaccinate his patients against smallpox, he understands that it works, that people who are vaccinated no longer get smallpox, but he doesn't understand why it works.

- 7.56** And it takes another 150 years before we have Louis Pasteur and the germ theory of disease. And we understand not just that smallpox vaccination works, but more importantly, why it works. And once we understand why it works, we can create many other vaccines. OK? So the early vaccinations—people didn't know why they worked. Now, they started to realize why they worked. So, no one knew why Jenner's vaccine worked. So vaccinations against other diseases had to wait until the germ theory of disease became widely accepted a century later.
- 7.57** Medical, scientific, and technological progress happened as a consequence of random experimentation or accidents. Likewise, advances in prescriptive knowledge had little to do, and had little impact on, propositional knowledge. And that included false propositional knowledge. Since the Greek Antiquity, medical practitioners, for example, had subscribed to the humoral theory of medicine, which held that the human body is composed of 4 basic humors—black bile, yellow bile, phlegm, and blood. And these 4 humors had to be in balance for a patient to be healthy. It was nonsense, but people believed in it for 2,000 years. And to the extent that prescriptive knowledge, such as medical treatments, was ever influenced by propositional knowledge, false propositions retarded or slowed down the advance of human understanding, and thus contributed to human suffering.
- 7.58** Another key aspect of premodern knowledge was its holistic nature, or comprehension of the parts as intimately interconnected and explicable only by reference to the whole. It relied totally on the belief in the creator deity. As such, human anatomy was seen as a microcosm of astronomy, meaning that any statement about the human body implied something about the cosmic order. That meant that ideas about the natural world had moral implications. If a new discovery, new innovation, undermined one part of the system, it threatened the entire order. So, if you delve too deep into, say, for example, human anatomy, you are not just revolutionizing medicine and our understanding of the human body, but also the relationships of humans to God, to religion. And therefore, it was better not to look. OK?
- 7.59** Since rulers typically justified their reign with theology, innovation directly threatened their authority. Therefore, scientific innovation was not only difficult, but also dangerous. To avoid the destabilizing effects of progress, the rulers often used traditional sources of knowledge as grounds for rejection of scientific discoveries. Now, occasionally there were periods during which reason prevailed over tradition. In these epochs, which scholars called "efflorescences," philosophical inquirers were permitted to freely question and debate established knowledge.
- 7.60** These ages of reason usually coincided with a loss of central authority, such as the Warring States period in China or the Hellenic period in Europe. And once central

authority was restored, these efflorescences were always stamped out by the ruling elite and replaced by orthodoxy and tradition. One of these ages of reason occurred during the Renaissance, when Europeans discovered ancient thinkers—or rather, rediscovered ancient thinkers—and began to revere and, eventually, to question an idealized version of the past.

- 7.61** There was also an increase in intellectual pluralism as people were exposed to ideas from other cultures that Europeans encountered during the Age of Discovery. And finally, it was also the first time that the individual in Western Europe was becoming a distinct notion. Identity became a more and more complete and self-contained concept, meaning that people were increasingly seen more than just a sum of their relationships with others. You were no longer just wife of some guy or a son of some person. You were an individual in your own right, OK?
- 7.62** And so, this cultural shift led to the desire for a more personal and direct relationship with God. The rise of individualism also leads to people wanting to have a more direct and personal relationship with God—one that wasn't filtered through the Church. And this underlying desire, along with the Catholic abuses, such as corruption and selling of indulgences, culminated in the Protestant Reformation that began in 1517.
- 7.63** Martin Luther called for a transfer of authority of the Catholic Church to the state. Protestants were supported by many ruling aristocrats across Europe, and the schism between Catholic and Protestants eventually led to religious wars in Europe, known as the Thirty Years' War. The Thirty Years' war was extremely destructive, killing millions and causing widespread famine and plague. And that war of religion led to the rise of skepticism, as the different religious schools competed and undermined each other—the Protestants and the Catholics—and the widespread chaos and destruction led many to question whether religion was really worth all the violence that it seemed to inspire. Furthermore, people became exposed to religions from other parts of the world and began to wonder if any religion is truly more legitimate than any other.
- 7.64** So, as noted, such intellectual turbulence would normally be replaced by the traditional system. The elite would reassert itself. But in Europe, political and religious divisions remained. That meant that there was no central intellectual authority, and thus no one system to appeal to. A plurality of thought, in other words, persisted at the national level.
- 7.65** Since intellectuals in Europe didn't revert back to trust, to traditional authority, they were faced with a philosophical dilemma. Either nothing can be proven to be true, as the skeptics argued, or there was a way to discover truth independent from revelation and from authority. A solution was eventually developed in which some types of propositional knowledge could be revised if they were shown to be false or insufficient. Furthermore, since knowledge was now discovered from concrete observations and not from assumptions about the whole system, individual discoveries were completely independent of each other.

- 7.66 As a consequence, prescriptive and propositional knowledge became connected. Prescriptive knowledge, such as how the human body worked, was now used to challenge and refine propositional knowledge. And since propositional knowledge was now more deeply rooted in reality, it could serve as an intellectual foundation for prescriptive innovation. Experimentation was no longer reliant on random mistakes, but could be consciously pursued using scientific inquiry. Science and technology, in other words, began to help one another.
- 7.67 That is not to say that religion or magic suddenly lost all influence. These were, and still are, powerful cultural forces, but they lost their monopoly on intellectual thought. Traditional systems of knowledge persisted, but now recent arguments were not just thrown out because they contradicted a holy text, at least in the European context. For the first time in history, reason was not stamped out by the elite. For the first time in history, innovation was not stamped out by the elite, and in this highly competitive Europe, the practical, military, and economic benefits of free thought were essential. The survival of the ruling elite depended on innovation, not on the destruction of innovation. That was the only way they could protect their countries against foreign predation.
- 7.68 The last bit I want to talk about in this lecture is the change in ethics. Concomitant with geopolitical competition, population growth, which was enabled by the Columbian Exchange, and the revolution in thought, was a change in ethics, which governed people's attitudes toward innovation and commerce.
- 7.69 Innovation and commerce are intimately connected—for, as McCloskey noted, innovation isn't enough. It has to be tested whether it is profitable to work. People innovate all sorts of things, but you have to decide somehow which innovations are actually going to be creating value, right? And that can only be accomplished in the market. So, innovation and commerce—trade—are intimately connected. Innovation is a trade-tested invention. OK?
- 7.70 So, generally speaking, Europeans were quite anti-innovation and anti-commerce until very recently. And so, let us look at some of the reasons why these negative attitudes toward innovation and commerce have changed. We can start in Ancient Greece. The Ancient Greeks and Romans actually didn't even have a word for innovation, but newness, any kind of deviation from the norm, tended to have negative connotations. Ancients were very suspicious of anything new. They were deeply, deeply conservative, steeped in their ways.
- 7.71 One example will perhaps suffice. Three Roman writers record a similar anecdote about a man who invented an unbreakable glass and brought it to Emperor Tiberius in—Tiberius reigned in the first half of the 1st century A.D.—and this inventor, who invented unbreakable glass, brought it to Tiberius in the hope of receiving a reward. Tiberius asked the inventor whether he had told anybody else of the innovation. And

when the inventor said no, the emperor had him executed because he worried that unbreakable glass would make precious metals valueless.

- 7.72** That does not mean that innovation didn't take place. It also doesn't mean that individual Europeans failed to appreciate innovations which improved their lives. Giordano da Rivalto, a Dominican friar from Pisa, for example, delivered a sermon in 1306 where he gloried in the discovery of spectacles. He said, "It is not 20 years since there was discovered an art of making spectacles which helped one to see the world, an art which is one of the best and most necessary in the world." So, innovations were taking place, but very slowly.
- 7.73** And the general societal attitude toward innovation was one of hostility. In the words of Donald Boudreaux, an American historian, American economist, "This negative attitude to innovation served as a dishonor tax. Like all taxes," he wrote, "the dishonor tax discouraged the activities on which it fell, while making alternative, untaxed activities relatively more attractive. Innovations that destroyed jobs were indeed held in contempt prior to the modern age."
- 7.74** That leads me to emperor Vespasian. Emperor Vespasian, according to ancient records, was supposed to have also been visited by an innovator who apparently invented a machine capable of carrying heavy columns. But emperor Vespasian feared that the machine would worsen Roman unemployment, and so he declined to use it. Right? So, there was innovation happening, as da Rivalto tells us. But the general attitude toward innovation was one of suspicion. And part of the suspicion was that you never knew what the consequences of innovation would be on the economy. Vespasian worried that, once you had these machines, they're going to put a lot of people out of work, and so it's much better to rely on human labor, right?
- 7.75** But if this was an attitude that we always had, if this was an attitude that ancient people had, living in caves, then we would never have made anything. We would always rely on human labor. We would never have machines to do anything. We wouldn't even have windmills or water wheels, because we worried that some people may be, may be put out of work. So once again, innovation was treated with suspicion and was generally avoided. And people who wanted to innovate were discouraged from doing so.
- 7.76** So let's turn to commerce. We are talking about innovation, now let's talk about commerce. Until relatively recently, most types of commerce or profit-making were universally derided. Homer's *Odyssey*, written 700 B.C.—the Greek hero Odysseus is insulted for resembling a merchant ship captain whose one concern is the greedy profit he makes when he sells his goods. So, 700 B.C., Homer already has a very negative attitude toward traders. What do traders do? They buy cheap, they sell expensive, they make profit.

- 7.77** People were always—especially intellectuals—were always opposed to buying low, selling high, and making a profit. Human history is full of opposition to commerce and treatment of commercial—of people who engaged in commerce as being very low and in fact, detrimental. Plato envisaged a society where everyone who engaged in retail and trade would be disparaged and subjected to contempt and insult.
- 7.78** He believed that anyone who engaged in trade would bring a shame on himself and also on his family. Roman Catholic theologians continued the old imperial hostility toward commerce and profit-making. Protestant theologians agreed. Martin Luther, for example, hated commerce and capitalism. He wrote, "Christians should earn their living by the sweat of their brow. The most admirable life was that of a peasant. The labor of the craftsman is also honorable, for he serves the community." Trade, banking, and credit—capitalist industry—seem to him to belong to the very essence of the kingdom of darkness.
- 7.79** Karl Marx, some few hundred years later, approvingly quoted Luther by saying, "The great wrong and unchristian thievery and robbery are committed all over the world by merchants." John Calvin, the great Protestant leader, said that "merchants closely resemble a prostitute full of tricks, traps, and deceits."
- 7.80** But as McCloskey documented in her books, European attitudes toward merchant class, now called the bourgeoisie, started to change sometime in the 17th century. She wrote, "Three centuries ago, in places like Holland and England, the talk and thought about the middle class began to alter." Middle class were the merchants. "Ordinary conversations about innovation and markets became more approving. The high theorists were emboldened to rethink their prejudice against the bourgeoisie, a prejudice by that time millennia old. In northwestern Europe around 1700, the general opinion shifted in favor of the bourgeoisie, and especially in favor of its marketing and innovating."
- 7.81** The result of this mental shift was modern economic growth. The change in attitudes toward commerce and innovation parallels the split between premodern and modern society. When innovator or merchant were hurled as curses, the main goal of the elite was the preservation of the feudal status quo. But to preserve their independence in a geopolitically competitive world, the elite was now forced to use innovation and trade to further their existential ends.
- 7.82** This shift birthed the modern world in which ordinary men and women were at last free to have a go—free to have a go at the previously unknown freedom, dignity, and prosperity. Thank you very much.

Lecture 8: Eco-Anxiety

At a glance — Eco-anxiety and environmental pessimism.

- 8.1** In this lecture, I will discuss the rise of environmentalism in 19th-century Europe in response to the economic growth and prosperity that started to spread in Western Europe, specifically, and then also in North America in 19th century. I want to turn to the rise of environmentalism in 19th-century Europe in general, and Germany in particular, and its eventual importation into the United States. As I will show, environmentalism contained within itself an anti-bourgeois and anti-capitalist sentiment from the beginning, eventually culminating in a comprehensive condemnation of modernity and civilization.
- 8.2** I will then turn to the rising strand of anti-humanism in the modern environmentalist movement that, in its mildest form, advocates in favor of anti-natalism, meaning having fewer children, and in its most destructive form, flirts with genocide. Next, I will discuss the rise of eco-anxiety that is causing depression and disillusionment. Finally, I will contend that environmentalism has acquired many aspects of traditional religion and is increasingly fulfilling the role of religion in societies where traditional beliefs have receded.
- 8.3** The earliest critic of modernity was the French philosopher Jean-Jacques Rousseau. According to Rousseau, civilization generates learning, wealth, and sophistication. But learning, wealth, and sophistication lead to moral degradation. Before reason—through which we were able to command the world, the planet, to increase our wealth, become prosperous—before reason was awakened, Rousseau conjectured, "People lived in an ideal state of simplicity, equality, and solitude. Their limited needs were easily met through hunting and gathering." To him, hunting and gathering, even though you are incredibly poor, is the ideal. "Then, civilization destroyed natural liberty of the hunters and gatherers and established for all time the law of property and inequality, and for the benefit of few ambitious men, subjected the human race henceforth to labor, servitude, and misery."
- 8.4** Duke University political scientist, Genevieve Rousseliere, observed that, "Rousseau consistently declared that commercial society"—which gave birth to the prosperity, the superabundance, that we enjoy—"that commercial society prevents us from being free because it makes us dependent on others and on endless desires in ways that we cannot control." So, the fact that I need to go to a shop in order to buy a glass of milk or a bottle of milk, that makes me dependent on somebody else, and therefore, I'm not properly free, according to Rousseau. Rousseau's critique of modernity inspired an entirely new political philosophy called Romanticism.
- 8.5** And by the end of the 18th century, attitudes to modernity had divided. To be sure, everyone agreed that modernity made humanity richer—superabundant. But that's

where the agreement ended. The thinkers of the Enlightenment believed that modernity made us friendly, thoughtful, and free—friendly because we cooperate. When I come to your shop, I give you something in exchange for that bottle of milk, right? So, people of the Enlightenment, the thinkers of the Enlightenment, believe that modernity made us friendly, thoughtful, and free. But Romantics, in contrast, inspired by Rousseau, thought that modernity made us jealous, weak, and enslaved.

- 8.6** The scholars of the Enlightenment thought that modernity could coexist in harmony with nature. But as the Romantics saw it, human tendency to dominate the natural world—here you can see the echoes of environmentalism—so, the human tendency to dominate the natural world would bring about humanity's downfall. The machine embodied the division between the two schools of thought. To many in the 18th and 19th century, the machine was a harbinger of progress. You no longer had to be bent over in the field, picking tomatoes. Maybe now you had a machine to help you to do that.
- 8.7** So, to the figures of the Enlightenment, machine was a harbinger of progress. But to Romantics, it meant the coming of an Apocalypse. The English poet William Blake, for example, lamented how the pastoral landscape of Britain was now pockmarked by "dark Satanic mills," by which he meant, of course, factories. He described London, the peak of modern development, as full of "marks of weariness" and "marks of woe." Worse, the hardworking bourgeoisie that pushed society forward was becoming spiritless. That's another thing that we will encounter during this lecture. It was becoming spiritless, empty, and Philistine, or Philistine, in the pursuit of profit. The overly specialized industrial economy created so much easy wealth and luxury that materialism became inescapable. Materialism—we will encounter that as well. In the future, Romantics believed, the spirit would wither, and society would fall into a trap of stagnation that characterized, at least in the European view, the Eastern civilizations at the turn of the 19th century.
- 8.8** In the writings of H. G. Wells, those writings show the Englishman's concern over both machines and spiritual degradation. In his 1895 science fiction novella, *The Time Machine*, Wells described how, almost 1 million years in the future, the industrial society had split humanity into two degenerate races— the brute cave-dwelling Morlocks, i.e., the workers, and the banal and weak Eloi, the upper classes. In another one of his novels, *The War of the Worlds*, in 1898, he described Martians taking over Earth with advanced machines—only to be struck down, these Martians were, by natural pathogens. To Wells, technology was not only at war with nature. It was, above all, a war that technology was not going to win. Nature was going to triumph over technology.
- 8.9** The word "ecology" was actually coined by a German biologist, Ernst Haeckel, who applied the term "ökologie" to the relation of animal, both to its organic as well as its inorganic environment. Haeckel's "ökologie" was central to his philosophy of Monism,

which dictated that the universe was one united, dynamic, and purposeless process. Haeckel wanted Monism to replace Christianity in Germany. He thought that his philosophy would help to spread communitarian values and end the biological hierarchy that is present in all Abrahamic religions and that gives humanity a privileged status over other creatures. So, even in 19th century, we have this strange feeling that humans who are in the center—established by the Abrahamic religion, right? Humans are special—that this specialty of human beings should be eliminated. That we should be brought down to the same level as other animals, and perhaps even plants. That we are not special at all.

8.10 So, while Haeckel never accomplished his goal of replacing Christianity, his Monism was very popular at the end of the 19th century in Germany, a country which was obsessed with the Romantic view of nature and deeply ambivalent about commerce, urbanization, and industrialization. The German Youth Movement, for example, which started in 1896, consisted of a number of organizations devoted to outdoor activities. The movement grew immensely popular with some 8 million German children affiliated with it in one form or another by 1938. It was for the German Youth Movement that the German philosopher Ludwig Klages wrote his 1913 essay, *Man and Earth*. I think it bears reading it in full.

8.11 So this is what Klages, one of the Romantics, one of the pro-environmentalist writers at the end of the 19th century, writes. "Today, we see ever-increasing hordes huddled together in our big cities where they grow accustomed to the soot belching from chimneys and thunderous turmoil of the streets, where the nights are as bright as days. These urban masses believe that they have had an adequate introduction to the world of nature as soon as they caught a glimpse of a potato field or a single starling perched on a branch of an emaciated roadside tree. But to anyone who recalls the sounds and sense of the German landscape 70 years ago"—German landscape before industrialization, the Romantic German landscape before factories, where people lived in this bucolic state of bliss—"but to anyone who recalls the sounds and sense of the German landscape of 70 years ago, from out of the words and images in which these memories are embodied, a wind would arise to pronounce a warning reproach to the lost souls of today as soon as they begin to regurgitate their weather-proof platitudes about economic development, necessities, and culture. An unparalleled orgy of destruction has seized humankind and its civilization that has unleashed this lust for murder so that Earth withers before its noxious breath. These are indeed the fruits of progress."

8.12 German Romanticism aligned with folkish nationalism that eventually was co-opted by National Socialists. The German philosopher and National Socialist Martin Heidegger distilled ideas of German Romanticism into a comprehensive critique of modern life. To that end, he identified two nature-destroying forces. First came technology, which gave

humanity the ability to dominate nature. Second came capitalism, which destroyed any connection between nature and the product of work or economic output.

- 8.13** According to some advocates of National Socialist worldview, the Aryan, with his connection to blood and soil, was the perfect man to reunite humanity with nature. Organic farming was encouraged by the members of the Nazi state machinery, such as the Minister of Agriculture, Richard W. Darré, for its harmonized relationship between humans and the land. Even the Reichsführer of the SS, Heinrich Himmler, thought that SS recruits should respect animal life and banned vivisections under the Nazi regime.
- 8.14** In the United States, environmentalism took a longer time to emerge. According to the Journal of the U.S. Environmental Protection Agency, many environmental ideas in America first crystallized in 1962. That year saw the publication of Rachel Carson's *Silent Spring*, first in a serial form in *The New Yorker*, and then as a bestseller. In her book, Carson attacked the indiscriminate use of pesticides, causing a revolution in public opinion. Within a year, the U.S. Congress passed the 1963 Clean Air Act, which gave the U.S. government more power to regulate the environment.
- 8.15** Five years later, Paul Ehrlich's book, *The Population Bomb*, caused a sensation of similar proportions. The book, which sold millions of copies and was translated into many languages, warned of the coming depletion of natural resources and put Malthusianism firmly back on the intellectual agenda.
- 8.16** The speed and the extent of environmental triumphs in the United States are noteworthy. The year 1970 kicked off with the celebration of first Earth Day on April 22nd. As Ronald Bailey from the *Reason* magazine remembers, "About 20 million Americans turned out for the first Earth Day on April 22, 1970. Lectures and rallies took place at more than 2,000 campus colleges, 10,000 elementary and high schools, and thousands of other places across the country. 42 states adopted resolutions endorsing Earth Day, and Congress recessed so that legislators could participate in the activities in their districts." It is sometimes described as, up to that time, the largest public demonstration in history. In September of the same year—again, we are talking about 1970—the U.S. Congress beefed up the 1963 Clean Air Act. By December, President Richard Nixon helped to create the Environmental Protection Agency. Private environmental organizations also flourished, among them Greenpeace, established in 1971.
- 8.17** Over 2,000 communes sprang up across the nation, striving to create real, ecologically balanced microsocieties. As the 1970s rolled on, American environmentalism became increasingly anti-capitalist. Arthur Herman from the Hudson Institute claims that it was the American writer Charles A. Reich, who brought German ideas to the United States with his book, *The Greening of America*. And Herman notes that "modern ecology in the United States, replayed the same enthusiasms that had animated every modern cultural regeneration movement since the German Romantics." Reich's book was a

bestseller, both in 1970 and 1971, topping the New York Times best seller list in December 1970.

- 8.18** According to Reich, "Work and living have become more and more pointless and empty. There is no lack of meaningful projects that cry out to be done, but our working days are used up in work that lacks meaning." His life obviously had a lot of meaning. It's just everybody else's didn't. "We are making useless and harmful projects for servicing of bureaucratic structures. For most Americans, work is mindless, exhausting, boring, servile, and hateful—something to be endured while life is confined to the time off." Times haven't changed. We constantly hear the same refrain, now 50 years later, again —mindless work, boring, servile, hateful.
- 8.19** "At the same time, our culture," Reich continues, "has been reduced to the grossly commercial. All cultural values are for sale, and those that fail to make a profit are not preserved. Our life activities have become artificial, vicarious, and false to our genuine needs —activities fabricated by others and forced upon us." This is Rousseau, just 200 years later. It is also the Marxist critique of capitalism as alienation of labor. But instead of acknowledging that, instead of acknowledging his dependency on Marxist ideas, Reich goes straight into environmentalism. And like a typical Marxist, he predicts a revolutionary turmoil. "There is a revolution coming," he prophesied. "And its ultimate creation will be a new and enduring wholeness and beauty—a renewed relationship of man to himself, to other men, to society, to nature, and to the land."
- 8.20** Other voices critical of capitalism's effect on the environment soon emerged. They included the American biologist Barry Commoner, who argued that modern society was unsustainable. Unlike Ehrlich, who focused on overpopulation, Commoner focused on capitalist production techniques—pollution-causing detergents and synthetic textiles, for instance —and advocated in favor of eco-socialism. In 1972, the British economist Barbara Ward and the French-American microbiologist René Dubos warned that the exponential economic growth of industrial society threatened the survival of the entire planet.
- 8.21** Where did we hear this before? Ehrlich saying that infinite growth on a finite planet is the growth of a cancer cell. Instead of what Gale and I are arguing about ability to exist on a planet, create ever more value, live in a superabundant world, where infinity of knowledge produces infinity of —reduces the importance of finitude of atoms, Barbara Ward and Dubos say, "Exponential growth of industrial society threatens the survival of the entire planet."
- 8.22** For these environmentalists, wealth generation was no longer the saving grace of capitalism—that it makes everybody better off. Wealth generation is a problem that needs to be tackled. Our message is the exact opposite. Wealth generation will enable us to save the planet because we have more wealth, we have more money, we have more resources, we have more ideas, we can save the planet. But for these

environmentalists in the 1970s, wealth generation is a problem that needs to be tackled.

- 8.23** By the 1980s, environmental demands had become even more radical. The Norwegian philosopher Arne Naess, for example, thought that reforming industrial society was not enough. Instead, he called a change of culture that allowed for any amount of ecological destruction to exist at all. So, to this Norwegian ecologist, it's no longer a trade-off between the well-being of a human being and the environment. Any environmental degradation, any environmental damage, must be rejected, avoided at all cost. Like Haeckel's Monism, Naess' philosophy of deep ecology concluded that the problem with modernity was that it placed humans above other life forms. To these ecologists, there is no difference between a snail and a human baby. So, modernity placed humans above other life forms, creating an inflated ego that enables our species to destroy nature.
- 8.24** By 1986, the American social theorist, Murray Bookchin, published *The Modern Crisis*, in which he called for the replacement of environmentally destructive capitalism. Embracing deep ecology, Bookchin's utopia was radically egalitarian, with men, women, plants, and animals living on equal terms and promoting each other's well-being. As he saw it, such utopia had existed for thousands of years in the form of primitive societies. Bookchin's ideas amounted to a total inversion of human progress. Civilization, he claimed, was just domination over nature, wrenching away the last remnants of the paradise that had existed amongst the aboriginal people of Africa and South America. Here we have another concept that was so dear to the heart of Jean-Jacques Rousseau—the idea of a noble savage. Before industrialization, before machines, before modernity, aboriginal people, native peoples in America, in Africa, in Asia, they live in perfect harmony with nature. Life is good. They are noble. Everything is working out. Everybody is happy.
- 8.25** In his 1991 book, *The Absence of Sacred*, American activist and author, Jerry Mander—that's his real name—claimed, likewise, that primitive societies are based on a rejection of modernity, not ignorance of it. In Mander's view, the subsistence lifestyle is a conscious cultural choice to avoid civilization. It's a conscious cultural choice. We reject civilization. We go back to be bent over in the fields, picking potatoes. And to them, this is an improvement on what we have today, where you can walk into a shop and get a bag of potatoes for a few cents, right? To this day, deep ecologists view primitive societies as not only ecologically harmonious, but free of the cultural desire to dominate and exploit nature.
- 8.26** In 1992, the U.S. Vice President, Al Gore, published his book, *Earth in the Balance*. Gore's critique of modern society is a fusion of the old ideas: modern society is ecologically destructive, materialist, shallow, and shields us from authentic experiences—once again, Rousseau. The culprit, for Al Gore, is new. This time, the fault is with human nature itself. In Gore's vision, culture, at its most basic level, represents control

over nature. The fact that we are in an air-conditioned room, that is culture, but it's bad because we control nature. We are not exposed to the heat and humidity outside, and this is bad.

- 8.27** Stone tools, to Al Gore, and cave paintings are also just rudimentary attempts to impose artificial order on the organic world. So, all those hominids with Acheulean or Oldowan axes in their hands—they too were wrong because they were using primitive tools in order to subjugate nature, in order to kill an animal and to skin it. They, too, were at fault. So the fault of humanity goes back 6 million years, right?
- 8.28** The West, capitalism, technology, and even sexism and racism are all extensions, according to Al Gore, of the innate human desire to dominate. Some ecologists even began to salivate at the thought of the end of the world. Writers like the American author Edward Abbey, and even U.S. domestic terrorists like the Unabomber Ted Kaczynski, dreamed of dams bursting and cities crumbling, forcing the last remnants of humanity to return to a primitive lifestyle.
- 8.29** The French explorer Jacques-Yves Cousteau called the idea of getting rid of human suffering and disease not altogether a beneficial one, and urged that we must eliminate 350,000 people per day. American environmentalist Christopher Manes called HIV/AIDS the necessary solution to environmental degradation. Paraphrasing Voltaire, he said that "if AIDS epidemic didn't exist, radical environmentalists would have to invent one." Finally, Richard Preston from the New York Times, wondered in his bestseller, *The Hot Zone*, a terrifying true story, whether "an extremely deadly virus like Ebola or Marburg may not be the biosphere's reaction to the human parasite and the cancerous rot-out of advanced industrial societies."
- 8.30** This anti-humanism has become firmly established within parts of the environmentalist movement. In May 2019, for example, CNN report, on the newly released report on the United Nations Intergovernmental Science-Policy Platform on Diversity and Ecosystem Services, asserted that we must act now, consuming less, polluting less, and having fewer children. The logical continuation of the Malthusian concern with population growth, as my colleague Chelsea Follett noted, is the voluntary human extinction movement called VHEMT. According to its American founder, Les Knight, VHEMT is gaining traction amongst the young. "In last year," Knight told a British newspaper in 2019, "I've seen more and more articles about people choosing to remain child-free or to not add more to the existing family than before."
- 8.31** He's right. Recent articles embracing the benefits of human extinction include *The New Yorker's* *The Case for Not Being Born*. *NBC News—Science Proves Kids Are Bad for Earth: Morality Suggests We Should Stop Having Them*. And the *New York Times* polemic, *Would Human [Extinction] Be a Tragedy?* The last piece from the *New York Times* muses that our species possesses inherent value, but we are devastating the Earth and causing unimaginable human suffering. It may well be, then, that the extinction of humanity would make the world better off.

- 8.32** In November 2019, the British newspaper The Guardian ran an article titled I Wish I Had Never Been Born: The Rise of Anti-Natalists. The article introduces anti-natalism with a story about an Indian man who, as a symbolic gesture, decided to sue his parents on the grounds that they should not have begotten him without his consent. The lawsuit's intent was to set an ethical precedent, according to which the net effect of bringing life into the world is, by definition, negative, thus rendering procreation objectively immoral. As the Guardian explained, the basic tenet of anti-natalism is that life, even under the best of circumstances, is a harm and an imposition.
- 8.33** Anti-natalism hinges on the idea that every human life involves some amount of suffering, and that suffering cannot be offset by happiness, and that reducing suffering regardless of happiness is the key moral imperative of society. It logically follows that the goal of eliminating suffering altogether requires the complete extinction of human beings. So, just because your life contains some suffering, and suffering must be avoided at all costs, there shouldn't be a life in the first place. These people do not stop and think that, yes, every life contains some suffering, but it also contains a lot of happiness, which is why people cling to dear life into an old age and when they are disabled and even when they are quadriplegic.
- 8.34** So, the article credits South African philosopher, David Benatar, with introducing the term anti-natalism and taking leadership of the modern anti-natalist movement. As with any philosophy, the definition of anti-natalism is a contentious subject. One group of anti-natalists is child-free. These people don't not want children themselves, but they do not consider all procreation to be unethical. Another group consists of de-natalists. These people disapprove of procreation only under certain conditions. True anti-natalists, in the meantime, oppose all births under all circumstances.
- 8.35** True anti-natalists, though united by their common overarching goal of human extinction, are split into two groups—those who prioritize human extinction, and those who advocate for the extinction of all sentient life. So the true anti-natalists are split into two groups. The first group just wants extinction of human beings. But it's not a group that wants extinction of all sentient life. Why? Well, think about the hundreds of thousands, possibly millions, of animals that die every day being devoured by other animals. The antelope happily eating grass somewhere in Africa, suddenly being devoured by the lion—does she not suffer? Of course she does. And therefore, to eliminate suffering, we should eliminate all life on Earth, including those of animals.
- 8.36** The article then explores the relationship between anti-natalists and climate activists, finding that the two groups have often similar concerns, practical lifestyles, and perspectives on the world, but they also differ, and very often, radically. Climate activists may fear having children for two main reasons. First, some activists are pessimistic enough to believe that the ecological state of the world today, and especially tomorrow, is so bad that inflicting it on the child would be unjust. Second, many climate activists believe that each new life brought into the world will consume

an unjust amount of scarce resources, generating emissions and, ultimately, endanger the life on the planet and the life of other fellow human beings.

- 8.37** So, whilst climate activists and true anti-natalists may both consider procreation immoral, they have different reasons for thinking so. Anti-natalists don't want humans because they don't want suffering. Climate activists don't want human beings because they believe that more human beings will lead to an ecological catastrophe, destroying the lives of others. That's the difference between the two. Or to put it differently, no matter how strongly climate activists oppose procreation, most of them do so because they fear human extinction. True anti-natalists, for them, extinction is the dream.
- 8.38** Most anti-natalists are content with voluntary reduction of birth rates. Others hope to achieve that goal through government enforcement. Prominent environmentalists, including Johns Hopkins University bioethicist Travis Rieder and a science popularizer, Bill Nye, have advocated in favor of special taxes or other state-imposed penalties on those with too many children. Bowdoin College philosopher Sarah Conly, in her book, *One Child: Do We Have a Right to Have More?*, noticed that we live in a world where burgeoning global population has started to have major and destructive environmental impact. The results include climate change and the struggle for limited resources. And future appears to be difficult. She acknowledged that many view procreation as an essential component of the right to personal happiness and autonomy—and she acknowledged that that is the dominant view.
- 8.39** She acknowledges that the government does not have the right to impose these restrictions on procreation, on its own citizens, for the sake of future people who have not yet been born. But ultimately, Conly decided—Conly ultimately decided that, "Not only is it wrong to have more than one child in the face of such environmental concerns, we do not even retain the right to do so." She said, "Personal autonomy and personal rights are not unlimited, especially if one's body may cause harm to anyone, and that the government has a moral obligation to protect both current and future citizens"—by limiting procreation.
- 8.40** Many anti-natalists believe that the world without humans, or with significantly fewer humans, would eventually revert to a pollution-free paradise with abundant natural resources. As one human extinction proponent put it in January 2020, "In approximately 20,000 years after human extinction, this magnificently resistant biosphere will return to its perfection." As COVID spread around the world in the early 2020, some environmental extremists rejoiced at the growing death toll. Stanford University Environmental Science graduate, Sierra Garcia, has compiled the horrifying instances of such rejoicement. One archetypical example is a tweet with about 300,000 likes, which says, "Wow, Earth is recovering. Coronavirus is the Earth's vaccine. We are the virus."
- 8.41** New York Times has noted that an upside of social distancing efforts may be that we are fighting climate change. And CNN ran the headline, *There's an Unlikely Beneficiary*

of Coronavirus: The Planet. The BBC's environmental correspondent has gleefully reported that air pollution and CO₂ emissions have been falling rapidly as the virus spread. Some environmentalists are very pleased with COVID-19 pandemic and the concomitant economic crisis because they have slashed the CO₂ emissions. They have worried that, once things got better and COVID was defeated, post-recession economies will see a surge of harmful emissions.

- 8.42** The most extreme adherents of anti-humanism are people who do not rely on persuasion or government action, but instead take matters into their own hands and start to kill the members of the human race, at least in part for the sake of the planet's environment. Just minutes before shooting 22 people in an El Paso Walmart on August 3, 2019, the shooter, Patrick Crusius, released a manifesto titled *The Inconvenient Truth*, a reference to the former US Vice President Al Gore's 2006 climate change documentary. "Our lifestyles are destroying the environment in our country," he wrote. "But God damn, most of you are just too stubborn to change your lifestyle. So the next logical step is to decrease the number of people in America using resources. If we can get rid of enough people, then our way of life can become more sustainable."
- 8.43** The people in the United States that Crusius wanted to kill were Hispanics, a group that Crusius believed would give the U.S. Democratic Party a permanent electoral advantage. In return for the Hispanic support, Crusius reasoned, the Democrats will keep the borders open, thus leading to further immigration. Crusius was clearly a white supremacist, but his actions were partly justified by Malthusian concerns.
- 8.44** He was not the first shooter to do so. In an opening statement of his manifesto, Crusius declared that he was inspired by Brenton Tarrant, an Australian man who killed 51 people at two mosques in New Zealand in 2019. Tarrant's manifesto, titled *The Great Replacement*, is similar to Crusius, although the former targeted Muslims in New Zealand, rather than Hispanics in the United States. In one section of the manifesto, Tarrant blamed climate change on higher birth rates in predominantly non-white countries, stating that "the invaders are the ones overpopulating the world. Kill the invaders, kill the overpopulation, and by doing so, save the environment." Later, he wrote, "There is no green future with never-ending population growth. The ideal green world cannot exist in a world of 100 billion, 50 billion, or even 10 billion people."
- 8.45** Tarrant himself seems to have been inspired by Anders Breivik, a right-wing extremist who killed 77 people in the bombing and shooting spree on the Norwegian Island of Utøya in 2011. Breivik, too, had a manifesto—a compendium that, at one point, called for global population cap of 2.1 billion people in order to avoid environmental destruction. Crusius, Tarrant, and Breivik represent a global mass killer phenomenon inspired by the new ideology dubbed "eco-fascism." Broadly speaking, eco-fascists combine white nationalism with environmental extremism, thus giving a new meaning to the concept of blood and soil.

- 8.46** Malthusian killers, though, are not limited to the far-right only. Beginning in the late 1970s, Ted Kaczynski, popularly known as Unabomber, sent bombs to prominent figures in academia industry with an objective of toppling industrial society itself and turning humanity into foraging existence. Kaczynski's manifesto focused on the effects of modern life on human culture and psychology, rather than environment. Still, when Kaczynski mentions environment in his manifesto, he laments the loss of Malthusian checks on population growth. "One of the effects of intrusion of industrial society," Kaczynski writes, "has been that controls on population have been thrown out of balance, hence the population explosion with all that that implies." Kaczynski continues, "No one knows what will happen as a result of ozone depletion, the greenhouse effect, and other environmental problems that cannot yet be foreseen, but the solution is relatively simple—fewer people."
- 8.47** So, what are we to make of this constant, gradual, and increasingly militant anti-human, anti-civilizational drive within the environmental movement? One possibility is to see extreme environmentalism as a new religion. Michael Schellenberger is an American environmentalist who won the Time magazine's Hero of the Environment Award in 2008 and founded a California-based think tank called Environmental Progress. For many years, Schellenberger observed, environmentalist rhetoric has grown increasingly apocalyptic, and so Schellenberger decided to write a book analyzing the environmentalist obsession with the apocalypse. In *Apocalypse Never: Why Environmental Alarmism Hurts Us All*, Schellenberger noted that nature is imbued with mysticism that can produce intense artistic and spiritual inspiration, but it can also make nature difficult to consider from a rational standpoint.
- 8.48** Schellenberger brings up the "appeal to nature" fallacy as an example. A person who appeals to nature argues that natural things, like composting, are better than artificial things, like synthetic fertilizer—just because they are natural. So it doesn't matter what the actual effect of these two types of fertilization is. What matters is that one is natural, the other one is unnatural, and therefore, the second is by definition worse.
- 8.49** Now, of course, artificial things are as much a part of nature as anything else, and most people would consider natural things like cancer and earthquakes to be pretty bad. The appeal to nature is obviously irrational, but it still pervades discourse around the environment. Genetically modified crops have the potential to return acres of farmland to wild animals, for example, but they are often criticized as being unnatural and, therefore, unhealthy for humans and the environment. So if our concern is with using more land in order to grow our crops in order to feed more people, maybe we can feed more people on fewer acres of land using genetically modified crops that would allow for the coexistence of human species and also the expansion of the area where animals live. But that, too, is unacceptable to the environmentalists. Why? Because GMOs are not natural. OK?

- 8.50** Another common myth about nature is that it is self-regulating. It is a self-regulating, harmonious whole with natural equilibrium that humans are destroying. When Earth's history is considered, though, it's clear that there is no ideal balance for humans to upset. Nature is, in fact, constantly changing, and the Carboniferous period was no better or worse than the Jurassic one. While it is true that some threats to the planet, such as asteroid impact, would certainly cause ecological collapse, Schellenberger noticed that environmental activists could not provide clear evidence that environmental changes, such as melting ice, deforestation, and potentially shifting ocean currents, will add up to apocalyptic sum greater than their parts.
- 8.51** What does Schellenberger mean by this? Well, the argument goes something like this. You're walking through nature, maybe you touch an animal, give them some sort of a human-led virus that spreads throughout the species. The species collapse, all the species that depend on that species to survive for food, they collapse. And before you know it, the whole world has collapsed. That's the idea behind this thinking. So Earth is this delicate balance, right? Delicate whole, in balance. And once you upset it, the whole thing goes kaboom. OK?
- 8.52** And what Schellenberger is saying that that's not true. Schellenberger points to religion. In Judaism and Christianity, the natural world is said to be the product of intelligent design, and every creature plays a part in God's grand design. Likewise, many environmentalists see each species as an essential part of the larger ecosystem or biosphere. Thus, when humans extinguish one species, the theory goes, they set up a chain reaction of extinctions that will eventually result in the extinction of humanity itself. There's absolutely no evidence that's true.
- 8.53** Environmentalists are not usually devout Christians, but the myths and morals of Judaism and Christianity are deeply rooted in Western culture. It is reasonable then that these religious traditions may be subconsciously shaping the foundational beliefs of environmentalism, thereby creating a new secular religion. Schellenberger explains how, in the Judeo-Christian tradition, human problems stem from our failure to adjust ourselves to God, while in the apocalyptic-environmental tradition, human problems stem from our failure to adjust ourselves to nature.
- 8.54** So nature, planet, Mother Gaia, has replaced God and Yahweh. In this secular religion, God is replaced by nature, as I said. Priesthood is replaced by scientists, who are tasked with interpreting the natural order of things. You have your saints—Greta Thunberg, right? You have your devils—CO₂-emitting power plants, fossil fuel companies, fossil fuel CEOs, etc.
- 8.55** To Schellenberger's observation, we can add Michael Crichton's analysis of the parallels between traditional religion and apocalyptic environmentalism. As Crichton noted 20 years ago, "If you look carefully, you see that environmentalism is in fact a perfect 21st-century remapping of traditional Judeo-Christian beliefs and myths. There is an initial Eden, a paradise, a state of grace, and unity with nature. There is a fall from grace into

a state of pollution as a result of eating from the tree of knowledge." Right? So, to environmentalists, eating from the tree of knowledge is a bad thing because it allows you to create this air-conditioned room. Knowledge is bad to the environmentalists. They prefer to live with Rousseau amongst the noble savages in hunter-gathering societies.

- 8.56** "There's a fall from grace into a state of pollution as a result of eating from the tree of knowledge. And as a result of our actions, there is a judgment day coming for us all." The Apocalypse, right? The environment collapsing. "We are all energy sinners, doomed to die unless we seek salvation, which is now called sustainability. Sustainability is salvation in the Church of the Environment, just as organic food is its communion, that pesticide-free wafer that the right people with the right beliefs imbibe."
- 8.57** In addition to being similar in structure, environmentalism fulfills the same psychological needs that religion does. Saving Earth is portrayed as a grand struggle for existence, turning environmentalists into righteous heroes and providing additional sources of meaning to their lives. Part of the reason why I think that environmentalism appeals so much to the young is because the young have this feeling that they can do anything. When you are in your late teens and your 20s, you feel like you are there on Earth to make the changes, to make the world anew. You're filled with this brimless optimism. You have a sense of the heroic.
- 8.58** Saving Earth is portrayed as a grand struggle for existence, turning environmentalists into righteous heroes and providing additional sources of meaning to their lives. These religious qualities are not necessarily bad. To be happy, humans need to find meaning somehow. But as Schellenberger notes, it has become increasingly apocalyptic, destructive, and self-defeating, leading its adherents to demonize their opponents, restrict power and prosperity, and spread anxiety and depression. How did that happen?
- 8.59** The apocalyptic side of environmentalism, Schellenberger argues, rises from anxieties about nature of human progress—superabundance that Gale and I talk about. After religion has separated from science during the Enlightenment, about 250, 300 years ago, knowledge was no longer derived from holy scriptures. Our understanding of the world became more accurate, but certainty and stability, which humans derived from religion, were by and large lost. Over time, scientific advances became the norm and seemed unstoppable. The positive effects of scientific progress, such as rising standards of living, became routine, and the explicit link between innovation and human betterment faded into background. Instead, new technologies became feared, with some people worrying that continued technological progress could lead to the end of civilization.
- 8.60** Rather than striving for a green utopia, as the environmentalists of yesteryear did, the contemporary green movement has become obsessed with fear, predicting the end of

the world by technology-induced climate change. Schellenberger recounted his experience at the Extinction Rebellion protest in London. He noted that, while the protesters were demographically similar to the members of the older Earth First group —white, wealthy, educated —the members of the Extinction Rebellion were uniquely obsessed with death. He wrote that when he saw large banners with the word death on them, women wearing black mourning veils, and dead-silent activists with blood-red gowns and ghost-white faces, he spotted an apocalyptic trend.

8.61 To explain this apocalyptic trend, Schellenberger borrows from *Denial of Death*, a 1973 book by the American anthropologist Ernest Becker. "Fear of death," Becker wrote, "is a core part of our subconsciousness. We realize our own mortality early in our lives and spend the latter part dealing with it, often subconsciously." Schellenberger mentions research showing that children who engage with climate activism have better mental health than children who know about climate change but don't do anything about it. He notes how the rise in climate alarmism has coincided with increased levels of anxiety, depression, and suicide in the United States and Europe, with 70% of American teenagers calling anxiety and depression a major problem.

8.62 Like any immortality project, environmentalism requires a narration to function properly. One such narrative concerns heroic environmental activists and scientists combating the villains of industry. But any story that contains villains also creates anger, and that can be dangerous. Carbon emissions, for example, are not born from evil inventions, but are the byproduct of energy consumption, which is a necessary part of eliminating global poverty. Were environmentally successful in reducing the use of fossil fuels in developing countries, the world's poorest and the most vulnerable people would suffer the most.

8.63 "Constructive anger," Schellenberger wrote, "can change the system, but nihilistic anger threatens a great deal of destruction." Besides, apocalyptic environmentalism creates a lot of sadness. It is a pessimistic worldview which holds that the pursuit of prosperity will drive our species extinct. He recalled from an experience that, "the more apocalyptic environmentalist books and articles I read," Schellenberger notes, "the sadder and more anxious I feel." Schellenberger notes that, while environmentalists can escape depression through their movement, the narrative that they push inflames anxieties that are already rising in modern society.

8.64 *Superabundance* is a secular book. Our values are humanistic. Gale and I value life because we value our lives and those of our friends and families. We neither welcome nor condemn a decline in organized religion as a source of meaning, but we acknowledge it as a fact of modern life. In a similar vein, we recognize that to cope with dying, human beings have evolved a strong desire for the transcendental. Finally, we observe that in the absence of traditional religion, the need for the transcendental is being filled by something else, including as Jason Josephson Storm found in his book,

The Myth of Disenchantment, including New Age spirituality, paranormal beliefs, pagan nature worship, and the like.

- 8.65** The decline of religion in rich countries and among the educated and increasingly prosperous global elite has created a void that is increasingly filled by environmentalism. As the University of Illinois at Chicago economist, historian, and Christian, Deirdre McCloskey, put it in her 2010 book, *Bourgeois Dignity: Why Economics Can't Explain the Modern World*, "Environmentalism is taught now as a civic religion in American schools, and with even more fevered rhetoric in Germany and the Netherlands, especially in Sweden. In Sweden, the worship starts at home and in the kindergartens, with stories of the beneficent troll, Mulle, a popular children's character, and is carried on in the rest of the school, taking up substantial parts of the curriculum in the manner of religious instruction. By adulthood, every Swede is a passionate nature worshiper." Remember how I was talking about the nature worship in Germany in the late 19th century? Hasn't gone away.
- 8.66** "By adulthood, every Swede is a passionate nature worshiper and spends her days picking berries in the woods. Humans need such contact with the transcendental — though the theologians observe that worshipping anything short of God has the problem of idolatry for things will pass," writes Deirdre McCloskey. "Sweden nowadays is no more secular a country than it was at the time of the Norse gods or of Lutheranism. The Swedes disdain Allah, yet worship passionately the transcendence of Mulle. The environmental left has now worshipfully adopted Malthus, not on fresh scientific evidence, but on the mathematical logic that resources must be limited. Such evidence-free logic, requiring no wearisome study of the social sciences or of social facts, might explain why a mechanical environmentalism appeals to so many physical, and especially biological, scientists." What is she telling us? She's saying, "Ehrlich, the biologist, doesn't really understand the special nature of human beings." And she concludes, "Forget about Marx, says the new left. Hurrah for Malthus."
- 8.67** Reason, Gale and I contend, is needed to reveal the quasi-religious role that the belief in the coming of the environmental apocalypse plays in the lives of many well-meaning, but increasingly unreasonable, individuals. And evidence, we insist, provides rational grounds for cautious optimism about the state of planet and the future of superabundance. Thank you very much.

Lecture 9: Enduring Superabundance

At a glance — Enduring superabundance, the conclusion.

- 9.1** In this course, we focused on one ancient source of despair that continues to resonate with people today—the fear of overpopulation and the concomitant exhaustion of natural resources. That fear was, of course, real for thousands of years. And Thomas Malthus, the most prominent proponent, deserves credit for being a decent historian. Paradoxically, Malthus penned his influential thesis —that human population grows geometrically, while food production can only grow arithmetically —just as the world started to undergo a fundamental change.
- 9.2** In the late 18th century, some Western European nations and their colonial offshoots started to break out from the Malthusian trap. Their mortality rates started to fall, incomes began to rise, and nutrition improved. By many measures, life became better. With time, much of the rest of the world followed in Western European success by adopting, however partially and imperfectly, liberal institutions—and by liberal, I mean, classically liberal institutions. I'm using liberal in the European sense, not in its current American sense —so, by adopting liberal or open institutions, liberal or open ethics, and liberal or open commitment to inquiry, science, and technology.
- 9.3** In this post-Malthusian world, our species flourished and multiplied. In 1800, there was only 1 billion people on Earth. But last year, in 2022, there were 8 billion of us. About 5 years ago, Gale and I started to work on this particular subject that, ultimately, has led us to publishing a book. We have developed a Simon Abundance Index, which we then turned into a paper that came out in 2018, called *The Simon Abundance Index: the New Way to Measure Availability of Resources*. As I said, sometime later, we were able to publish this book, *Superabundance: The Story of Population Growth, Innovation, and Human Flourishing on an Infinitely Bountiful Planet*.
- 9.4** In our work, what has featured prominently was the intellectual battle between the Malthusian, Paul Ehrlich, and the Cornucopian, if you will, superabundant Julian Simon. The wager between the two men inspired thousands of books and articles. And in this course, Gale and I offered a new perspective on the relationship between population and resources. We believe that the most meaningful way to estimate the abundance of resources is through the prism of hourly wages —time prices. How much time you have to work in order to buy something. Simon, of course, himself recognized that "a more personal—but often relevant —test of scarcity is whether you and I and others feel that we can afford to buy the material. That is, the relationship between price and income may matter. If the price of food stays the same but income falls sharply, then we feel that food is more scarce. By a similar test, if our wages rise while the price of oil remains constant, our fuller pockets lead us to feel that oil is getting less scarce."

- 9.5 So, to that end, we have estimated the abundance of resources at a personal level and also at a population level, which Dr. Gale Pooley talked about. We looked at many different resources and wage rates in different countries over different periods, the longest being 168 years. On average, we estimate that personal resource abundance grew by about 3% per year, thereby doubling every two decades or so. We also estimate that population resource abundance grew at more than 4% per year on average, thereby doubling every 16 years or so. Moreover, we found that, in all the data sets that we have analyzed, abundance grew at a faster rate than population did. Once again, abundance was growing at a faster rate than population did. We call that state of affairs "superabundance." Based on those findings, we have concluded that every additional human being tends to benefit rather than impoverish the rest of humanity.
- 9.6 There is, in other words, some inherent value in human beings, precisely because it is the human beings which produce ideas, which then leads to inventions, which then leads to innovations, which then leads to productivity gains and increased standards of living. The question is, "Can superabundance continue?" Superabundance, we believe, depends on two main components—people and freedom. People who are rational and knowledgeable, but also people who are free to think, speak, read, publish, and interact with others to generate new ideas, to market-test those ideas, and then achieve progress. The more people the planet has, the more freedom they enjoy, the greater the likelihood that the new ideas will be generated to tackle the problems that still remain and those that will arise in the future.
- 9.7 We have not tackled all the problems. There will be problems that will arise in the future. Problems that we may foresee already—new pandemics—and problems that we may not foresee. And it's for that reason that having a richer and more technologically sophisticated planet matters. We want to have the wealth and we want to have the technological know-how to be able to tackle problems that arise in the future. So, the more people we have, the more ideas we have, and we can tackle the problems that remain and that will arise.
- 9.8 Consider one such problem. Fossil fuels have freed humanity from reliance on human and animal muscle and ushered in a modern era dominated first by the steam engine, and later by the combustion engine. But our use of fossil fuels has also contributed to warming of the planet that we are currently experiencing. Nuclear fission is a very safe and reliable source of energy that emits no CO₂ into the atmosphere. But most countries refuse to build new nuclear reactors, partly because of highly publicized accidents in places like Chernobyl and Fukushima. Nuclear fusion could be safer still. During the nuclear fusion process, smaller atoms fuse into larger ones, releasing huge amounts of energy. Alas, nuclear fusion is difficult and, to make it viable, humanity needs to go on innovating.

- 9.9** By analogy, during the innovation process, elements of capital are fused into new creations, generating potentially enormous amounts of value. Since ideas are not made of matter, the laws of thermodynamics do not apply. There may be a finite number of atoms, but there may well be an infinite number of new ideas. So innovations that arise from those new ideas can create exponentially new value. But will we continue to innovate?
- 9.10** Dr. Gale Pooley and I can see a number of problems on the horizon. The first that I will discuss is negativity bias, which makes our brains an easy target for demagogues who are trying to scare us with apocalyptic predictions of doom and gloom to achieve particular ends, such as limits on consumption, innovation, procreation, and freedom of action. The second is the environmental panic-induced decline in population and its deleterious consequences for innovation, growth, and prosperity, not to mention fiscal stability and fulfilment of our promises to future generations in terms of social security and health care.
- 9.11** The third is a potentially serious decline in freedom of expression. As Jordan Peterson maintains, freedom of speech is deeply connected to freedom of thought. Thinking and reasoning are not solitary activities, but gain sustenance from dialogical exchanges with others. Free thinking requires questioning and interlocutors. In this way, censoring speakers affect the freedom of thought of actual and potential listeners. Obviously, if you are not able to express your views, I cannot learn from you. My freedom of thought is impaired by my inability to listening to the points—the ideas that you may have wanted to express.
- 9.12** The fourth problem is the omnipresent danger of further restrictions on the freedom of the market. Only a market where prices are allowed to rise and fall in accordance with the law of supply and demand, where new products and services are allowed to displace old ones, where people are incentivized to invent and innovate, to benefit or profit from their efforts, can we learn what works and what doesn't work. The Soviet Union didn't have a market. North Korea didn't have a market. Cuba doesn't have a market. That means that the market, which is really just a knowledge-generating mechanism, cannot function. You cannot learn. It is only in the marketplace that you can learn. It is in the marketplace that you discover value, new knowledge. So markets are very important.
- 9.13** So let's look at these threats in turn. Negativity bias refers to a well-documented array of psychological traits that have accumulated in our brains over a span of human evolution, a period of hundreds of thousands of years, during which time the Homo sapiens navigated what was a much less hospitable world. "Our modern skulls house a Stone-Age mind," noted Leda Cosmides and John Toobey from University of California, Santa Barbara. And that mind is often better at dealing with problems encountered by our foraging ancestors, rather than the problems that we face today.

- 9.14** Think of it this way. The latest archeological discoveries—if the latest archeological discoveries are correct, Homo sapiens is about 300,000 years old. Now, from that historical perspective, material abundance, which surrounds us today, certainly in the Western countries, is only 0.08% of our time on Earth old. We've only been prosperous, at least in the West, for about 150, maybe 200 years. But we are a species that's 300,000 years old. So we have experienced prosperity only for something like 0.08% of the time. For the rest of it, we constantly experienced hunger. We experienced much higher levels of violence. The world was a much more dangerous place. And that means that our brains have negative biases because those were real for most of our evolution.
- 9.15** So let's take a closer look at the mysteries of human mind. According to Daniel Kahneman, a Nobel Prize-winning psychologist, organisms that treat threats as more urgent than opportunities have a better chance to survive and to reproduce. Put differently, humans have evolved to prioritize bad news. If what we desired was survival, it must be the bad news which gets our attention first because it is that which could eliminate our ability to survive and to pass our genes on to our progeny.
- 9.16** And these negativity biases are true both for hardware and software of our brains—the physical structure of our brains, as well as the software upon which it runs. Human beings are constantly bombarded with information. Because our brains have limited computing power, humans have to separate that which is paramount—let's say, a lion running toward us—from that which is mundane—maybe a bed of flowers. So because survival is more important than all other considerations, most information is first sifted through the amygdala, the part of the brain that is responsible for primal emotions like rage, hate, and fear. Information relating to those primal emotions gets our attention first because amygdala is always on the lookout for something to fear.
- 9.17** Now that is a very powerful impulse that can deceive even the most dispassionate observers. A recent psychological experiment showed that even when people expressly said that they were interested in the good news, the eye-tracking experiments show that, in fact, they were much more likely to be interested in bad news. Regardless of what participants of this study said—they said, "We are interested in good news"—in fact, their eyes immediately gravitated toward the bad news. So people's express preference is for negative news content.
- 9.18** As Steven Pinker said, "The nature of cognition and the nature of news interact in ways that make us think that the world is much worse than it really is. News, after all, is about things that happen. Things that do not happen are not reported. You never see a reporter saying to the camera, 'Here we are, live from a country where nothing bad is happening.'" Newspapers and other media, in other words, tend to focus on the negative. If it bleeds, it leads. You put the worst story on the front page because that's what buys the newspaper. You lead your newscast with the most horrific story because that's what will get people glued to the screen.

- 9.19** Furthermore, media seldom provide comparative analysis—putting terrible events in their proper context. According to one study, for example, in spite all the improvements that Gale and I talked about, and that you are aware of —we live longer, we are better educated, women have equal rights—in spite all of these tremendous improvements that have happened, certainly in the Western world over the last 200 years, both articles and broadcasts have become increasingly gloomy over time. So the better the world gets, the more gloomy the reporting in the newspaper are.
- 9.20** Why should that be so? Well, once again, if you are trying to sell newspapers, you have to emphasize the negative. If you have competition between different media outlets, it's not just the TV stations and newspapers fighting for eyeballs. It's now the Internet and myriad of different websites. What do you do? You increase the negative content. In fact, we do have a study recently showing that the more negative words you can put into a title of a story, the more likely it is that somebody will click on it.
- 9.21** The reasons for pessimism change, but the stream of apocalyptic warnings continues to gush forth. Here's what Matt Ridley said in his 2010 book, *The Rational Optimist*. "In my own adult lifetime, I have listened to the implacable predictions of growing poverty, coming famines, expanding deserts, imminent plagues, impending water wars, inevitable oil exhaustion, mineral shortages, falling sperm counts, thinning ozone, acidifying rain, nuclear winters, mad-cow epidemics, Y2K computer bugs, killer bees, sex-changed fish, global warming, ocean acidification, and even asteroid impacts that would presently bring this happy interlude to a terrible end. I cannot recall a time when one or other of these scares was not solemnly espoused by sober, distinguished, and serious elites, and hysterically echoed by the media."
- 9.22** The human brain also tends to overestimate danger due to what psychologists call an "availability heuristic"—that is, a process of estimating the probability of an event based on the ease with which relevant instance comes to mind. Unfortunately, human memory recalls events for reasons other than their rate of occurrence. When an event turns up because it is traumatic, the brain will overestimate how likely it is to occur. That's just an academic way of saying that the more horrific something is in your brain, the more likely you are going to pull it out of that memory file, and then you think that it is much more likely to occur than in reality it is.
- 9.23** Consider our fear of terrorism. In the years since 9/11, Islamic terrorists have managed to kill about 7 people a year in the United States. All those deaths were tragic, of course, but some comparisons are warranted. Lightning kills about 46 people a year, accident-causing deer, another 150, and drownings in bathtubs, around 300. Nonetheless, we continue to fear terrorism much more than drowning in a bathtub. Why? Because we've never seen anybody drown in a bathtub. But all of us remember 9/11 and falling of the Twin Towers.
- 9.24** Another problem with the human brain is that bad is stronger than good. Consider how much happier you can imagine yourself feeling, and then consider how much more

dejected you can imagine yourself to be. The answer to the latter question is "infinitely." Research shows that people fear losses much more than their delight in gains. They harp on setbacks more than they relish successes. They resent criticism much more than being encouraged by praise. I could leave my annual meeting with my boss, where he spends 95% of the time telling me what a great job I did, but when I walk out of that office, I will remember that 5% where I messed up.

- 9.25** Good and bad things tend to happen on different timelines. Basically, what that means is that to create something good to improve the world is gradual. It takes a very long time. It took us thousands of years. Well, it certainly took us hundreds of years to get away from absolute poverty in rich countries to superabundance. But it took hundreds of years, incremental progress every year. Bad things happen on a different timeline. A house can burn down in hours. A person can be killed in a car accident in a second.
- 9.26** There's another problem called—once again, in not-very-friendly terms—prevalence-induced concept change in human judgment. You don't have to remember that. But here is what it basically means. According to the authors, people judge each new instance of a concept in the context of previous instances. So, as we reduce the prevalence of a problem—such as, for example, discrimination—we judge each new behavior in the improved context that we have created. So, overdiscrimination against, for example, racial minorities or against women, has been overcome. That has become the new floor, and we have replaced it with microaggressions, right? Overt aggression has disappeared, overt discrimination has disappeared, so now we focus on micro.
- 9.27** So, according to these scholars, again, "As we reduce the prevalence of a problem such as discrimination, we judge each new behavior in the improved context that we have created. Another way to say this is that solving problems causes us to expand our definition of problems." OK? Slavery. Everybody used to understand that by the word slavery, we meant chattel slavery—one person's ownership of another person. Today, we talk about wage slavery—the fact that you had to wake up early in the morning and get to the office to earn your money. We reference that as being modern-day wage slavery. We have redefined the concept from something very serious to something that I would consider much less serious.
- 9.28** So when problems become rare, we count more things as problems. Our studies—that is to say, according to these scores—suggest that when the world gets better, we become harsher critics of it. And this can cause us to mistakenly conclude that it hasn't actually gotten better at all. "Progress, it seems, tends to mask itself." Other psychological effects apply, such as reminiscence bump, for example. We tend to develop rosy nostalgia for the days of our youth when we are young, virile, full of potential. And by feeling good about the past, dissatisfied with the present, and gloomy about the future, we fail to reflect on the real improvements of the world around us. Instead, we superimpose our own physical and mental decline on the world around us,

which is why every generation of older people thinks that the world going to hell, and the young people have never been so immoral or so without structure, and what have you.

- 9.29** Now, finally, humans suffer from a psychological quirk known as pessimism extrapolation, or the end-of-history illusion. Even people who are aware of the progress that humanity has made underestimate our ability to change in the future. "If you underestimate our ability to adapt to unsustainable situations, you'll find all kinds of things that currently look bad and that can be extrapolated into disasters. Extrapolate college tuition increases, and they will be prohibitively expensive in 10 years. Extrapolate government deficits, and we'll go bankrupt in 30 years. Extrapolate a recession, and we'll be broke before long. All of these could be reasons for pessimism if you assume that no future change or adaptation will take place. And to assume that no future change or adaptation will take place is crazy, given that our history has been defined by change and adaptation."
- 9.30** This is what we talk about when we say the end-of-history illusion: "All the people in the past, they were able to solve our problems, but we are so special that we are not going to be able to solve our problems in the future. We are that special."
- 9.31** As we have shown in this course, or tried to show, humans are unlike other species. We have developed sophisticated forms of cooperation that increase our chances of survival and our chances to prosper. We have, in other words, shown that there are rational grounds for optimism about the future. Or, as Thomas Babington Macaulay, a great British historian and statesman, said, "On what principle is it that with nothing but improvement behind us, are we to expect nothing but deterioration before us?"
- 9.32** So, what's the upshot? As you reflect on this course —indeed, as you go through life—remember the different ways in which your mind is playing tricks on you. Recognize that you are a member of a species that's always on the lookout for something to fear. And recognize that your predisposition toward negativity provides a market for purveyors of bad news, be they doomsayers who claim that overpopulation will cause mass starvation, or scaremongers who claim that we are running out of natural resources. The negativity bias is deeply ingrained in our brains. It cannot be wished away. The best that we can do—but we can do it —is to realize that we are suffering from it.
- 9.33** Next, I want to talk about population collapse. Now, to start with, Dr. Pooley and I don't take a particular position on the optimal size of the world's population, except to note that population growth has, in the past, dramatically enhanced rather than retarded economic growth and standards of living. As we already explained and extensively documented, coercive policies and other incentives aimed at reducing family size are not only not helpful, but they are deeply immoral. They harm economic growth, but much more importantly, they cause a great deal of human suffering. In the first lecture,

I pointed out to one-child policy in China, which is responsible for either eliminating or preventing the birth of some 400 million people.

- 9.34** But of course, neither Dr. Pooley nor I want to coerce women into have babies that they don't want, and so we feel that population growth should reflect the free choices of men and women. That being said, even in free societies, choices that parents make are not made in an intellectual vacuum. Even in free societies, men and women who decide on the family size freely, very often do not have the full information—or rather, information can often be skewed, in order to channel their parental decisions in a certain direction. And what we claim in *Superabundance*, in our book, is that the nostrums, the beliefs of extreme environmentalism, can reduce population growth when they are thoughtlessly repeated by policymakers, academics, social media influencers, religious leaders, celebrities, and above all, by the news media.
- 9.35** The impact of persistent fearmongering is readily discernible. Since we started writing *Superabundance*, the British newspaper *Guardian* ran an article: "I feel like I can't in good conscience bring a child into this world and force them to try and survive in what may be apocalyptic conditions," said a 27-year-old woman. The BBC wrote about a couple that doesn't want to have children because they want to mitigate their carbon footprint. The *New York Times* told the story of a woman suffering from terrifying anxiety, recurring nightmares about her daughter's safety, because she assumes the world is going to end. The *New York Post*, in the meantime, opined, "Life is already exhausting enough. The world is broken and burning. Who would want to bring a new innocent life into criminally unequal society situated on a planet with catastrophically rising sea levels?"
- 9.36** These are not just mere anecdotes. A 2020 study found that 60% of Americans between the ages of 27 and 45 reported to be very or extremely concerned about carbon footprint of procreation. And 97% of Americans between the ages of 27 and 45 said that they were very or extremely concerned about the well-being of their hypothetical children. Analysts at Morgan Stanley found that the movement to not have children, owing to fears over climate change, is growing and impacting fertility rates quicker than any other preceding trend in the field of fertility decline. So this is Morgan Stanley, an investment banking company, with real money in the game, saying, "The movement not to have children is growing and impacting fertility rates quicker than any preceding trend in the field of fertility decline," driven by apocalyptic concerns about the future of humanity, future of the world, and so on.
- 9.37** So, as we have noted, in order for population to be stable, you need 2.1 children per woman per lifetime. Out of the 200 countries and territories in the world, 90 had a total fertility of 2 or less in 2018. It is our belief that parents are being unduly influenced by pervasive anti-natalist sentiments, driven primarily by Malthusian concerns. What we are calling for is not an enforcement of higher birth rates, but what we are asking for is a more balanced and factual debate about population growth that

includes a discussion of the benefits of population growth and the dangers inherent in population decline.

- 9.38** And that leads to the second, or rather the third, threat to superabundance — restrictions on the freedom of speech. Governments have generally curtailed freedom of expression much more than not, and speech continues to be restricted in many countries today. Certain subjects, such as the existence of a particular deity or the movement of heavenly bodies, have been a taboo, and in some places they remain a taboo today. People proclaiming these taboos, such as priests and shamans, spoke from authority, but without logic or evidence. They enforced these taboos with ostracism, exile, imprisonment, torture, and even death.
- 9.39** The great contribution of the Enlightenment to human progress was to recognize that the right to free speech was important. No subject should be off-limits to discussion. Free expression should not be punished, and that logic and evidence were required before a claim was accepted as truthful. That set of enlightened values helped Western Europe gradually to break free from deep ignorance and superstition, and embark on a journey toward global scientific and technological pre-eminence—and a historically unprecedented degree of prosperity. Unfortunately, the light of the Enlightenment has not been being lit in some parts of the world. Just think about Afghanistan under Taliban, or think about North Korea under the Kim dynasty.
- 9.40** What's worse is that the light of the Enlightenment seems to be shining much less brightly in some advanced countries on which we rely for a good measure of economic growth and economic innovations. And that includes the United States, Britain, and many other advanced Western democracies. Speakers deemed as controversial and problematic are being banned from speaking on university campuses—the very places that are supposed to be devoted to the pursuit of free inquiry by a minority of noisy and sometimes violent protesters.
- 9.41** Inconvenient questions about sensitive issues, such as the extent of climate change—nobody denies climate change, it's the extent that we are talking about—in other words, the distinction between environmental problems as opposed to an environmental catastrophe or collapse or crisis. People who talk about these sensitive issues are being condemned as denialists. Instead of relying on evidence such as that contained in big data and long-term trends, we are asked to prioritize and generalize from individual people's lived experience. Instead of seeing logic and math as quintessentially objective, we are being asked to see those two as best subjective and, at worst, racist. Instead of searching for the objective truth, we are asked to believe his truth or her truth, not the truth, but his truth and her truth. In other words, many of the values of the Enlightenment appear to be in a bit of a retreat.
- 9.42** Now, as a consequence of that trend, a growing share of the population in some Western democracies are increasingly self-censoring what they say. Orwellian doublethink, by which individual men and women say one thing publicly while

reserving their real feelings for the privacy of their homes, is emerging in countries where people used to feel free and publicly say exactly what they thought, and used to say publicly what they thought privately.

- 9.43 Now, some argue that limits on free expression of some people are tolerable if they prevent hurting other people's feelings. And to be sure, politeness and consideration are welcome traits. But self-censorship in matters of public importance can have far-reaching consequences. Lest we forget, many facts commonly accepted today, such as heliocentrism and natural selection, started as intellectual heresies that offended, discomfited, and hurt the feelings of many people when they were first proposed.
- 9.44 Moreover, self-censorship is particularly difficult for people on whom we disproportionately rely to advance technological and scientific progress. Steve Jobs, for example, was a famously difficult boss. James Watson, the co-discoverer of the structure of DNA, is a provocateur who likes to shock and offend people. In fact, some of our most brilliant minds belong to people who are, to put it lightly, quirky, eccentric, a little bit off. Go to Silicon Valley—it's full of them. Imagine if Steve Jobs was fired from his position because he was too rude to an intern. Would we have the Apple company? So, we ask, should humanity forego additional discoveries simply because future inventors and innovators fail the test of niceness, or if they decide to censor themselves? We believe that the answer to that question should be a resounding "no".
- 9.45 Finally, I want to turn to the freedom of the market. Further restrictions on the free functioning of the market is a threat to superabundance. As we already noted, inventors must test their ideas in the marketplace. It is in the marketplace that inventors discover if their ideas create additional value or not. To reveal a value of an idea, markets must be free. Buyers and sellers must come and go as they please, and prices must be allowed to freely increase or decrease. Market-generated profits and losses tell us a great deal about value creation because rising and falling prices capture the personal preferences of billions of buyers and sellers.
- 9.46 Conversely, restrictions on the function of the market, such as limits on profits and the socialization of losses, prevent the emergence of valuable knowledge. George Gilder, an American economist, argues that, "Wealth is knowledge, and growth is learning." Communists and socialist economies have failed to generate wealth because they did not have markets to help them distinguish between what's valuable and what's not valuable. In other words, they failed to learn. Now, as far as we can tell, there has never been a completely free economy anywhere in the world. In every country, the government has always interfered with market freedom to some extent. But we do have measures, such as the Economic Freedom of the World Report by the Fraser Institute in Canada, which measures economic freedom around the world on a scale from 0 to 10. And what we can say, based on evidence, that economic freedom has been declining, even in Western democracies. It peaked around the new millennium,

and since then, it's on a downward trend in the United States and in many other places all over the world.

9.47 So the free market—or to use a more loaded term, capitalism—produces more wealth and higher standards of living than any other system that humanity has conceived and implemented. The differences in economic performance between North and South Korea, East and West Germany, Chile and Venezuela, Botswana and Zimbabwe, not to mention the United States and the Soviet Union, speak for themselves. Yet in spite of that generally recognized fact—that free markets work—capitalism has never enjoyed anything close to the long-term support that you would expect. In fact, quite the opposite is true.

9.48 As one commentator put it, "There have been innumerable political parties called socialist. In the history of the world, there has never been a single political party called capitalist." There is not even a name for a supporter of capitalism. A socialist champions socialism. A democrat champions democracy. But a capitalist is someone who manipulates capital. And that's because capitalism rubs against certain, very deeply ingrained parts of human nature.

9.49 Again, remember that we have developed in the Pleistocene—hundreds of thousands, perhaps millions of years ago, at the time when people didn't know about agriculture, pastoralism, governments, police, sanitation, medical care, education, armies, transportation, and so on. These are all novel ideas that our ancestors—Homo erectus, Homo habilis, Australopithecines—they never had to encounter. And this is important, because the length of time that we have spent before the agriculture revolution, let alone before modernity, that defines the last two or 300 years, we have developed a certain mind. Our minds have developed to deal with problems that we encountered millions, certainly thousands or hundreds of thousands, of years ago—not problems that we often encounter today.

9.50 Now, among the relevant psychological characteristics that we have developed in the Pleistocene, that we have developed, say, anything older than 12,000 years —among these psychological characteristics is our propensity toward tribalism, egalitarianism, and zero-sum thinking. Let's look at them in turn.

9.51 We evolved in small bands comprising between 25 and 200 individuals. We all knew each other, and we were often related to each other in these small bands. Everyone knew who contributed to the band's survival and who shirked his or her responsibilities. Cheaters and free riders were targets of anger and sometimes punishment. Just as consequentially, cheaters and free riders lost valuable cooperative partners. And once they lost cooperation of their partners, they were solitary, self-reliant, self-dependent, and usually dead within a short while.

9.52 In such bands, where once again, we were often related to each other, sharing of food was common. Storing food for future consumption was not practical—obviously, no

refrigerators —and so, when hunter-gatherers brought in food, such as a slaughtered antelope or a giraffe or whatever, they shared it with other members of the group and stored it in the form of social obligation. "Today, I got lucky. I killed a giraffe. I'm going to share it with you. You don't give me anything in return, but you have a social obligation at the back of your mind, so the next time, when you bring in a killed giraffe, you are going to share it with me." OK?

9.53 Moreover, the volume of personal possessions was limited by what our ancestors could carry on their backs as they moved from one location to another. So the accumulation of property and wealth inequality was not really a major concern. And finally, sharing and cooperation amongst hunter-gatherers ended at the band's edge. I already talked about small bands that we evolved in. We were fiercely loyal to each other. We were related to one another often. But that's where the loyalty ended. That's where cooperation ended, was at the edge of the band. In a world without specialization and trade, disproportionate gains by one band often came at the expense of another band. By forming aggressive coalitions, men could expand their bands' foraging territories or gain more wives by cooperating to kill men from other bands.

9.54 So, to summarize, the psychology that has evolved when our ancestors lived in these small hunter-gathering groups prepared us to cope with a world of personal cooperation and exchange in small communities. It did not prepare us to cope with a world of impersonal cooperation and exchange amongst millions of people, such as is a typical advanced economy, or billions of people, such as the global economy. In a way, you could argue that the complexity of the modern economy outran the ability of our Stone Age minds to understand it. Yet it is that transition from personal simplicity to impersonal complexity that makes capitalism so effective at producing great wealth.

9.55 To complicate matters further, the extended marketplace of millions or even billions of people enables enterprising individuals with value-creating ideas to amass great wealth —much greater wealth than they would be able to amass if they served only a small community. They are now selling to the entire world. The reason why Steve Jobs died a billionaire is because everybody in the world wanted his product. And that wealth inequality rubs against our egalitarian predispositions and zero-sum thinking. And finally, our tribalism helps to explain why we continue to resent other nations. We suspect them of cheating us when we are involved in free trade relations with them. Our tribalism makes it difficult for us to trade with foreigners. The size of the global economy allows certain people to accumulate much more wealth, which then leads to resentment. Then we have evolved to deal in person with one another, rather than dealing with somebody on the other side of the globe. And that leads to problems as well.

9.56 To understand capitalism, let alone to appreciate its benefits, requires all of us to distinguish between the personal and the impersonal, between the simple and the complex, between the limited and the extended. Or as the Austrian Nobel Prize-

winning economist F. A. Hayek put it, "Part of our present difficulty is that we must constantly adjust our lives, our thoughts, and our emotions in order to live simultaneously within different kinds of orders according to different rules. If we were to apply the unmodified, uncurbed rules of the microcosmos—which is to say, the small band or troupe or our family—to the macrocosmos or the wider civilizations, as our instincts and sentiment yearnings make us wish we could do, we would destroy it. Yet if we were always to apply the rules of the extended order to our more intimate groupings, we would crush them. And so we must learn how to live in two worlds at once." The rules that apply to how you run your family, how you interact with your children, are different rules how you interact with the global economy.

9.57 You cannot expect to treat your children or your husband as a simply market mechanism or means to an end. Families are different. Economies are different. And so we constantly struggle to live in these two worlds. If we apply the rules of the global economy to our families, we destroy them. But if we apply the rules of the family, from each according to his ability to each according to his needs—if we embrace socialism, in other words, then we destroy the global economy, our prosperity, and superabundance.

9.58 So, we hope that those of you who have made it to the end of this course will appreciate how far humanity has come and the sheer improbability of what we have accomplished. Struggling against conflict, hunger, and disease, we have managed to gradually gain the upper hand in a world trending toward entropy. We have succeeded in creating complex and superabundant civilization. We have done so because humans are a very unique kind of animal, one that has developed a cooperative culture that allows for the accumulation and sharing of knowledge. That knowledge, in turn, helps to make our society more prosperous. Population growth and freedom are crucial part of that positive feedback loop.

9.59 It is free people, not machines, who generate new ideas, and it is free people who test those new ideas against other people's ideas in the marketplace. The process of knowledge and value creation is at the heart of humanity's progress. It is what enables us and our civilization to bend toward goodness and superabundance. Let us try to keep it that way. Thank you very much.

9.60 I just had one question, or one thought real quick, I failed to mention in my previous lecture —and that was, how many Steve Jobs are in Syria today? Now, I think there's a lot. Yeah, so the issue is how many Steve Jobs are all over the planet because of the culture they have not been able to flourish. So part of what we have to recognize is that we have these individuals around us, but because of some kind of, maybe, typically, a cultural or a government constraint, they're not able to manifest their potential.

9.61 And it suggests that people look at China, for example, and say, "How did we go from China like this, and they suddenly do this?" Well, they had the entrepreneurs there the whole time. What they did is they changed the policy, and the policy then allowed

them to activate their inactive, or their dormant, entrepreneurs. So, we think about, think about it that way, too. We've got these people on the planet that could make these contributions, and what's preventing them from doing that? Policy in those countries—that culture needs to get tuned, to be open to innovation.

- 9.62 Thank you for this opportunity. I'm going to ask a question, but I want to mention first—I had a lot of questions that possible answers came to my mind after reading the book and also coming to this class. I'm going to put myself as an example. When I started, maybe 15 years ago, I wanted to have 4 kids. And then that went from finishing medical school and residency, starting to live in life, and I said, "Whoa, the world is going crazy." Things are going, are worse, because I was always looking at, from the point of view, the perspective or feeling, that the world was getting worse, that, you know what? I have one, maybe two.
- 9.63 That is something that is shared among a lot of people that I know, and I don't know if you have heard also that female physicians usually have less kids, either one or zero. I was thinking of that, and that's how I decided, "Yes, we're going to have one." But it wasn't until I come here and I read the book that I realized that I was making choices without not knowing that I was wrong. So this has opened my eyes. I was like, "Oh, my God, I was wrong." And then you have shown us evidence that things are getting better. And I'm very grateful for the work that you guys have done because I see that takes a lot of time because it's opened us, our awareness and our knowledge and understanding what's going on. It's changed completely my point of view about the world, and now I can share with others.
- 9.64 Thank you very much.
- 9.65 I think you also made really one good little statement there. You said you made choices without knowing what's going on. That's our hope, is that people give themselves an opportunity to figure out, "What's going on? What is really going on here?" So many of us, we fail to do the work necessary to figure out for ourselves what is going on here. We're hopeful that the book will give people an opportunity to see these facts, and then the theory of explaining these facts, and then being able to say, "This is what we could have in the future if we were to change our perspective and our understanding." Then our choices will be informed choices, and we don't look back at our lives and say, "I made a mistake there, and I can't really correct it at this point." So, thank you.
- 9.66 Hi. I had a question, maybe even asking for advice more than anything, but coming from the younger generation, I have a lot of friends that definitely are environmentalists, like climate activists, and I'm wondering if you have advice for how to share some of this with them in a way—maybe meeting them where they're at, because I know a lot of people my age are so paralyzed with the anxiety of just the world, feeling like the world's going to end, feeling like this, just weight of that, that it's hard for them to think through some of this clearly and maybe understand some of what you're saying. I have a lot of friends that have just made a lot of major life choices,

like their career, what they're going to do, based off of how they feel the world is because of what's being touted in the media.

- 9.67** Look, I think that the entirety of human knowledge is now online. Anybody who is interested in finding out what's really happening in the world can do so. There are websites filled with data about the real state of the world. I run one, HumanProgress.org. There is Our World in Data, which is also excellent. There is Gapminder. There are all of these websites which are providing people with objective data on the state of the world. People can read it. People can see it for themselves. All they have to do is take the first step, which is to say, "I am actually interested in finding out what's really going on. I'm not just going to consume the news that I'm seeing on TV or on radio or whatever." It's out there, but they have to take the first step.
- 9.68** What you can do is to say, "Hey, look, are you aware of the fact that, in the last 100 years, the number of people who have died due to extreme weather has decreased by 99%?" So, the point being is that, if we are going to face an environmental crisis, or if we are facing an environmental crisis, everybody says, "It's an environmental crisis." Not just that, they say, "It's an existential crisis." Well, what does it mean, existential crisis? To whom is it an existential crisis? "Well, it's existential to the human beings. If we don't do something about it, human beings are going to die." OK, so what's happening to death of human beings due to climate change? And it turns out that the number of dead people have actually—the rate of dead people has actually decreased by 99%. All of this information is out there. They just have to make the first step.
- 9.69** You used an interesting term. You said "paralyzed by anxiety." It's like, where did the anxiety come from? Why are you so anxious? Part of what we're asking or challenging people to do is to think about how you can persuade another person. There's some people that aren't persuadable. They're just so committed to an ideology. They're not going to be persuadable. But there are people that are open and just, a lot of times, it's just, "Please look at the numbers." Elon Musk says, "Please look at the numbers." What he's saying is, "Look at the facts, and then draw your conclusions about what these facts mean." It could be contrary to what you've been told all of your life. Then be open to this idea that—are you willing to take the risk to be optimistic? You know, it's very easy to be pessimistic. Much easier to be pessimistic than optimistic.
- 9.70** We've got to be able to take these facts, change our perspective, and then communicate that in our lives, that we're rationally optimistic because we know what the facts are. We've got these conclusions. We believe that this is what we're capable of doing, and we live our lives according to that. Then we hope that we have influence on other people and our ability to persuade them with the facts, with the logic, and then with this emotion of, "We have this little secret that we know about now, about how the world works, and we can live a life that's going to be much better with this knowledge that we have." And hopefully people will ask us, "Why do you live like this?" "Well, let me share with you what, what I've discovered about how this world really

works." I invite you to watch a podcast, read this book, go see what these people are saying here, and it's going to be different. But I think it's going to be much better for us to live with this knowledge than to live this pessimistic paralysis of anxiety. Because there's really never been a better time on this planet to be alive and to have a family. There really has never been a better time to do that.

- 9.71** You mentioned earlier the statistical improbability of the time period we're in and how humans are more used to egalitarianism and everyone being equal in their, not natural state, but how they are as cavemen. Assuming all of your fears came to fruition—that speech was on the decline, market was restricted, and things regressed—how do you think, if at all, that the world could return to the state that we're currently in, where we're prospering and ideas are freely shared and all of this is only getting better?
- 9.72** Well, things like that have happened before. Great civilizations such as Song China experienced tremendous amount of economic growth, the discovery of gunpowder and compass and the printing press, and then they declined for about 6- 700 years. And then until 1978, and then they started growing back up again, until the moment where they are now—the second largest economy in the world. So China, which was the center of global economic output, that was really the superpower, economic superpower of its day for a very long time, was able to decline but then bring itself up again. Europe, after the fall of the Roman Empire, went to Dark Ages. It took a very long time for Europe to bounce back and start growing again. We have seen things like that happen before. But what we are arguing is, "Why go through that Dark Age if we can avoid it, right?" Why not simply continue this process, and keep on getting more prosperous and more technologically advanced and safer from all those future problems that are going to face us?
- 9.73** I would just add to that that people, human beings, are extremely adaptable. Adaptation is this part of innovation, and we've got to trust one another that we're going to be able to adapt and come up with these ideas. Trust the other 8 billion people on the planet that we're going to be able to have this ability to accommodate and to deal with these problems, and to seek new opportunities, and to innovate around these challenges that we have today, because look at how far we've come. Can we take one more question? We got a guy here that's been wanting to ask a question all day long.
- 9.74** Well, this question was more directed at Marian in relation to what he mentioned about social class and environmentalism. I'm just curious what you make of the tendency for contemporary environmentalists to be increasingly upper-class bourgeois, contrary to the historical tendency for them to be the anti-elite, anti-bourgeois, anti-materialist. I'm thinking Al Gore in his carbon-footprint mansion or Leonardo DiCaprio in his emissions' private jets. Then also, do you attribute this to the increasingly irreligiosity of the upper class? And that contemporary environmentalism is merely an

expression of the religious instinct in the form of a neo-Romantic paganism, like a secular religion?

- 9.75** It's not necessarily just the upper class. Throughout the West, people who are maybe even poor, or maybe they are middle class, or maybe they are really rich—for everybody, environmentalism means much more than it used to to our ancestors. That's simply the result of the fact that we are much richer. We are much more distant from farming, for example, or from poverty. We don't really see true poverty unless we go and search for it in underdeveloped parts of the world. The entire Western world is a little bit more distant from the existential struggle for survival that you will still see in, for example, parts of Africa, like Democratic Republic of the Congo or Burundi or whatever. So, I think that's part of the reason why environmentalism has taken off, is that life is so good that now we have decided to resolve another problem that we didn't think was a problem before.
- 9.76** Now, the other thing that we talk about in the book is called the Maslow Hierarchy of Needs that you guys should look at. Basically, people in poor countries, they tend to be primarily concerned with shelter, food, clothing, that sort of thing, personal security, water. At the top of the pyramid is self-actualization, right? Once you have taken care of all of those needs at the bottom of the Maslow Hierarchy of Needs, and you become much more prosperous—when you have all of those base needs covered, you end up at the top of the pyramid, which is about self-actualization. That's when a rich Western person is looking for ways of expressing themselves. One way to express yourself can be by embracing this extreme form of environmentalism.
- 9.77** I do think that environmentalists have some good things to say, and I generally applaud what they are trying to do. Extreme environmentalists, however, they tend to be completely off the charts when it comes to their misanthropy, when it comes to their attitudes to economic development and so forth. You might very well say that, in the past, that self-actualization would probably materialize through an expression of a religious belief or whatever. But in a secular age, environmentalism basically steps into that role. And people pursue environmentalism, as I explained, with a very religious fervor, including beliefs in Garden of Eden, life of the planet before industrialization, saints like Greta Thunberg, devils like CO₂-emitting companies, and so forth.
- 9.78** So, as I explained in my lecture, I think that I see a lot of parallels between religious beliefs and environmental beliefs. And this is people partly are determined to believe in something, then once they stop believing in traditional religion, they have embraced a new one. Thank you very much.